ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

SONAR Operator (SONAR Op) 00324
Rank Qualification – SONAR Op –S3

Junior SONAR Operator (Jr SONAR Op)
Rank Qualification – SONAR Op –S2

RQ S3: ALFT
RQ S2: ALFU

MITE Course ID:
RQ S3: 122418
RQ S2 OJPR: 122419

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. QSP versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents.

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FOREWORD

1. This training is based on the specification found at :

a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-003 Non-Commissioned Member (NCM) General Specification dated 09 Feb 11;

c. A-PD-055-002/PP-002 Job Based Specification (JBS) for the SONAR Operator (SONAR Op) Change 8 – 2019-08-26;

d. Naval Orders – Royal Canadian Navy Individual and Collective/Operational Training Policy (NAVORD) 4500-0; and

e. Naval Personnel Training Order (NPTORD) 4500-22.

2. A trainee will achieve the qualification National Qualification (NQUAL) ALFT for in house course and ALFU for OJPR when the standards listed in all Performance Objectives have been met.

3. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / NFS(P)	RQ S3	122418	ALFT
NFS(A) / NFS(P)	RQ S2	122419	ALFU

CHAPTER 1: TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Junior SONAR Operator/ RQ S3 with associated RQ S2 OJPR. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with IAW NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration. The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of trainees enrolled on the course.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	452
Enabling Checks (ECs)	125
Performance Checks (PCs)	62
Actual Training Time (ATT) Sub-total	**639 pds** **78.87 trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	20
Training Administrative Time (TAT)	20
Training Travel Time (TTT)	0
Physical Fitness (PT)	54
Sub-total	**94 pds** **11.75 trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**733 pds** **91.6 trg days** **92 days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Course Officer	Lt(N)	UWWO (ALBV)	1	All
Senior Instructor	PO1	SONAR Op (ALBV)	1	All
ADAC Instructor	PO2/MS	Passive Acoustic Signature Analysis (AIXJ, ALBC)	1	PO 007
Instructor	PO2/MS	Oceanography/Sound Propagation Theory (ALBD)	1	PO 005
Instructor	PO2/MS	RQ MS Instructor (ALBC)	1 2 for PO 008	All
Instructor / Ship Staff	PO2/MS	RQ S1 WENG / EQ – WENG – SONAR Sub-Occupation Maintainer	1	PO 009 Only

Critical Training Resources	No of Days
PO 002 – On HMC Ship (12 periods)	2
PO 003 – NITES2R Trainer or HMC Ship (17 periods)	3
PO 003 – UWT Trainer or HMC Ship (19 periods)	3
PO 005 – DWAN workstations (86 periods)	13
PO 006 – SEACOT Trainer or HMC Ship (99 periods)	14
PO 007 – TRINITY ADAC support (217 periods)	31
PO 008 – SPS On HMC Ship or SEACOT (12 periods)	2
PO 008 – CANTASS Mission Simulator or SEACOT (78 periods)	12
PO 009 – On HMC Ship or Torpedo Tubes Trainer (8 periods)	1
PO 009 – SEACOT or HMC Ship or Pullen Building (NFS(A) only) (31 periods)	5

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 12;

   b. Minimum – 4; and

   c. Optimum – 8.

PREREQUISITES

5. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Valid FORCE Test for duration of course; and

b. Minimum level II security clearance (PO 001-005 enhanced reliability only).

COURSE REPORTING

6. Upon completion of the course, trainees shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

7. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

8. Upon completion of the RQ-SONAR OP-S3 course and achievement of the NQUAL ALFT, trainees will be issued the RQ-SONAR OP-S2 OJPR found in Annex K.

9. Trainees will have 12 months to complete their OJPR.

10. Upon successful completion of their respective RQ-SONAR OP-S2 OJPR, they will be granted the following MITE code 122419 and National Qualification ALFU.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 – COMPLY WITH SECURITY

1. **Performance**: Comply with Security

2. **Conditions**:

   a. Given:

      (1) Emergency Destruction Plan/Material;
      (2) Shredder;
      (3) Secure containers;
      (4) Access to restricted areas;
      (5) Secure publication accounting sheets;
      (6) Classified Standard Documents (CSD);
      (7) References;
      (8) Supervision; and
      (9) Assistance.

   b. Denied: Nil.

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall comply with security, IAW refs, by maintaining physical security IAW Ship Standing Orders (SSOs), National Defense Security Orders and Directives (NDSOD), and NAVORD 2115-1 Control and Accounting Procedures for Classified Standard Documents.

   a. Maintaining physical security; and
   b. Mustering Classified Standard Documents (CSD).

4. **Training Limitations**: Nil.

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EO 001.01 – Maintain Physical Security

001.01.01 – Physical Security Protocols

Conditions		Sequence
Unclassified Classroom with DWAN		Conducted after Course Introduction
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Understand Physical and technical Security requirements to include: a) Personnel Security b) Need-to-know c) Individual Security Screening d) Objectives and requirements e) Access to control and zoning f) Building safeguards g) Security containers and locks h) Sensitive discussion areas	Lecture Confirmation questions

Activity Steps	
	1) Instructor will present a Power Point on Physical Security requirements to include: (70 mins) a) Personnel Security b) Need-to-know c) Individual Security Screening d) Objectives and requirements e) Access to control and zoning f) Building safeguards g) Security containers and locks h) Sensitive discussion areas 2) Instructor will ask trainees confirmation questions and provide examples. (30 mins)

Resources	References
1) DWAN for Instructor with projector 2) Security Power Point presentation	1) SSOs Ships Standing Orders 2) NDSODs National Defence Security Orders and Directives 3) NAVORD 2115-1 Control and Accounting Procedures for Classified Standard Documents

Notes to Instructor
Nil

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EO 001.01 – Maintain Physical Security

001.01.02 – Mustering CSD Material

Conditions		Sequence
Unclassified Classroom with DWAN		Conducted after EO 001.01.01
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Mustering Classified Standard Document (CSD) material: a) Daily Musters b) Page checks	EDI Handout and practice

Activity Steps	
	1) Instructor will demonstrate how to muster a publication using a List of Effective Pages (LEP) IAW NAVORD 2115-1 to include: (10 mins) a) Mustering a publication / document using LEP; and b) Completing the publication checklist/muster. 2) Instructor will demonstrate how to muster CSD material IAW NAVORD 2115-1 to include: (10 mins) a) Mustering a Classified Standard Document(CSD) safe; and b) Completing the daily CSD Muster Sheets. 3) Instructor will distribute a publication with LEP and page muster sheet, and a daily CSD muster sheets. Trainees will be directed to conduct a muster of the publication using the LEP and fill out the page muster sheet. Trainees will practice conducting a CSD material muster and fill out the CSD muster sheet. (65 mins) 4) Instructor will ask trainees confirmation questions and provide examples. (15 mins)

Resources	References
1) Publication or Document with LEP and Muster sheet 2) CSD daily muster handout 3) Blank and example CSD muster sheets 4) Blank and example List of Effective Pages (LEP)	1) SSOs Ships Standing Orders 2) NDSODs National Defence Security Orders and Directives; and 3) NAVORD 2115-1 Control and Accounting Procedures for Classified Standard Documents

Notes to Instructor

1) Teaching point will require handouts for CSD daily muster, and List of Effective Pages (LEP) to be distributed in classroom; and
2) Instructor will demonstrate and act as a mentor during practice time to ensure trainees are on the right track.

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EO 001.01 – Maintain Physical Security

001.01.03 – Securing CSD Material

Conditions		Sequence
Unclassified Classroom with safe and computer		Conducted after EO 001.01.02
Time	**Classification**	**Instructor**
250 min	UNCLASSIFIED	RQ-SONAROP-MS or higher

Learning Objectives	Method
1) Securing CSD Material to include: a) Sergeant Greenleaf Lock b) Combination Safe 2) Securing a room/space to include: a) Windows b) Computers c) CSD Material d) Safes e) Filing cabinets f) Doors	EDI

Activity Steps	
	1) Instructor will demonstrate how to lock and unlock a sergeant Greenleaf lock. (10 mins) 2) Instructor will now have trainees practice locking and unlocking a Sergeant Greenleaf lock. (40 mins) 3) Instructor will demonstrate how to lock and unlock a combination safe. (10 mins) 4) Instructor will have trainees practice locking and unlocking a combination safe. (40mins) 5) Instructor will set up the room to be intentionally unsecured (Windows, safes, locks, CSD Material, filing cabinets, computers, doors). (5 mins) 6) Instructor will demonstrate the process of securing a room. (10mins) 7) Instructor will have trainees practice securing a room. (10 mins per trainee, 135 mins total, includes set up time)

Resources	References
1) Classroom with computer 2) Combination safe 3) Sergeant Greenleaf lock x 12	1) SSOs Ships Standing Orders 2) NDSODs National Defence Security Orders and Directives; and

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	3) NAVORD 2115-1 Control and Accounting Procedures for Classified Standard Documents.

Notes to Instructor

1) Instructor to set up unsecured classroom between each trainee
2) Instructor shall take trainees to a room with a safe if the provided room has none

EC 001.01 - Maintain Physical Security

Scope	Conditions	Sequence
EO 001.01	Unclassified Classroom	Conducted after EO 001.01.03
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will enter a room setup as follows: 1) Locked safe 2) Simulated classified publications on the desk 3) Open window(s) 4) Unlocked/unsecure computer **Performance:** Trainees must remedy the security concerns of the room IAW EC 001.01 checklist within 15 minutes includes setup and debrief **Output:** Trainees must achieve 5/5 SATs **Feedback:** immediate written and verbal debrief IAW EC 001.01 checklist upon completion of the "run"

Resources	References
1) Sergeant Greenleaf lock 2) Safe 3) Classified materials 4) Classroom with computer and window	1) SSOs Ships Standing Orders; 2) NDSOD – National Defense Security Orders and Directives; and 3) NAVORD 2115-1 - Control and Accounting Procedures for Classified Standard Documents.

Notes to Instructor

Setup for each trainee: Instructor will have to have a room set up with a locked safe, combination written down for the trainee, simulated classified publications on the desk space, open windows and unlocked computer.
Room windows can be simulated if no room with a window is available.

Limitations	Nil
Impact of Failure	Additional practice or retest

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Item	EC 001.01 MAINTAIN PHYSICAL SECURITY	SAT	UNSAT	Remarks
1. Survey Prescribed Area				
a.	Demonstrate how to open a Safe or Sergeant Greenleaf lock	☐	☐	
b.	Ensure Classified material secured in classified safe or container	☐	☐	
c.	Lock Safe(s) or container(s)	☐	☐	
d.	Close window(s)	☐	☐	
e.	Log off Computer(s)	☐	☐	

Score: /5

The trainee must achieve 5/5 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 001.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 001.01 with the trainee.	**Trainee** I have been debriefed on the result of EC 001.01.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 001 – Comply with Security

Scope	Conditions	Sequence
PO 001	Unclassified Classroom	Conducted after EC 001.01
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will enter a room setup as follows: 1) Locked safe 2) Classified publications on the desk 3) Open window 4) Unlocked computer Part 2-3: trainees will be provided a pub and muster sheets **Performance:** Trainees must remedy the security concerns of the room, and conduct a publication muster IAW PC 001 checklist Part 1&2: conducted with instructor 15 minutes per trainee, 180 minutes total Part 3: conducted in classroom 1 x 20 minutes. **Output:** To pass this PC, the trainee must achieve 8/10 SATs. **Feedback:** written feedback provided using PC 001 checklist upon completion of the assessment.

Resources	References
1) Unclassified Classroom 2) Computer (DWAN or standalone) 3) Safe or container 4) Pub Muster Sheets 5) Publication w/ List of Effective Pages 6) Confidential or Secret Lock	1) SSOs Ships Standing Orders; 2) NDSOD – National Defense Security Orders and Directives; and 3) NAVORD 2115-1 Control and Accounting Procedures for Classified Standard Documents.

Notes to Instructor
Nil

Limitations	
Impact of Failure	IAW NPTORD 4500-54.

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Item	PC 001 MAINTAIN PHYSICAL SECURITY	SAT	UNSAT	Remarks
1. Survey Prescribed Area				
a. Demonstrate how to open a Safe or Sergeant Greenleaf lock		☐	☐	
b. Ensure Classified material secured in classified safe or container		☐	☐	
c. Lock Safe(s) or container(s)		☐	☐	
d. Close window(s)		☐	☐	
e. Log off Computer(s)				
f. Explain the difference between each security zones to include; • Reception Zone • Access zone • Working zone • Restricted area		☐	☐	
2. Muster safe to include;				
a. Conduct publication muster IAW muster sheet		☐	☐	
b. Account for publications on muster sheets		☐	☐	
3. Muster a publications to include;				
a. Conduct a page muster IAW List of Effective Pages		☐	☐	
b. Sign the page muster sheet		☐	☐	

Score: /10 The trainee must achieve 8/10 SATs to pass this PC.

Positive Feedback	Improvement Feedback

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PC 001 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 001 with the trainee.		**Trainee** I have been debriefed on the result of PC 001.	
PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 002 – PERFORM NOISE ROUNDS (C0135)

1. **Performance**: Perform Noise Rounds (C0135).

2. **Conditions**:

   a. Given:

      (1) HMC ships;
      (2) Sound sources;
      (3) Noise monitoring organisation;
      (4) Logs, records and reports;
      (5) References; and
      (6) Assistance.

   b. Denied: Supervision.

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall perform noise rounds (C0135), IAW CFCD 109 Naval Vessel Signature Control Manual, by:

   a. locating prescribed area;

   b. surveying prescribed area;

   c. identifying noise shorts;

   d. fixing noise shorts using available resources (if possible);

   e. maintaining operational logs (Noise Rounds Log) (B0055); and

   f. reporting noise shorts to DNM/recommending noise remedies.

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4. **Training Limitations**:

a. If not HMC Ship is unavailable, conducting noise rounds will have to be simulated in classroom environment.

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EO 002.01 - Perform Noise Rounds

002.01.01 – Ship Noise, Quiet States and Noise Monitoring Organization

Conditions		Sequence
Unclassified Classroom with DWAN		
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Introduction to sources of noise to include: a) Ambient Noise; b) Self-Noise; and c) Radiated Noise. 2) Identifying SONAR Quiet states 3) Identifying the Noise Monitoring Organization and their duties and responsibilities. 4) Maintaining Noise Log to include: a) Noise Rounds Log; b) Noise Defect Log; and c) Artifact Log.	Reading Assignment 3 Star activity Floating Flipchart Best Summary EDI Confirmation Quiz

Activity Steps	
	1) Instructor will divide the class into 3 groups. Each group will conduct a reading assignment of CFCD 109 and be assigned one of the following chapters: a. Chapter 2 - Ambient Noise b. Chapter 3 – Self-Noise c. Chapter 4 – Radiated Noise. While each group is conducting the reading assignment they will conduct a 3 Star activity identifying; 3 specific sources of noise, what the Threat to the ship is, and how to counter the threat. (30 min) 2) Instructor will have each group present their findings from the previous activity. Trainees will be allowed to ask questions to each group and instructor will provide feedback. (20 min) 3) The instructor will then have the 3 groups conduct a reading assignment of CFCD 109 Chapter 23 on Quiet Running Bill and Quiet States. (20 mins) 4) The instructor will set up 3 floating flipcharts/whiteboard stations labelled: a. NORMAL Quiet State b. SONAR Quiet State c. ULTRA Quiet State. Each group will be assigned a Quiet State flipchart/whiteboard station and asked to fill out the restrictions or no restrictions during the Quiet State.

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Trainee may use CFCD 109 as reference. Each group will then be asked to explain the findings of one of the stations. Instructor will provide additional comments / feedback on the Quiet States. (20 mins)

5) Instructor will display the Noise Monitoring Organization Power Point Slide. Instructor will have the trainees conduct a Brief Summary exercise. Instructor will assign trainees or groups one or more of the following positions within the Noise Monitoring Organization:
    a. Unit Signature Reduction Officer (USRO);
    b. Noise Control Officer (NCO);
    c. Noise Engineering Officer (NEO);
    d. Noise Detection Officer (NDO);
    e. Senior SONAR OP (A/NDO);
    f. Departmental Noise Detection Officer (DNDO); and
    g. Departmental Noise Control Monitor (DNCM).

    Each trainee will conduct a reading assignment in CFCD 109 Chapter 23 Signature Monitoring and Control on their assigned position listed above. From the reading assignment, they will identify 3 responsibilities for their position. (10 min)
6) Instructor will then have each trainee / group present their Noise Monitoring Organization position to the class. Trainees will be allowed to ask questions and instructor will provide feedback. (20 min)
7) The instructor will provide an example of a Noise Rounds log and explain how and when it is to be filled out correctly. (10 min)
8) Instructor will have a hard copy printed out of a blank Noise Rounds Log for the class. The instructor will assign each trainee or group a Department, a SONAR Quiet State, a date and time, and have them to fill out the Noise Rounds log IAW with the example provided. Instructor will take up and provide feedback on the Noise Rounds Log. (15 min)
9) The instructor will provide an example of a Noise Defect log and explain how and when it is to be filled out correctly. (10 min)
10) Instructor will have a hard copies printed out of a blank Noise Defect Log for the class. The instructor will assign each trainee or group a Department, a date and time, a Noise defect/deficiency, and an Action to be taken. Trainees will have to fill out the Noise Defect log IAW with the example provided. Instructor will take up and provide feedback on the Noise Defect Log. (15 min)
11) Instructor while onboard ship will show and explain what the ship's Artifact Log is, where it can be found and why it is held. (5 min)
12) Instructor will conduct a Confirmation Quiz. Each trainee will be given a blank copy of the Noise Monitoring Organization diagram. Trainees will be asked to fill in all the position, both the short and long form names, of the organization. (20 min)

	13) Instructor will have the trainee pass their Confirmation Quiz to another trainee and take up the answers. Instructor will provide feedback on any incorrect answers. (5 min)

Resources	References
1) DWAN for each trainee and instructor with projector 2) CFCD 109 3) Flipchart / Whiteboard with markers 4) Noise Monitoring Organization Diagram 5) Noise Rounds Logs Example 6) Noise Rounds Log Template 7) Noise Defect Log Example 8) Noise Defect Log Template 9) HMC Ships Artifact Log (held onboard ship) 10) Noise Monitoring Organization Confirmation Diagram	1) CFCD 109

Notes to Instructor

Instructor can demonstrate Ship's Artifact Log while onboard ship.

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EO 002.01 - Perform Noise Rounds

002.01.02 – Conducting Noise Rounds

Conditions		Sequence
HMC Ship or Unclassified Classroom		Conducted after EO 002.01.01
Time	**Classification**	**Instructor**
200 min (with travel time)	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Identifying onboard ship noise defects a) areas of responsibility b) types of noise defects c) causes of noise defects d) effects of noise defects 2) Rectifying noise defects 3) How to survey prescribed area using investigation techniques	Tour of ship EDI

Activity Steps	
	1) Instructor will take the class on a tour of a HMC Ship. While on the tour the instructor will point out common noise defects observed onboard ship. They will also identify the following: (100 min) a) locations of SONAR Quiet State signs; b) doors / hatches; c) rubber/shock mountings; d) rubber pucks / wedges; e) athletic equipment; f) broadcast / piping system; g) workshops/maintenance areas; h) recreation equipment (TV's/stereos); i) Galley equipment. Instructor will demonstrate what causes these noise defects, explain how they can negatively impact the ships noise signature and provide recommendations on how to easily rectify the noise defects. 2) Instructor will select a space/compartment onboard ship that is available and has noise deficiencies present and/or able to set up with noise deficiencies and will demonstrate to the class how to survey a prescribed area using noise investigation techniques to include: (20 min) a) Listening for moving/rattling objects; b) Using proper door and hatch procedures; c) Ensuring loose objects are secured and/or shock mounted. 3) Instructor will now conduct a practical exercise, one at a time, trainees will be asked to enter the prescribed space onboard ship and conduct noise

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	rounds identifying a minimum of 3 potential noise defects and writing them down on a notepad handing it to the instructor before leaving the space. (60 mins) 4) Once all trainees have done so, the instructor will discuss each identified noise defects indicating which ones were common among the class, and which were unique. The class will then discuss as a group how to rectify some of the more commonly identified noise defects. (20 mins)

Resources	References
1) HMC Ship 2) Sticky Pad or Paper	1) CFCD 109

Notes to Instructor

Any class of HMC Ship can be utilized for this instruction. Instructor may utilize ships passage ways, common areas or multiple spaces to conduct activities.
If no ship is available, instructor may set up classroom to conduct surveying of spaces for noise.

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EC 002.01 - Perform Noise Rounds

Scope	Conditions	Sequence
EO 002.01	HMC Ship	Conducted after EO 002.01.02
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Scenario	Trainees are conducting noise rounds
Assessment Steps	**Input:** Trainees are taken to a space onboard an HMC Ship (any class) with noise faults, provided a Noise Defect log and resources to rectify the noise faults (if required). **Performance:** Trainees are given 15 min each to conduct rounds IAW EC 002.01 checklist and fill out the noise defect log. Additional 20 minutes for instructor setup / travel time. **Output:** the trainee must achieve 7/9 SATs. **Feedback:** Verbal and written feedback IAW EC 002.01 checklist upon completion of the assessment

Resources	References
1) HMC Ship (any class) 2) Noise Defect Template 3) Noise dampening material (i.e. rubber pucks, tape, etc.)	1) CFCD 109

Notes to Instructor
1) Assessment will occur on a ship 2) Trainees will be given an area to search and identify the noise shorts and the measures to correct the faults 3) Instructors must review associated resources and references to prepare for this assessment

Limitations	If no HMC Ship is available, instructor will have to simulate in classroom.
Impact of Failure	Additional practice or remedial instruction IAW NPTORD 4500-54.

Item	EC 002.01 PERFORM NOISE ROUNDS	SAT	UNSAT	Remarks
1. Survey prescribed area				
Verbally describe areas of responsibility		☐	☐	
Carries out search / survey of area space		☐	☐	
2. Identify noise shorts				
Verbally identify types of noise		☐	☐	
Identify and describe types of noise shorts		☐	☐	
Identify and describe causes of noise shorts		☐	☐	
Identify and describe effects of noise shorts		☐	☐	
3. Fix noise shorts				
Fix noise shorts using available resources		☐	☐	
4. Maintain operational logs				
Recording noise shorts		☐	☐	
Reporting to Departmental Noise Monitor		☐	☐	

Score: /9

The trainee must achieve 7/9 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 002.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002.01 with the trainee.		**Trainee** I have been debriefed on the result of EC 002.01 with the trainee.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 002 - Perform Noise Rounds

Scope	Conditions	Sequence
PO 002	HMC Ship (any class)	Conducted after EC 002.01
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Trainees are taken to a space onboard an HMC Ship (any class) with noise faults, provided a Noise Defect log and resources to rectify the noise faults (if required). **Performance:** Trainees are given 15 min each to conduct rounds IAW EC 002.01 checklist and fill out the noise defect log. Additional 20 minutes for instructor setup / travel time. **Output:** the trainee must achieve 7/9 SATs. **Feedback:** Verbal and written feedback IAW PC 002 checklist upon completion of the assessment.

Resources	References
1) HMC Ship (any class) 2) Noise Defect Template 3) Noise dampening material (i.e. rubber pucks, tape, etc.)	1) CFCD 109

Notes to Instructor

1) Assessment will occur on a ship
2) Trainees will be given an area to search and identify the noise shorts and the measures to correct the faults
3) Instructors must review associated resources and references to prepare for this assessment

Limitations	If no HMC Ship is available, instructor will have to simulate in classroom.
Impact of Failure	IAW NPTORD 4500-54.

Item	PC 002 PERFORM NOISE ROUNDS	SAT	UNSAT	Remarks
1. Survey prescribed area				
Verbally describe areas of responsibility		☐	☐	
Carries out search / survey of area space		☐	☐	
2. Identify noise shorts				
Verbally identify types of noise		☐	☐	
Identify and describe types of noise shorts		☐	☐	
Identify and describe causes of noise shorts		☐	☐	
Identify and describe effects of noise shorts		☐	☐	
3. Fix noise shorts				
Fix noise shorts using available resources		☐	☐	
4. Maintain operational logs				
Recording noise shorts		☐	☐	
Reporting to Departmental Noise Monitor		☐	☒	

Score: /9

The trainee must achieve 7/9 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 002 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 002 with the trainee.		**Trainee** I have been debriefed on the result of PC 002 with the trainee.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 003 - OPERATE ANCILLARY ACOUSTIC EQUIPMENT

1. **Performance**: Operate Ancillary Acoustic Equipment (C0025).

2. **Conditions:**

   a. Given:

      (1) Ancillary acoustic equipment;
      (2) References;
      (3) Chronoscope; and
      (4) Logs and records.

   b. Denied:

      (1) Assistance; and
      (2) Supervision.

   a. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall operate ancillary acoustic equipment (C0025) by:

   a. Performing pre-sail checks on equipment (B0005) IAW NCPM 600 by:

      i. flashing up equipment IAW Digital Underwater Communications System UWT (DUWCS TUUM-6);
      ii. setting up oceanography equipment (C0012) IAW the MK8-F Surface Ship Bathythermograph Data Acquisition System manual;
      iii. completing all operator functional checks IAW DUWCS TUUM-6, the MK8-F Surface Ship Bathythermograph Data Acquisition System manual, and NCPM 600; and
      iv. shutting down equipment IAW DUWCS TUUM-6, the MK8-F Surface Ship Bathythermograph Data Acquisition System manual, and NCPM 600.

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b. Launching Bathythermograph (XBT) IAW the MK8-F Surface Ship Bathythermograph Data Acquisition System manual;

c. Analyzing XBT/XSV forms (C0040), IAW the MK8-F Surface Ship Bathythermograph Data Acquisition System manual, the Naval Integrated Tactical Environmental System (NITES) II Redesign (N2R), Standard Operating Procedures (SOP), and the Ocean Navigator Website, by:

   i. operating oceanographic equipment (C0030) IAW the Naval Integrated Tactical Environmental System (NITES) II Redesign (N2R), Standard Operating Procedures (SOP);
   ii. recording XBT/XSV information IAW the Naval Integrated Tactical Environmental System (NITES) II Redesign (N2R), Standard Operating Procedures (SOP);
   iii. completing standard Bathymetric forms (IGOSS) (C0045) IAW NCPM 600; and
   iv. drafting Bathy messages IAW NCPM 600.

d. Communicating externally on Underwater Telephone, IAW DUWCS TUUM-6, NCPM 600 and Allied Exercise Publication (AXP)-01, by:

   i. employing Mark-Snap ranging procedures IAW NCPM 600 and AXP-01;
   ii. conducting Gertrude check IAW AXP-01;
   iii. recognizing Low grade codes IAW AXP-01;
   iv. identifying Electronic Sound Underwater Signal (ESUS) codes IAW AXP-01; and
   v. Performing Search and Rescue (SAR)/Distress/SUBSMASH communication procedures (A0020).

e. Maintaining operational logs (B0055).

4. Training Limitations

a. Environmental conditions that exist at sea between ships is simulated and cannot be reproduced in the classroom.

EO 003.01 - Initialize Ancillary Acoustic Equipment

003.01.01 – Operate Oceanographic Equipment

Conditions		Sequence
Oceanography (NITES2R) Trainer or HMC Ship		
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) How to launch a Bathy procedure 2) Flashing up and shutting down equipment to include: a. flashing up MK-8F Surface Ship Bathythermograph Data Acquisition System (MK8-F) b. shutting down MK-8F 3) Completing all operator function checks to include: a. Setting up oceanographic equipment b. Initiating a new probe drop; c. Selecting the probe type; d. Verifying the data file name; e. Updating or verifying the setup parameters f. Loading the probe; g. Launching the probe; and h. Waiting for terminal depth or stopping data acquisition.	Video EDI

Activity Steps	
	1) Instructor will show the class the video “Bathy CPF” on the MK-8F Bathythermograph and how to set up to launch a bathy. (10 min) 2) Instructor will explain and demonstrate how to turn on / flash up and shut down the MK-8F Surface Ship Bathythermograph Data Acquisition System (MK8-F) IAW EC 003.01 Checklist. (10 min) 3) Trainees will practice turning on / flashing up and shut down the MK-8F Surface Ship Bathythermograph Data Acquisition System (MK8-F) IAW EC 003.01 Checklist. (20 min) 4) Instructor will explain and demonstrate how to set up MK-8F operating parameters IAW EC 003.01 Checklist to include: (20 min) a. Setting up oceanographic equipment b. Initiating a new probe drop; c. Selecting the probe type; d. Verifying the data file name; e. Updating or verifying the setup parameters f. Loading the probe;

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	g. Launching the probe; and h. Waiting for terminal depth or stopping data acquisition. 5) Trainees will practice setting up MK-8F operating parameters IAW EC 003.01 Checklist to include: (60 min) a. Setting up oceanographic equipment b. Initiating a new probe drop; c. Selecting the probe type; d. Verifying the data file name; e. Updating or verifying the setup parameters 6) Instructor will explain and demonstrate a new probe drop IAW C-69-894-001/MS-001 to include: (20 min) a. Loading the probe; b. Launching the probe; and c. Waiting for terminal depth or stopping data acquisition. d. Review data to ensure successful launch 7) Trainees will explain and demonstrate a new probe drop IAW C-69-894-001/MS-001 to include: (60 min) a. Loading the probe; b. Launching the probe; and c. Waiting for terminal depth or stopping data acquisition. d. Review data to ensure successful launch

Resources	References
1) 1 x Instructor NITES2R laptop with MK-21 software installed with Projector 2) 12 x NITES2R laptops with Mk-21 software installed for trainees 3) MK-8F Bathythermograph recorder/processor; 4) Launcher 5) XBT/XSV probes 6) “Bathy CPF” video	1) C-69-894-001/MS-001 MK8-F Surface Ship Bathythermograph Data Acquisition System manual

Notes to Instructor

During Activity #1, Instructor to ensure trainees know how to open/close the Ball Valve. Instructor will use the NITES2R laptops with the MK-21 XBT software loaded on them to conduct this training or onboard HMC Ship if available.

EC 003.01 – Initialize Ancillary Acoustic Equipment

Scope	Conditions	Sequence
EO 003.01	NITES2R Classroom or HMC Ship	Conducted after EO 003.01.01
Time	**Classification**	**Instructor**
150 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Trainees will be asked to Operate Oceanographic Equipment. **Performance:** Trainees will be required to initialize the Oceanographic Equipment, launch Bathythermograph (XBT/XSV), and shutdown the Oceanographic Equipment. Each trainee will have 10 mins to conduct this assessment. Additional 30 mins for instructor setup / debrief time / travel time. **Output:** The trainee must achieve 11/14 SATs to pass this EC. **Feedback:** Written feedback upon completion of the assessment IAW EC 003.01 checklist.

Resources	References
1) NITES2R Classroom or HMC Ship 2) WIN MK21 Program 3) XBT/XSV data set 4) XBT/XSV Probes	1) C-69-894-001/MS-001 MK8-F Surface Ship Bathythermograph Data Acquisition System manual

Notes to Instructor
None

Limitations	If conducting assessment in classroom, launching probe will need to be simulated.
Impact of Failure	Retrain and retest

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Item	EC 003.01 Operate Oceanographic Equipment	SAT	UNSAT	Remarks
1. Flash up MK-8F Bathythermograph:				
a. turn on the computer		☐	☐	
b. double-click the WinMK21 icon		☐	☐	
2. Set up operating parameters				
a. open WinMK21 Main Window		☐	☐	
b. ensure WinMK21 is in Data acquisition mode		☐	☐	
3. Initiate a new probe drop				
a. click the New Drop button or choose New Drop from the File menu		☐	☐	
4. Select the probe type				
a. choose Probe from the Options menu, and make appropriate probe selection		☐	☐	
5. Verify the data file name				
a. Verify the correct file name for the data and the folder in which the file will be		☐	☐	
6. Load/Launch the probe				
a. Load the probe		☐	☐	
b. Pull the release pin out of the canister		☐	☐	
c. Wait for the probe to complete its descent or Stop data acquisition, by choosing *Terminate* from the *Actions* menu		☐	☐	
7. Record XBT/XSV information IAW METOC standards				
a. Select correct profile type		☐	☐	
b. Review data to ensure a successful launch		☐	☐	
8. Shut down MK-21 Bathythermograph:				
a. Close WinMk21 program		☐	☐	
b. turn off the computer		☐	☐	

Score: /14

The trainee must achieve 11/14 SATs to pass this EC.

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Positive Feedback	Improvement Feedback

EC 003.03 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 003.03 with the trainee.		**Trainee** I have been debriefed on the result of EC 003.03.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 003.02 - Analyze XBT/XSV Forms

003.02.01 – Draft XBT/XSV IGOSS Forms

Conditions		Sequence
Unclassified Classroom with DWAN or NITES2R Classroom		
Time	**Classification**	**Instructor**
250 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Bathy Probe types 2) Recording XBT/XSV information; 3) Completing Standard Bathymetric forms 4) Discuss Bathy messages	EDI

Activity Steps	
	1) Instructor will hold guided discussion and review XBT/XSV information, placing emphasis on the accuracy of the measurements IAW NCPM 600. (30 min) 2) Instructor will explain and demonstrate completing an IGOSS form, placing emphasis on the accuracy of the measurements. (25 min) 3) Trainees will practice completing an IGOSS form, placing emphasis on the accuracy of the measurements. (50 min) 4) Instructor will explain and demonstrate completing a VELO form, placing emphasis on the accuracy of the measurements. (25 min) 5) Trainees will practice completing a VELO form, placing emphasis on the accuracy of the measurements. (45 min) 6) Instructor will explain drafting a bathy message. (15 min) 7) Trainees will practice completing bathy log. (15 min) 8) Instructor will explain and demonstrate completing the monthly expenditure log. (15 min) 9) Trainees will practice completing the monthly expenditure log. (15 min)

Resources	References
1) Bathythermograph recorder; 2) Example bathy profile data. 3) Bathy launcher; 4) XBT/XSV probes; 5) Instructor DWAN access; 6) Templates for digital IGOSS, VELO, Bathy message forms; 7) Example bathy message.	1) NCPM 600 2) Expendable Bathythermograph / Expendable Sound Velocitmeter (XBT/XSV) Expendable Profiling Systems 3) MILOC 4) TRINITY METOC Website

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Notes to Instructor

None

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EC 003.02 – Analyze XBT/XSV Forms

Scope	Conditions	Sequence
EO 003.02	Unclassified Classroom with DWAN or NITES2R Classroom	Conducted after EO 003.02.01
Time	**Classification**	**Instructor**
50 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be asked to complete XBT/XSV forms **Performance:** Trainees will be required to complete XBT/XSV forms and bathy log. Each trainee will have 50 min to conduct this EC. **Output:** The trainee must achieve 10/13 SATs to pass this EC. **Feedback:** Written feedback upon completion of the assessment IAW EC 003.02 checklist.

Resources	References
1) Classroom or Ship 2) WIN MK21 Program 3) XBT/XSV data set 4) XBT / XSV Forms 5) XBT/XSV Probes	1) NCPM 600 2) Expendable Bathythermograph / Expendable Sound Velocitmeter (XBT/XSV) Expendable Profiling Systems 3) MILOC 4) TRINITY METOC Website

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and Retest

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Item	EC 003.03 ANALYZE XBT/XSV FORMS	SAT	UNSAT	Remarks
1. Complete IGOSS form				
a. Date		☐	☐	
b. Time		☐	☐	
c. Quadrant		☐	☐	
d. Latitude		☐	☐	
e. Longitude		☐	☐	
f. Select correct probe		☐	☐	
g. Correct bathy points		☐	☐	
8. Complete VELO form				
a. Date		☐	☐	
b. Time		☐	☐	
c. Quadrant		☐	☐	
d. Latitude		☐	☐	
e. Correct VELO points		☐	☐	
9. Bathy log				
a. Complete bathy log		☐	☐	

Score: **/13** The trainee must achieve 10/13 SATs to pass this EC.

Positive Feedback	Improvement Feedback

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EC 003.03 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 003.03 with the trainee.		**Trainee** I have been debriefed on the result of EC 003.03.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 003A - Operate Ancillary Acoustic Equipment

Scope	Conditions	Sequence
EO 003.01-003.02	Unclassified Classroom with DWAN or NITES2R Classroom	Conducted after EC 003.02
Time	**Classification**	**Instructor**
200 mins	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be asked to initialize, launch and analyze XBT/XSV **Performance:** trainees will be required to operate ancillary acoustic equipment and correctly draft a Bathy Message IAW PC 003 checklist. Trainees have 10 min each for the equipment operation and 50 min to conduct analysis and write up. Additional 30 minutes is for instructor setup / debrief time. **Output:** The trainee must achieve 15/19 SATs to pass this EC. **Feedback:** Written feedback IAW PC 003 checklist within 48hrs.

Resources	References
1) Classroom or Ship 2) WIN MK21 Program 3) XBT / XSV Forms	1) C-69-894-001/MS-001 MK8-F Surface Ship Bathythermograph Data Acquisition System manual 2) NCPM 600

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

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Item	PC 003 OPERATE ANCILLARY ACOUSTIC EQUIPMENT	SAT	UNSAT	Remarks
1. Flash up MK-8F Bathythermograph:				
a. Turn on the computer		☐	☐	
b. Initialize WinMK21 program		☐	☐	
2. Set up operating parameters				
a. Open WinMK21 Main Window		☐	☐	
b. Ensure WinMK21 is in Data acquisition mode		☐	☐	
3. Initiate a new probe drop				
a) Click the New Drop button		☐	☐	
4. Select the probe type				
a. Choose Probe from the Options menu		☐	☐	
5. Verify the data file name				
a. Verify the correct file name for the data and the folder in which the file will be		☐	☐	
6. Load/Launch the probe				
a. Load a probe		☐	☐	
b. Pulling the release pin out of the canister		☐	☐	
c. Wait for the probe to complete its descent or Stop data acquisition, by choosing *Terminate* from the *Actions* menu		☐	☐	
7. Record XBT/XSV information IAW METOC standards				
a. Select correct profile type		☐	☐	
b. Review data to ensure a successful launch		☐	☐	
8. Complete standard Bathymetric forms (C0045)				
a. Complete a IGOSS form		☐	☐	
b. Complete a VELO form		☐	☐	
a. Fill in Bathy log		☐	☐	
10. Shut down MK-21 Bathythermograph:				
a. Close WinMK21 program		☐	☐	
b. Turn off the computer		☐	☐	

Score: /19

The trainee must achieve 15/19 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 003 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 003 with the trainee.		**Trainee** I have been debriefed on the result of PC 003	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 003.03 - Communicate Externally on Underwater Telephone

003.03.01 – Operate and Communicating Underwater Telephone

Conditions		Sequence
Classified Classroom / HMC Ship or Pullen Building (NFS(A) Only)		
Time	**Classification**	**Instructor**
450 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Operate UWT 2) Recognize Low Grade Codes 3) Identifying ESUS codes (US Navy video, YouTube) 4) Conducting Gertrude check; 5) Employing Mark-Snap ranging procedures; 6) Employing Uncle Joe procedure; 7) Employing Hooker procedure;	EDI Video

Activity Steps	
	1) Instructor will show video on identifying different ESUS codes (10 min) 2) Instructor will explain and demonstrate operating UWT IAW UWT SOP to include; (50 min) a) Flash up b) Operating Procedures to include; i) Channel Selection ii) Voice Operation iii) Message Function (record and playback0 iv) TTY Function v) Replay Data vi) Modem Function (text message) vii) Distance viii) Pinger c) Adjust Receiver Gain and TX Power d) Shut Down 3) Instructor will provide each trainee with UWT SOP, trainees will then practice individually operating UWT. (200 min) 4) Instructor will explain and demonstrate Gertrude check (UWT) (20 min) 5) Trainees will practice Gertrude check individually or in pairs (30 min) 6) Instructor will explain and demonstrate Mark-Snap ranging procedures to include; (20 min) a) Measuring time; and b) Calculating ranges. 7) Trainees will practice Mark-Snap Procedures to include (30 min)

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	a) Measuring time; and b) Calculating ranges. 8) Instructor will explain and demonstrate Uncle Joe procedure. (15 min) 9) Trainees will practice Uncle Joe procedures individually or in pairs. (30 min) 10) Instructor will explain and demonstrate Hooker procedures. (15 min) 11) Trainees will practice Hooker procedures individually or in pairs. (30 min)

Resources	References
1) Underwater Telephone 2) Chronoscope/Stopwatch 3) UWT SOP 4) Gertrude (UWT) code words and phrases handouts 5) ESUS codes handout	1) DUWCS TUUM-6 2) NCPM 600 3) AXP 01

Notes to Instructor
Nil

EC 003.03 – Communicate Externally on Underwater Telephone

Scope	Conditions	Sequence
EO 003.03	Classified Classroom / HMC Ship or Pullen Building (NFS(A) Only)	Conducted after EO 003.03.01
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainee is provided with code words and voice procedures used on the Underwater Telephone. **Performance:** Each trainee will have 15 min where they are required to communicate externally on Underwater Telephone and fill in the log IAW EC 003.04 Checklist **Output:** The trainee must achieve 25/33 SATs to pass this EC. **Feedback:** Immediate written and verbal feedback IAW EC 003.04 checklist upon completion of the assessment.

Resources	References
1) Classroom or Trainer 2) Underwater telephone (headset and microphone) 3) References 4) ESUS Audio files 5) Pen and Paper	1) DUWCS TUUM-6 2) NCPM 600 3) AXP 01

Notes to Instructor
Nil

Limitations	Trainer availability
Impact of Failure	Retrain and retest

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Item	EC 003.03 COMMUNICATE EXTERNALLY ON UNDERWATER TELEPHONE	SAT	UNSAT	Remarks
1. Conduct Flash-up				
a.	Ensure power is on and green light is illuminated.	☐	☐	
b.	Turn key "ON" in OPS and Bridge.	☐	☐	
c.	Plug in headset.	☐	☐	
d.	Selected channel is in voice mode.	☐	☐	
2. Conduct messages transmission				
a.	Recording a message and TX to include;			
	• Press RECORD	☐	☐	
	• Push-to-Talk on microphone	☐	☐	
	• Press REPLAY than PLAY	☐	☐	
	• Press FILE than SAVE	☐	☐	
	• Choose DELAY	☐	☐	
	• Push CALL to transmit.	☐	☐	
b.	Choose Pre-set messages to TX to include;			
	• Push MESSAGE mode than OK	☐	☐	
	• Press FILE to choose message than press OK	☐	☐	
	• Choose DELAY x2	☐	☐	
	• Choose a DELAY value between 30 and 300	☐	☐	
	• Press PLAY to confirm msg	☐	☐	
	• Press OK than CALL to TX	☐	☐	

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3. Conduct Gertrude check			
a. Conduct communications check with another unit	☐	☐	
4. Employ Mark-Snap ranging procedures			
a. Measure time between TX's IAW procedure	☐	☐	
b. Calculate ranges	☐	☐	
5. Employ Uncle Joe procedures			
a. Employ Uncle Joe procedures	☐	☐	
6. Employ Hooker procedures			
a. Employ Hooker procedures	☐	☐	

Score: /33

The trainee must achieve 25/33 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 003.03 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 003.03 with the trainee.	**Trainee** I have been debriefed on the result of EC 003.03.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 003B - Communicate Externally on Underwater Telephone

Scope	Conditions	Sequence
003.03	Classified Classroom / HMC Ship or Pullen Building (NFS(A) Only)	Conducted after EC 003.03
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: trainee is provided with code words and phrases used on the Underwater Telephone. **Performance:** Each trainee will have 15 min where they are required to communicate externally on Underwater Telephone and fill in the log IAW EC 003.04 Checklist **Output:** The trainee must achieve 22/29 SATs to pass this EC. **Feedback:** Immediate written and verbal feedback IAW EC 003.04 checklist upon completion of the assessment.

Resources	References
1) Classroom or Trainer 2) Underwater telephone (headset and microphone) 3) References 4)ESUS Audio files 5) Pen and Paper	1) DUWCS TUUM-6 2) NCPM 600 3) AXP 01

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and retest

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Item	PC 003 COMMUNICATE EXTERNALLY ON UNDERWATER TELEPHONE	SAT	UNSAT	Remarks
1. Conduct Flash-up				
a.	Ensure module is powered and push button "Power on".	☐	☐	
b.	Turn key "ON" in OPS and Bridge.	☐	☐	
c.	Plug in headset.	☐	☐	
d.	Ensure Programmable Channel recording is active.	☐	☐	
e.	Selected channel is in voice mode.	☐	☐	
f.	Ensure "Status On" is green.	☐	☐	
2. Conduct messages transmission				
a.	Recording a message and TX to include;			
	• Press RECORD	☐	☐	
	• Push-to-Talk on microphone	☐	☐	
	• Press REPLAY than PLAY	☐	☐	
	• Press FILE than SAVE	☐	☐	
	• Choose DELAY	☐	☐	
	• Push CALL to transmit.	☐	☐	
b.	Choose Pre-set messages to TX to include;			
	• Push MESSAGE mode than OK	☐	☐	
	• Press FILE to choose message than press OK	☐	☐	
	• Choose DELAY x2	☐	☐	
	• Choose a DELAY value between 30 and 300	☐	☐	

• Press PLAY to confirm message	☐	☐	
• Press OK than CALL to TX	☐	☐	
3. Conduct Gertrude check			
a. Conduct communications check with another unit	☐	☐	
4. Employ Mark-Snap ranging procedures			
a. Measure time between TX's IAW procedure	☐	☐	
b. Calculate ranges	☐	☐	
5. Employ Uncle Joe procedures			
a. Employ Uncle Joe procedures	☐	☐	
6. Employ Hooker procedures			
b. Employ Hooker procedures	☐	☐	
7. Communicate using Low grade codes			
a. Use appropriate 3 letter codes	☐	☐	
b. De-Code 3 letter codes IAW AXP-1	☐	☐	
8. Identify Electronic Sound Underwater Signal (ESUS) code			
a. Identify Setting	☐	☐	
9. Log UWT communications			
a. Enter time of communication	☐	☐	
b. Enter unit(s) conducting communication	☐	☐	
c. Enter communication / response(s)	☐	☐	

Score: /29

The trainee must achieve 22/29 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 003 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 003 with the trainee.		**Trainee** I have been debriefed on the result of PC 003.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 004 – LAUNCH UNDERWATER EXPENDABLES

1. **Performance**: Launch Underwater Expendables (C0035).

2. **Conditions**:

   a. Given:

      (1) Sonobuoys;
      (2) Expendable Mobile ASW Training Targets (EMATT);
      (3) Portable Target Programmer (PTP);
      (4) Portable PC loaded with appropriate software;
      (5) Acoustic Counter Measures (ACMs);
      (6) ESUS;
      (7) Maritime Command Ammunition Allowance List (BAAL);
      (8) Expenditure logs and inventory records;
      (9) Tools and safety equipment;
      (10) Internal communications equipment;
      (11) References;
      (12) Mentoring; and
      (13) Assistance, for Naval Acoustic Electromechanical Beacon (NAEB) preps.

   b. Denied: Supervision

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall launch underwater expendables (C0035), IAW Sonobuoy Reference Guide, NCPM 600 and the Operator's Manual MK39 EMATT Advanced Acoustic Threat by:

   a. Safely storing acoustic expendables;

   b. Safely handling acoustic expendables;

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c. launching acoustic expendables, by preparing acoustic expendables for launch (C0050);

d. maintaining logs and records (B0055); and

e. verifying the shipboard stock against the BAAL entitlement (C0055).

4. **Training Limitations**: Nil

EO 004.01 – Underwater Expendables

004.01.01 - Safe Storage and Handling of Underwater Expendables

Conditions		Sequence
Unclassified Classroom with DWAN		
Time	**Classification**	**Instructor**
150 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Safely storing Underwater Expendables to include: a) Sonobuoys b) Expendable Mobile ASW Training Target (EMATT) c) Acoustic Device Countermeasures (ADCs) d) Naval Acoustic Electromechanical Beacons (NAEBs) e) Mk 84 ESUS 2) Safely Handling Underwater Expendables to include: a) Sonobuoys b) Expendable Mobile ASW Training Target (EMATT) c) Acoustic Device Countermeasures (ADCs) d) Naval Acoustic Electromechanical Beacons (NAEBs) e) Mk 84 ESUS 3) Maintaining operation logs 4) Verifying the shipboard stock against the BAAL entitlement.	Lecture 3 Star EDI Ship tour (If Available)

Activity Steps	
	1) Instructor will present PowerPoint Presentation on Safe Stowage and handling of Underwater expendables to include: (20 mins) a) Sonobuoys b) Expendable Mobile ASW Training Target (EMATT) c) Acoustic Device Countermeasures (ADCs) d) Naval Acoustic Electromechanical Beacons (NAEBs) e) Mk 84 ESUS 2) Instructor will break the class into groups (1-3 Trainees), each group will be given the technical document for one of the expendables listed above. 3) Instructor will have groups read the technical references' safety sections. Trainees will conduct a 3 Star Activity where they will identify 3 important safety considerations when storing or handling their expendable. (15 mins) 4) Trainees will present their 3 star points. Instructor and trainees may provide comments/feedback. (15 mins) 5) Instructor will show the trainees the expenditure log(s) and demonstrate how to fill them out, using the example provided. (10 mins)

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	6) Instructor will hand out log(s) to trainees Instructor will provide a scenario, and trainees will fill out the logs. (20 mins) 7) Trainees will exchange their logs with one another and provide feedback. (5 mins) 8) The instructor will present the Basic Ammunition Allocation List for a CPF and explain the expendable requirements based on readiness level of the ship. (10 mins) 9) Each group will be assigned an expendable along with a ship's readiness level IOT determine their unit's allocation, they will then present their answers to the class. (5 mins) 10) Instructor will take class to CPF and do a tour to include expendable stowage spaces. (50mins)

Resources	References
1) Instructor DWAN with Projector 2) PowerPoint presentation on safe handling of expendables 3) Example expenditures log handouts 4) Example shipboard stock log handouts. 5) BAAL	1) C-59-007-006/MB-001, Sonobuoy Reference Guide 2) C-69-789-000/MB-002, Operator's Manual MK39 EMATT Advanced Acoustic Threat. 3) Intermediate Maintenance and Handling Procedures for ADC MK2 MOD 3 4) Intermediate Maintenance and Handling Procedures for NAEB MK 3 MOD 3 5) NCPM 600

Notes to Instructor
Will be tested in future ECs

EO 004.01 – Underwater Expendables

004.01.02 – Prepare Sonobuoys for Launch

Conditions		Sequence
Unclassified Classroom with DWAN		Conducted after EO 004.01.01
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Preparing underwater expendables (Sonobuoys) for launch 2) Launching a sonobuoy	EDI

Activity Steps	
	1) Instructor will explain and demonstrate how to prepare the sonobuoy for launch utilizing training shapes. (15 mins) 2) Instructor will have trainees practice preparing the sonobuoy using instructor determined settings, and verify the trainee's settings. (80 mins) 3) Instructor will show video on proper sonobuoy launching procedures. (5 mins)

Resources	References
1) Sonobouy training shapes 2) DWAN with projector 3) Sonobuoy launch video	1) C-59-007-006/MB-001, Sonobuoy Reference Guide

Notes to Instructor
Instructor will allow each trainee the opportunity to program a Sonobuoy twice.

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EO 004.01 – Underwater Expendables

EC 004.01A – Prepare Sonobuoys for Launch

Scope	Conditions	Sequence
EO 004.01	Unclassified Classroom	Conducted after EO 004.01.02
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Each trainee will conduct the assessment individually, they will enter the room, be provided a Sonobuoy (or training Sonobuoy) that they must prepare for launch. **Performance:** Trainees will be required to launch a sonobuoy IAW EC 004.01 checklist, 8 min is allocated per trainee. Additional 4 mins for instructor setup/debrief. **Output:** The trainee must achieve 5/6 SATs to pass this EC. **Feedback:** Verbal and written feedback IAW EC 004.01 checklist.

Resources	References
1) Sonobuoy	1) C-59-007-006/MB-001, Sonobuoy Reference Guide

Notes to Instructor
Nil

Limitations	Some items may require simulation, and are noted on the EC checklist.
Impact of Failure	Retrain and reassess

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Item	EC 004.01A PREPARE SONOBUOY FOR LAUNCH	SAT	UNSAT	Remarks
1. Safely Handle Sonobuoy				
a.	Handle Sonobuoy with two hands	☐	☐	
2. Prepare Sonobuoy For Launch				
a.	Describe process for removing sonobuoy from over pack	☐	☐	
b.	Input ordered parameters	☐	☐	
c.	Confirm parameter inputs	☐	☐	
d.	Simulate deploying sonobuoy	☐	☐	
3. Maintain operation logs (B0055)				
a.	Complete Ship Expenditure Log	☐	☐	

Score: / 6 The trainee must achieve 5 / 6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.01A – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 004.01A with the trainee.		**Trainee** I have been debriefed on the result of EC 004.01A.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 004.01 – Underwater Expendables

004.01.03 – Prepare ADC for Launch

Conditions		Sequence
Unclassified classroom with DWAN and Projector		
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Preparing underwater expendables (ADCs) for launch 2) Launch Acoustic Device Countermeasures	EDI

Activity Steps	
	1) Instructor will explain and demonstrate how to prepare the ADC for launch utilizing training shapes. (15 mins) 2) Instructor will have trainees practice preparing the ADC using instructor determined settings, and verify the trainee's settings (80 mins) 3) Instructor will show video on proper ADC launching procedures (5 mins)

Resources	References
1) ADC's training shapes 2) ADC Launching video 3) DWAN with Projector 4) Flat head screwdriver	1) NCPM 600 2) Intermediate Maintenance and Handling Procedures for ADC MK2 MOD 3 3) DAEME ASSB Tech letter 10-70 Trial use of ADC MK 2 MOD 3 on HMC ships and submarines

Notes to Instructor
Instructor will allow each trainee the opportunity to program an ADC twice.

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EO 004.01 – Underwater Expendables

EC 004.01B – Prepare ADC for launch

Scope	Conditions	Sequence
EO 004.01	Unclassified classroom	Conducted after EO 004.01.03
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Each trainee will conduct the assessment individually, they will enter the room, be provided an ADC (or training ADC) that they must prepare for launch. **Performance:** Trainees will be required to launch an ADC IAW EC 004.01B checklist, 8 min is allocated per trainee. Additional 4 mins for instructor setup/debrief. **Output:** The trainee must achieve 6/7 SATs to pass this EC. **Feedback:** Verbal and written feedback provided IAW EC checklist.

Resources	References
1) ADC's training shapes 2) Flat head screwdriver	1) NCPM 600 2) Intermediate Maintenance and Handling Procedures for ADC MK2 MOD 3 3) DAEME ASSB Tech letter 10-70 Trial use of ADC MK 2 MOD 3 on HMC ships and submarines

Notes to Instructor
Nil

Limitations	Plastic packaging may not be available on training shapes and will have to be simulated.
Impact of Failure	Retrain and reassess

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Item	EC 004.01B PREPARE ADC FOR LAUNCH	SAT	UNSAT	Remarks
1. Safely store ADCs				
a.	Explain Safe storage requirements of ADCs to include; • Storage requirements • Firefighting considerations	☐	☐	
2. Safely handle ADCs to include;				
a.	Safely remove plastic and pin	☐	☐	
b.	Using two hands transport ADC to launch point.	☐	☐	
3. Prepare ADCs for launch to include;				
a.	Remove shear-pin	☐	☐	
b.	Enter settings as directed	☐	☐	
c.	Rotate propeller with tool.	☐	☐	
4. Maintain operational logs (B0055)				
a.	Complete Ship Expenditures Log	☐	☐	

Score: /7 The trainee must achieve 6/7 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.01B – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 004.01B with the trainee.	**Trainee** I have been debriefed on the result of EC 004.01B.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 004.01 – Underwater Expendables

004.01.04 – Prepare NAEB for Launch

Conditions		Sequence
Unclassified classroom with DWAN and Projector		
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Preparing underwater expendables (NAEBs) for launch 2) Launch NAEB	EDI

Activity Steps	
	1) Instructor will explain and demonstrate how to prepare the NAEB for launch utilizing training shapes. (15 mins) 2) Instructor will have trainees practice preparing the NAEB using instructor determined settings, and verify the trainee's settings (80 mins) 3) Instructor will show video on proper NAEB launching procedures (5 mins)

Resources	References
1) NAEB training shapes 2) NAEB training video 3) Classroom with DWAN and projector 4) Hammer	1) NCPM 600 2) Intermediate Maintenance and Handling Procedures for NAEB MK 3 MOD 3 3) DAEME ASSB Tech letter 10-71 Trial use of NAEB MK 3 MOD 3 on HMC ships and submarines

Notes to Instructor
Each trainee shall have 2 opportunities to prepare the NAEB for launch

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EO 004.01 – Underwater Expendables

EC 004.01C- Prepare NAEB for Launch

Scope	Conditions	Sequence
EO 004.01	Unclassified classroom	Conducted after EO 004.01.04
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Each trainee will conduct the assessment individually, they will enter the room, be provided an NAEB (or training NAEB) that they must conduct the launch process for. **Performance:** Trainees will be required to launch an NAEB IAW EC 004.04 checklist, 8 min is allocated per trainee. Additional 4 mins for instructor setup/debrief. **Output:** The trainee must achieve 5/6 SATs to pass this EC. **Feedback:** Verbal and written feedback provided IAW EC checklist.

Resources	References
1) NAEB training shape 2) Hammer or mallet	1) NCPM 600 2) Intermediate Maintenance and Handling Procedures for NAEB MK 3 MOD 3 3) DAEME ASSB Tech letter 10-71 Trial use of NAEB MK 3 MOD 3 on HMC ships and submarines

Notes to Instructor
Nil

Limitations	Plastic packaging may not be available on training shapes and will have to be simulated.
Impact of Failure	Retrain and reassess

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Item	EC 004.01C PREPARE NAEB FOR LAUNCH	SAT	UNSAT	Remarks
1. Safely store NAEBs				
a.	Explain Safe storage requirements of NAEBs to include; • Storage Requirements • Firefighting considerations	☐	☐	
2. Safely handle NAEBs to Include;				
a.	Safely remove plastic and pin; and	☐	☐	
b.	Using two hands transport NAEB to launch point.	☐	☐	
3. Prepare NAEBs for launch to include;				
a.	Remove shear-pin;	☐	☐	
b.	Enter settings as directed: and	☐	☐	
4. Maintain operational logs (B0055)				
a.	Complete Ship Expenditures Log	☐	☐	

Score: **/6** The trainee must achieve 5/6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.04 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 004.01C with the trainee.	**Trainee** I have been debriefed on the result of EC 004.01C.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 004.01 – Underwater Expendables

004.01.05 – Prepare ESUS for Launch

Conditions		Sequence
Unclassified classroom with DWAN and projector		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-SONAROP-MS

Learning Objectives	Method
1) Preparing underwater expendables (ESUS) for launch	EDI

Activity Steps	
	1) Instructor will show video demonstrating parts and function of ESUS. (15 mins) 2) Instructor will explain and demonstrate how to prepare the ESUS for launch utilizing training shapes. (5 mins) 3) Instructor will have trainees practice preparing the ESUS using instructor determined settings, and verify the trainee's settings. (30 mins)

Resources	References
1) ESUS training shapes; 2) DWAN with projector 3) US Navy YouTube video	1) NCPM 600

Notes to Instructor
Nil

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EO 004.01 – Underwater Expendables

EC 004.01D – Preparing ESUS for Launch

Scope	Conditions	Sequence
EO 004.01.05	Unclassified classroom	Conducted after EO 004.01.04
Time	**Classification**	**Instructor**
100 min	UNCLASSIFIED	RQ-SONAR OP-MS

Assessment Steps	**Input:** Each trainee will conduct the assessment individually, they will enter the room, be provided an ESUS (or training ESUS) that they must prepare for launch. **Performance:** Trainees will be required to launch an ESUS IAW EC 004.01.05 checklist, 8 min is allocated per trainee. Additional 4 mins for instructor setup/debrief. **Output:** The trainee must achieve 4/5 SATs to pass this EC. **Feedback:** Verbal and written feedback provided IAW EC checklist.

Resources	References
1) ESUS	1) NCPM 600

Notes to Instructor
Nil

Limitations	Plastic packaging may not be available on training shapes and will have to be simulated.
Impact of Failure	Retrain and reassess

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Item	EC 004.01D PREPARE ESUS FOR LAUNCH	SAT	UNSAT	Remarks
1. Safely store and handle ESUS				
a. Explain Safe storage and handling requirements of ESUS to include; • Storage requirements • Firefighting considerations • Handling with care		☐	☐	
2. Prepare ESUS for launch				
a. Remove ESUS from packaging		☐	☐	
b. Ensure desired code selected		☐	☐	
c. Ensure salt water seal is punctured		☐	☐	
3. Maintain operational logs (B0055)				
a. Complete Ship Expenditures Log		☐	☐	

Score: /5

The trainee must achieve 4/5 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.01D – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 004.01D with the trainee.		**Trainee** I have been debriefed on the result of EC 004.01D.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 004 – Launch Underwater Expendables

Scope	Conditions	Sequence
PO 004	Unclassified classroom	Conducted after EC 004.01A-D
Time	**Classification**	**Instructor**
250 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	**Input:** Each trainee will conduct the assessment individually, they will enter the room, be provided underwater expendables (SONOBUOY, ADCs, NAEBs AND ESUS) (or training expendables) that they must prepare for launch. **Performance:** Trainees will be required to prepare all of the following for launch: SONOBUOY, ADCS, NAEBS and ESUS IAW PC 004 checklist, 20 mins is allocated per trainee. Additional 10 mins for instructor setup/debrief. **Output:** The trainee must achieve 4/4 SATs to pass this PC. **Feedback:** Verbal and written feedback provided IAW PC checklist.

Resources	References
1) Sonobuoy training shapes 2) ADC training shapes 3) NAEB training shapes 4) ESUS training shapes	1) C-59-007-006/MB-001, Sonobuoy Reference Guide 2) Intermediate Maintenance and Handling Procedures for ADC MK2 MOD 3 DAEME ASSB Tech letter 10-70 Trial use of ADC MK 2 MOD 3 on HMC ships and submarines 3) Intermediate Maintenance and Handling Procedures for NAEB MK 3 MOD 3 DAEME ASSB Tech letter 10-71 Trial use of NAEB MK 3 MOD 3 on HMC ships and submarines 4) Ultra Electronics NAE BEACON SSV Quarry hand launch demo plan 5) NCPM 600

Notes to Instructor
Nil

Limitations	Plastic packaging may not be available on training shapes and will have to be simulated.
Impact of Failure	IAW NPTORD 4500-54

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Item	PC 004 LAUNCH UNDERWATER EXPENDABLES	SAT	UNSAT	Remarks
1. Prepare launch for:				
a. Sonobuoy: • Input and verify parameters accordingly		☐	☐	
b. ADCs to include; • Remove shear-Pin; • Choose settings; and • Rotate propeller.		☐	☐	
c. NAEBs to include; • Remove shear-Pin and; • Choose settings;		☐	☐	
d. ESUS to include; • Puncture sea water seal; • Select code as directed.		☐	☐	

Score: **/4** The trainee must achieve 4/4 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 004 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 004 with the trainee.	**Trainee** I have been debriefed on the result of PC 004.

PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 005 – DISPLAY INTELLIGENCE VIA UNDER WATER WARFARE (UWW) BRIEF

1. **Performance**: Display Intelligence via UWW Brief (B0035).

2. **Conditions**:

   a. Given:

      (1) Contact (CTC) of Interest (platform and mines) information;
      (2) Oceanographic Data in digital format, to include:
         i. Track/ship's position;
         ii. Ocean Feature Analysis (OFA);
         iii. Bottom topography;
         iv. Bottom composition;
         v. XBT/XSV profiles;
         vi. Calculated acoustic ranges;
         vii. Optimal sensor depth; and
         viii. Recommendations.
      (3) References;
      (4) Assistance; and
      (5) Supervision.

   b. Denied: Nil

   c. Environment: Any

3. **Standard**: The Jr. SONAR Operator shall display intelligence via UWW brief (B0035), by:

   a. Contributing to intelligence gathering Ops (B0030);

   b. Identifying capabilities and limitations of platforms (B0045);

   c. delivering a platform brief; and

   d. Interpreting oceanographic conditions data (F0045) IAW METOC MILOC Manual and RP 33 to include interpreting oceanographic environment conditions.

4. **Training Limitations**: Nil

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EO 005.01 - Deliver a Platform Brief

005.01.01 – Platform Briefs

Conditions		Sequence
Unclassified classroom with DWAN		
Time	**Classification**	**Instructor**
550 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Prepare platform brief to include: a) Gathering content material for brief b) Use presentation tool (Power Point software) 2) Present brief	Flash card game EDI Presentation

Activity Steps	
	1) Instructor will break class into teams and distribute submarine recognition flash cards evenly. Teams will study their assigned cards and trade cards with another group. Each member of the team will ask 1 question to the opposing team.(50 min) 2) Instructor will Present the Example brief, demonstrating and explaining good presentation techniques to include: (15 min) a) Body Language b) Mannerisms c) Verbal ticks 3) Instructor will explain gathering all the content material for brief (20 min) 4) Instructor will use Submarine Recognition flash cards and have each trainee select 1 nuclear or conventional submarine. (5 min) 5) Instructor will demonstrate open source websites for information collection to include(not exclusively): (10 min) a) Wikipedia b) Google c) Jane's fighting ships d) Other open source websites 6) Instructor will demonstrate how to use and edit template platform brief on PowerPoint to include: (20 min) a) Create new slide; b) Format slide; c) Proper order of slides d) Insert images; e) Select correct font; and f) Save presentation.

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	7) Trainees will gather all applicable open source information and edit template brief accordingly (200 mins) 8) Trainees will present their brief, Instructors and trainees will give feedback (180 mins) 9) Trainees will implement corrections from feedback given (50mins)

Resources	References
1) Computer with DWAN for each trainee and instructor with projector 2) Submarine recognition cards 3) Briefing topics 4) Researched material 5) Example Platform Brief 6) UWW intelligence brief template	1) Jane's Online 2) Open source intelligence websites

Notes to Instructor
This practice brief will be the same one the trainees will present on the EC, so ensure feedback is implemented and saved.

EC 005.01- Deliver a Platform Brief

Scope	Conditions	Sequence
EO 005.01	Unclassified classroom with DWAN	Conducted after EO 005.01.01
Time	**Classification**	**Instructor**
250 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will use the brief they developed in 005.01.01 **Performance:** Trainees have 15 min to conduct their brief and respond to instructor queries. **Output:** The trainee must achieve 7/9 SATs to pass this EC. **Feedback:** 5 min per trainee has been allocated for feedback. Verbal and written feedback to be provided IAW EC 005.01 checklist.

Resources	References
1) UWW intelligence brief template 2) Computer with DWAN and projector 3) Brief developed in 005.01.01	1) Jane's Online 2) Open source intelligence websites

Notes to Instructor
Nil

Limitations	Open source websites may not have all the required information on the platforms, instructor to make note of information not available on EC Checklist and give trainee SAT.
Impact of Failure	Retrain and Retest

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Item	EC 005.01 DELIVER A PLATFORM BRIEF	SAT	UNSAT	Remarks
1. Present Brief				
Briefing subject material to include:				
a.	Assigned platform	☐	☐	
b.	Describe role of platform	☐	☐	
c.	Identify type of propulsion	☐	☐	
d.	Identify type of weapons	☐	☐	
e.	Identify type of sensors	☐	☐	
f.	List 3 platform capabilities/limitations (Endurance/speed/Max depth)	☐	☐	
Briefing techniques, to include:				
a.	Capability to expand on subject matter	☐	☐	
b.	Layout of brief IAW template	☐	☐	
c.	Responds to questions	☐	☐	

Score: **/9** The trainee must achieve 7/9 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 005.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 005.01 with the trainee.	**Trainee** I have been debriefed on the result of EC 005.01.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.01 – Introduction to Oceanography / Military Oceanography

Conditions		Sequence
Unclassified classroom with DWAN with projector		Conducted after EC 005.01
Time	**Classification**	**Instructor**
50 min	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Define Oceanography 2) Define Military Oceanography	Reading Assignment Video Think Pair Share Brainstorm

Activity Steps	1) Instructor will begin the class with an Initial Question, "What is Oceanography?" Trainees will be asked to think of an answer to this question and will provide it later in this lesson. (2 mins) 2) Trainees will be directed to complete a reading assignment found in the MILOC Manual Chapter 1 Sections 1.1 Origins of the Oceans and 1.2 Ocean Dimensions. (15 mins) 3) Instructor will indicate this upcoming video is from the National Oceanic and Atmospheric Administration and provides examples of why oceanography is important. Instructor will then show the video "NOAA - Know your Ocean" (5 mins total includes set up time) 4) On completion, instructor will direct the trainees to conduct a Think-Pair-Share activity. Trainees will be asked to come up with their own answer to the Initial Question, then pair up with another trainee and compare their answers, trainees will then be asked to share their answers with the class. Instructor will provide feedback on trainee's answers. (10 mins) Possible expected answers: - Study of the Oceans - Study of life in Oceans 5) Instructor will indicate this upcoming video is from the United States Navy and shows some of the practical applications of oceanography that are used for military purposes. Instructor will then show the video "METOC - Navy Meteorology & Oceanography." (3 mins total includes set up time) 6) Instructor will conduct a Brainstorm activity where they will ask the class the question, "What is Military Oceanography?" The instructor can write the question on the whiteboard and have a trainee write down the answers provided by the class. Upon completion, the instructor will summarize the answers the trainees discussed. Below is a list of possible answers: (15 mins)

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	Possible expected answers: - How to apply knowledge of the Oceans to exploit for military applications? - How submarines & ships can use the oceans to advantage/disadvantage? - How sound travels in sea water? - How ASW sensors will work in ocean?

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) Video "What is Oceanography?" 6) Video "What is Military Oceanography?" 7) Video "NOAA – Know Your Ocean" 8) Video "METOC - Navy Meteorology & Oceanography" 9) notepad per trainee	1) MILOC Manual 2) RP 33

Notes to Instructor
Nil

EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.02 – Ocean Bottom Topography & Sediments

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.01
Time	**Classification**	**Instructor**
300 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Define the following Bottom Topography a. Shore b. Continental Shelf c. Continental Slope and Rise d. Deep Ocean e. Abyssal Plains f. Seamounts g. Ocean Trenches 2) Define Military Significance of Marine Geology to include: a. Sediment Types and Origin b. Bottom Composition c. Shallow Water Sediment and Topography 3) Demonstrate how Ocean Bottom Topography and Sediments affect ASW	Distributed Learning Video Reading Assignment Team Quiz Diagram Draw T-Chart Card Sort Case Study

Activity Steps	
	1) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Ocean Bottom Topography and Composition. (40 mins) 2) Instructor will show a video on: “The Ocean Floor”. (15 mins) 3) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 1 Sections 1.4 Bottom Topography and all subsections of 1.4. During the reading assignment they will be required to develop 5 questions (with corresponding answer key) on the content they read that will be used later in a team quiz. (30 mins) 4) Instructor will break the class up into 2 teams and conduct a Team Quiz, each trainee will be asked to present at least 1 of their questions developed during the reading assignment to the other team. The other team will be allotted 2 mins too answer the question. The instructor will determine the accuracy of the answers. Each team will be allowed a maximum of 6

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questions per team. Instructor will determine the winner of the quiz. (30 mins)

5) Instructor will use a diagram of the Ocean Floor (or draw one on the whiteboard) with the following features: (5 mins)
    a. Shore
    b. Continental Shelf
    c. Continental Slope and Rise
    d. Deep Ocean
    e. Abyssal Plains
    f. Seamounts
    g. Ocean Trenches
6) Instructor will ask a trainee to come up to show and demonstrate where these features are on the Ocean Floor diagram. Instructor will critique the trainee's demonstration. (20 mins)
7) The instructor will now place the trainees into 2 groups and conduct a Brainstorm and T-Chart comparison. One group will be asked to provide a submarines perspective, the other group a warships perspective. Both groups will be asked to provide ways that their platform may use/exploit each of the Ocean Floor features previously discussed. Each group will have 10 mins to Brainstorm and when ready, each group will have 5 mins to provide their answers to the other group, instructor will provide critique as required. (30 mins)
8) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 1 Sections 1.5 Military Significance of Marine Geology and all subsections of 1.5. (20 mins)
9) Instructor will show a video on: "Ocean Sediments". Instructor will pause the video and ask the trainees the 3 questions from the video. Allow the trainees determine the answer and then ask 1 trainee per question for their answer. Restart the video and show the results. (15 mins)
10) Instructor will give the trainees an Ocean Sediments handout taken from MILOC Manual Figure 1.6 Absorption and Reflection of sound in different sediments with parts of the chart missing. Trainees will need to fill in the missing answers. This is a closed book activity. (15 mins)
11) The instructor will then take up the handout as a class and the trainees will fill in a classroom T-Chart on the whiteboard. (20 mins)

Bottom Type	**Bottom Reflectivity**
Coarse Sand	Some Scattering / Good Reflection
Fine Sand	Good Reflection for Bottom Bounce
Calcareous Mud/Ooze	Sound Absorbed, but better reflectivity

	due to calcium content
Siliceous Mud/Ooze	Sound Highly Absorbed
Rough Rock	High Scattering, Good Reflection
Smooth Rock	Low Scattering, Good Reflection
Clay	Poor Reflection, Sound Absorbed, Low Scattering
Silt	Some Reflection, Low Scattering

12) Instructor will now conduct a Card Sort activity, they will create a scale with Good Reflection (Low Absorption) on one side and Poor Reflection (High Absorption) on the other side on the Whiteboard. Trainees will be given a Card or 3M Sticky with one of the Bottom types on it. Trainees will be asked to place the Care/Sticky on the scale where they believe it applies. The instructor will provide critique(s) on the trainees' answers. (15 mins)
13) The instructor will now place the trainees into 2-3 groups and each group will conduct a Case Study. (25 mins)
14) Instructor will have the groups present their findings to the class and allow trainees to discuss the answers. (20 mins)

Resources

1) Whiteboard with markers
2) DWAN terminals (1/trainee)
3) Projector & Speakers
4) Ocean Bottom Topography and Composition DLN module
5) MILOC Manuals
6) video “The Ocean Floor”
7) video “Ocean Sediments”
8) Ocean Sediments Handout
9) diagram of Ocean Floor
10) Reflection Cards or 3M Stickies
11) Case Study Cards
12) notepad per trainee

References

1) MILOC
2) RP 33

Notes to Instructor

Instructor may also use individual bottom topography diagrams given to trainees to understand the bottom topography.

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.03 – Characteristics of Seawater

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.02
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Understanding the 3 states of water: a) Solid b) Liquid c) Gas 2) Identify the 3 characteristics of Seawater to include: a) Salinity b) Pressure c) Temperature 3) Demonstrate how the characteristics of Seawater affects ASW.	Distributed Learning Videos 3 Star Reading Assignment Card Sort Case Study

Activity Steps	
	1) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN/DLN on Chemistry of Seawater. (50 min) 2) Instructor will show an "Ice Breaker" video on "Opening a Soda Can on the Ocean Floor." (1 min) 3) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 2 Seawater Characteristics and sections 2.1 Salinity, 2.2 Pressure and 2.3 Temperature. During the reading assignment they will conduct a 3 Star activity where they will identify one item they deem important from the characteristics of; Temperature, Salinity and Pressure. (20 mins) 4) Instructor will then ask the trainees to present their 3 Star points and provide feedback on their answers. (10 mins) 5) Instructor will show a videos on: "Salinity of Sea Water" and "Fresh Water meets Sea Water." (14 mins for both) 6) Instructor will conduct a Card Sort activity. Instructor will display a large scale World Map with major water masses displayed. Trainees will be given Salinity Cards/Stickies labelled; Average Salinity, High Salinity, and Low salinity. Trainees will be asked to place the appropriate Salinity card on the water masses or indicate the type of salinity. Instructor should have the trainees identify the salinities for at minimum on the following geographical water masses: (10 mins) a) Middle of North Atlantic Ocean;

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b) Middle of North Pacific Ocean;
c) Arctic Ocean;
d) Fjords of Norway;
e) Baltic Sea;
f) Mediterrean Sea;
g) Red Sea;
h) Black Sea;
i) Persian Gulf;
j) Mouth of Fraser River;
k) Indian Ocean; and
l) Sea of Japan.

Instructor will critique trainees' answers and provide feedback. Instructor may create more area water masses for activities.

7) Instructor will conduct a Card Sort activity. Instructor will display a large scale World Map with major water masses displayed. Trainees will be given Temperature Cards/Stickies labelled; Average Temperature, High Temperature, and Low Temperature. Trainees will be asked to place the appropriate Temperature card on the water masses or indicate the type of temperature. Instructor should have the trainees identify the temperature areas for at minimum on the following geographical water masses: (10 mins)
   a) Arctic Ocean;
   b) Sargasso Sea (Tropical North Atlantic Ocean)
   c) Winter in North Atlantic Ocean;
   d) Summer in South Pacific Ocean;
   e) Mediterrean Sea;
   f) Caribbean Sea
   g) Persian Gulf;
   h) Indian Ocean; and
   i) Gulf Stream.

   Instructor will critique trainees' answers and provide feedback. Instructor may create more area water masses for activity.
8) The instructor will now place the trainees into groups of 2-3 trainees and each group will conduct a Case Study. (20 mins)
9) Instructor will have the groups present their findings to the class and allow trainees to discuss the answers. (15 mins)

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Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) Chemistry of Seawater DLN module 5) MILOC Manuals 6) video “Salinity of Sea Water” 7) video “Fresh Water meets Sea Water” 8) video “Opening a Soda Can on the Ocean Floor” 9) Salinity Cards or 3M stickies 10) Laminated Map of the World 11) Temperature Cards or 3M stickies 12) Case Study cards 13) notepad per trainee	1) MILOC 2) RP 33

Notes to Instructor

Instructor should explain to the trainees what a “3 Star” activity is prior to Activity #3.
Instructor can generate more water mass areas for the Card Sort activity.
During the Case Studies, the instructor can give more than one case to each group.

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.04 – Ocean Thermal Structure

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.03
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Define the characteristics of the 3 Layers of the Ocean Thermal Structure to include: a) The Surface Layer i) Surface Mixed Layer ii) Seasonal Mixed Layer b) The Permanent Thermocline c) The Deep Layer 2) Discuss the factors that affect The Surface / Mixed Layer depth to include: a) Atmospheric conditions b) Heating / Cooling cycles c) Wind Mixing d) Evaporation / precipitation e) Acoustic implications to include: i) Afternoon Effect ii) Seasonal Variability iii) Geographic Variability iv) Ekman Spiral v) Down welling & Upwelling 3) Demonstrate how the characteristics of the Ocean Thermal Structure affects ASW.	Distributed Learning Module Team Quiz Draw diagram Team Pictionary Quiz Case Study

Activity Steps	
	1) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Ocean Thermal Structure (60 mins) 2) During the DLN lecture, trainees will develop 5 questions each regarding the Ocean Thermal Structure. These questions will be used in a Team Quiz. 3) Instructor will divide the class into 2 teams and conduct a Team Quiz activity. Instructor will have the trainees use the questions they developed during the Ocean Thermal Structure DLN to ask the other team. Instructor will alternate questions between the teams and provide feedback on the answers. (20 mins)

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4) Instructor will handout a diagram of the Ocean Thermal Structure to each trainee. Trainees will be asked to fill in the names and depths that apply. Trainees may use the MILOC Manual for reference. (10 mins)
5) Instructor will draw on the whiteboard the Ocean Thermal structure and ask the trainees to provide the answers to the labels. Instructor will provide feedback on the answers. (10 mins)
6) Instructor will now break the class into 2 groups and conduct a Team Pictionary Quiz where each group will have a member of their team demonstrate by drawing how a factor affects the Surface Layer and what affect it will have on ASW operations. The instructor will be the judge of the responses accuracy and provide feedback. These affects will include: (25 mins)
   a) Season variability to include:
      i) Winter
      ii) Summer
      iii) Spring
      iv) Fall
   b) Afternoon Effect (Diurnal)
   c) Geographic Variability to include:
      i) Arctic regions
      ii) Mid-Latitudes
      iii) Equator/Tropical regions
   d) Weather variability to include:
      i) Wind
      ii) Precipitation (Rain)
      iii) Evaporation
      iv) Convective Mixing
      v) Advection Mixing
7) The instructor will now place the trainees into groups of 2-3 trainees and each group will conduct a Case Study. (25 mins)

Resources

1) Whiteboard with markers
2) DWAN terminals (1/trainee)
3) Projector & Speakers
4) MILOC Manuals
5) Ocean Thermal Structure DLN module
6) Ocean Thermal Structure Handout
7) notepad per trainee

References

1) MILOC Manual

Notes to Instructor

Ensure TDL instructor is identified early to ensure availability.

EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.05 – Sound Velocity Profile

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.04
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Understanding Sound Velocity mediums 2) Understanding Sound Velocity 3) Understanding Sound Speed profiles & gradients 4) Measurement Probes & Databases	Reading Assignment Distributed Learning Module Domino Demonstration Video Card Sort Math Quiz Best Summary Show & Share Interactive Lecture

Activity Steps	
	1) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 2 Seawater Characteristics section 2.5 Sound Velocity Profile. (10 mins) 2) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Sound Velocity. (40 mins) 3) Instructor will have the class conduct a Domino Sound Medium demonstration activity. Trainees will be given 3 sets of dominos coloured and labelled Solid, Liquid and Gas. They will be required to set them up into 3 rows, in the following manner: (30 mins) a) Solid Medium domino row will be placed closer together, approximately 2 cm (1 inch) apart. b) Liquid Medium domino row will be placed slightly farther apart than the solid medium, approximately 3-4 cm (1.5 inches) apart; c) Gas Medium domino row will be placed farther apart than the liquid medium, approximately 5 cm (2 inches) apart. d) A ruler with measurements may be used to assist in setting this demonstration up. 4) Instructor will then have the trainees use a ruler to touch all 3 mediums simultaneously and watch as the dominos fall. Trainees will then be asked to indicate which sound medium travelled the fastest to the end. If necessary, the trainees will be allowed time to reset the demonstration and conduct it again. The instructor may use the “Speed of Sound” video as a reference for the activity and show the trainees the video upon completion of the demonstration. (30 mins)

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5) Instructor will now conduct a Card Sort activity; he will create a scale representing Low to High Sound Velocity in meters per second (m/s). Trainees will be given a Card or 3M Sticky with different medium types on them. Trainees will be asked to place the Care/Sticky on the scale where they believe it applies, Low, Middle or High part of the Sound Velocity Scale. The instructor will provide critique(s) on the trainees' answers. (15 mins)
6) Instructor will display via Power Point Slide or drawn on Whiteboard the Sound Speed formulas for: (10 mins)
   a) Average Speed of Sound in Water = 1500 m/s
   b) Temperature: +/- 1 degree Celsius = +/- 4.5 m/s
   c) Salinity: +/- 1 part per thousand = +/- 1.4 m/s
   d) Pressure: +/- 100 meters in depth change = +/- 1.7 m/s
7) The instructor will conduct a Sound Velocity math quiz, trainees will be asked to individually calculate the change in the Speed of Sound using the Average Speed of Sound as a control speed using the formulas displayed from the Speed of Sound Math questions provided.
   Instructor will take up the questions and provide feedback to the trainees (35 mins)
8) Instructor will break the class into 3 groups, each group will be given one of the Sound Speed gradients: Positive, Negative, and Iso-velocity/thermal. The groups will be asked to provide a Best Summary of the following information for their Sound Speed gradient: (30 mins)
   a) Draw the profile on the board
   b) Provide a correlation of which Ocean Thermal Structure (Temperature, Salinity, and Pressure) would apply to their gradient
   c) Give a geographical example of where in the world you may see your Sound Speed gradient
   d) Explain / demonstrate how this gradient will affect Sound Speed
9) Instructor may conduct a Show & Share of Expendable Bathy Thermograph (XBT) and Sound Velocity (XSV) probes. (10 mins)
10) Instructor will show the trainees the link to the Ocean Navigator webpage used by METOC and how to use it. (http://navigator.oceansdata.ca/public/) Trainees will be allowed to explore the site and then present a Sound Speed profile and indicate which gradient(s) are present and why? (40 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) Physics of Sound in Seawater - Sound Velocity DLN module	1) MILOC Manual 2) RP 33 3) Ocean Navigator (http://navigator.oceansdata.ca/public/) 4) METOC Website

6) Video "Speed of Sound through solids, Gasses and Liquids" 7) Sound Mediums Cards or 3M stickies 8) Speed of Sound Math Questions 9) Calculator per trainee 10) notepad per trainee 11) Speed of Sound Formulas (Power Point)	

Notes to Instructor

Instructor may use video in lieu of dominos

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.06 – Ocean Bottom & Seawater Characteristics Confirmation

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.05
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Confirmation of understanding Ocean Bottom Topography 2) Confirmation of understanding Ocean Bottom Sediments 3) Confirmation of understanding Sea Water Characteristics 4) Confirmation of understanding Ocean Thermal Structure 5) Confirmation of understanding Sound Gradient Profile	Diagram Written Confirmation

Activity Steps	
	1) Instructor will conduct a confirmation check of the trainees comprehension of the subject material covered in EO 005.02.01 to 005.03.05. The confirmation check will comprise of questions and diagrams that the trainees will be required to perform on the following items: (50 mins) a) Label the Ocean Topography diagram; b) Match the Ocean Bottom Sediments to the effects they cause on sound; c) Answer the questions on Sea Water Characteristics; d) Label the Ocean Thermal Structure diagram; e) Label the Sound Gradient Profile and explain what factors affect the different parts of the SVP. 2) Once the trainees have completed the confirmation check, they will pass it to the instructor who will mark them later and provide feedback to the class during EO 005.02.13, 20 mins of time has been allocated.

Resources	References
1) Whiteboard with markers 2) MILOC Manuals 3) RP 33 Manual 4) Confirmation Check Questions & Diagrams 5) Confirmation Check Questions & Diagrams Instructor Answer Key 6) notepad per trainee	1) MILOC Manual 2) RP 33 Manual

Notes to Instructor
Nil

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.07 – Basic Sound Theory

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.06
Time	**Classification**	**Instructor**
300 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Why Use Sound in ASW? 2) Understanding Basic Sound Theory to include: a) Sound Waves b) Wave fronts & Sound Rays c) Decibels & dB math d) Doppler	Distributed Learning Module Reading Assignment Videos Floating Flipchart Card Sort Diagram Math Quiz

Activity Steps	
	1) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 3 Foundations of Acoustics section 3.1 Why Use Sound in ASW? (20 mins) 2) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Physics of Sound. (50 mins) 3) Instructor will show video(s) on: "How SONAR Works" and "How Submarines Can "See" Underwater" by Covert Cabal (accessed through Facebook, approved for use) (10 mins) https://www.facebook.com/watch/?v=715258582546309 4) Instructor will now conduct a Floating Flipchart activity with the class, the class may be broken into groups if preferred. Trainees will be asked to provide the Electro Magnetic and Sound characteristics for the following: (15 mins) a) Light b) Magnetics c) Sound d) Heat All answers can be found in Figure 3.1 of the MILOC Manual. 5) Instructor will now conduct a Card Sort activity; they will use the Military of Applications of Sound Scale Diagram and will give the trainees 15 Sound Application Cards. Trainees will be required to place each Sound Application in the correct Frequency Zones on the poster/diagram. Instructors may use the MILOC Manual Figure 3.2: Military Applications of Sound to confirm the trainees' answers. (20 mins) 6) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 3 Foundations Of Acoustics section 3.2 Basic

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Sound Theory and RP 33 Chapter 1 The Nature of Underwater Sound subsection 1.1.1 Wave Motion (30 mins)

7) Instructor will show a video(s) on: “Transmission of Sound” (15 mins)
8) Instructor will give trainees a diagram of a sound wave, the trainees will be asked to label the following: (20 mins)
   a) Source
   b) Medium
   c) Receiver
   d) Compression Phase
   e) Refraction phase
   f) Amplitude
   g) Wavelength (1 full cycle)
9) Instructor will then take the diagram up with the class and provide feedback. (5 mins)
10) Instructor will conduct a Frequency Math Quiz; the instructor will write on the board the equation for Wavelength (Wavelength = Speed of Sound (1500 m/s) / Frequency). The instructor will conduct a demonstration on how the calculation works then ask the trainees to calculate the Wavelengths for the questions in the Frequency Math Quiz. (15 mins)
11) The instructor will then ask the trainees rank the wavelengths that will travel farthest to the shortest distance in water. (5 mins)
12) Instructor will show a video(s) on: “Wave fronts & Sound Rays” (10 mins)
13) Instructor will give trainees a diagram of a Wave front, they will be asked to label/draw the following: (10 mins)
   a) Sound Source
   b) Wave fronts
   c) Rays (ensuring they are perpendicular to the wave fronts.
14) Instructor will then take the diagram up with the class and provide feedback. (5 mins)
15) Instructor will now conduct an Interactive (Power Point) Lecture on Decibels. (25 mins)
16) Using the Figure 3.4 Nomo gram for Combining Sound Levels in the MILOC manual, trainees will conduct a Decibel Math quiz included in the lecture. Instructor can walk trainees through the first question and provide feedback on the remaining questions. (20 mins)
17) Instructor will show a video(s) on: “The Doppler Effect” (10 mins)
18) Instructor will show/draw the trainees a diagram of an object generating noise in the water with the sound waves emanating outward from the object. The trainees will then be asked to show / draw how the sound waves will react when the object begins to move in the water. The trainees should be able to understand the following:
   a) Compressed Wave Front that show higher frequency
   b) Refracted Waves that show lower frequency

	c) Closest Point of Approach (CPA) Zero Doppler d) Instructor will then take the diagram up with the class and provide feedback. (15 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) Physics of Sound in Seawater DLN module 6) Video "How SONAR Works" 7) Video "How Submarines Can "See" Underwater" by Covert Cabal (accessed through Facebook, approved for use) https://www.facebook.com/watch/?v=715258582546309 8) Military of Applications of Sound Poster/Diagram 9) 15 Sound Application Cards 10) Video "Transmission of Sound" 11) Wavelength diagram 12) Frequency Math Quiz 13) Video "Wave fronts & Sound Rays" 14) Wave front diagram 15) Power Point Lecture on Decibels 16) Nomo gram for Combining Sound Levels 17) Video "Doppler Effect" 18) Closest Point of Approach Diagram 19) Calculator per trainee 20) notepad per trainee	1) MILOC 2) RP 33

Notes to Instructor

Nil

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.08 – Sound Propagation

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.07
Time	**Classification**	**Instructor**
350 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Snell's Law (HALT Rule) 2) Propagation / Sound Paths to include: a) Propagation Path Terminology b) Direct Path c) Bottom Bounce d) Surface Duct e) Sound Channel / Deep Sound Channel f) Convergent Zone g) Cut-off Frequency	Reading Assignment Distributed Learning Module Group / Share Best Summary 3 Star Confirmation Activity

Activity Steps	
	1) Instructor will give the trainees back their completed EO 005.02.06 Confirmation Check #1 and take up the questions with the class. (20 mins) 2) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Sound Paths (Propagation). (50 mins) 3) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 4 Foundations of Acoustics section 4.1 Snell's Law and Refraction of Sound Rays. During the reading assignment, the trainees will be required to find the answers to the following questions: (30 mins) a) What does the term HALT mean? b) What does a consistently positive sound speed gradient do to sound? c) What does a uniformly negative sound speed gradient do to sound? d) What does an iso-velocity profile do to sound? e) In your own words, why does sound react this way to sound speed gradients? 4) The instructor will now take up the questions with the class. They will ask trainees to provide the answers to their questions and provide feedback on their answers. (15 mins) 5) Instructor will break the class into 3 groups. The instructor will give each group a diagram of the 3 different sound gradients: Positive, Negative, Iso-velocity. Using Snell's Law (HALT rule) each group will draw on their diagram(s) how the sound will travel for each gradient. The instructor will then call on each group to present and explain 1 of the 3 different sound

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gradients to the class. Instructor may use Figure 4.2 in the MILOC manual for reference. (25 mins)

6) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 4 Foundations Of Acoustics sections 4.1.1 Ray Traces and 4.2 Propagation Paths (15 mins)
7) Instructor will give each trainee a copy of Figure 4.3 Propagation Path Terminology diagram with no labels. Instructor will then conduct a Think / Pair / Share activity where the trainees will be asked to label the diagram on their own and then pair up with another trainee(s) and compare their answers to the following terminology: (20 mins)
   a) Label Depth axis (Z)
   b) Label Sound Velocity axis (SV)
   c) Label and explain Sonic Layer Depth (SLD)
   d) Label and explain Limiting Ray
   e) Label and explain Shadow Zone
8) The instructor will then ask the paired trainees to share their answers with the class. Instructor may use Figure 4.3 in MILOC manual as reference. (10 mins)
9) The instructor will now break the class down into 5 groups (if possible) and ask each group to conduct a Reading Assignment and then provide a Best Summary of the following Sound Paths:
   a) Direct Path - MILOC - Section 4.2.1 and RP 33 Section 7.2.1
   b) Bottom Bounce - MILOC - Section 4.2.2 and RP 33 Section 7.2.5
   c) Surface Duct - MILOC - Section 4.2.3 and RP 33 Section 7.2.2
   d) Sound Channel - MILOC - Section 4.2.4 and RP 33 Section 7.2.3
   e) Convergence Zone - MILOC - Section 4.2.5 and RP 33 Section 7.2.6
   f) Trainees will also be asked to draw and explain their Sound Path on the whiteboard and share their findings with the class. (50 mins)
10) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 4 Foundations of Acoustics subsection 4.2.3.1 Surface Duct Cut-off Frequency. During the readings, trainees will be conduct a 3 Star activity. Each trainee will be asked identify 3 factors that are important to the Surface Duct Cut-off Frequency. (20 mins)
11) Instructor will then ask the trainees to provide 1 of their 3 answers, no repeat answers are accepted. (10 mins)
12) Instructor will then draw / show a diagram of how Cut Off Frequency works and explain how some sound sources will be trapped within a sound layer, while others will not. (15 mins)
13) Instructor will conduct a Confirmation Activity. Instructor will give each trainee a Sound Velocity Profile diagram and ask the trainees to label the following information: (15 mins)
   a) Surface
   b) Bottom

	c) Sonic Layer Depth (SLD) d) Deep Sound Channel Axis (DSCA) e) Critical Depth f) Depth Excess 14) Instructor will then take up the Sound Velocity Profile diagram with the class (15 mins) 15) Instructor will then place a Submarine contact at different depths within the Sound Velocity Profile. Trainees will be asked to draw which sound paths exist and explain why? Instructor will repeat this activity until all the Sound paths are demonstrated. (40 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) Sound Paths (Propagation) DLN module 6) Snell's Law Questions handout 7) Snell's Law Sound Gradient diagram 8) Propagation Path diagram 9) Sound Velocity Diagram 10) Cut-off Frequency Diagram 11) notepad per trainee	1) MILOC 2) RP 33

Notes to Instructor

If there is not enough trainees for the 5 Sound Path groups, then instructor can assign more than one Sound Path per group. Instructor should pose the question; What is Cut Off Frequency mean?

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.09 – Sound Propagation Loss

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 005.02.08
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Propagation Loss to include: a) Spreading Loss to include: i) Spherical Spreading ii) Cylindrical Spreading iii) Refractive Spreading iv) Dipolar Spreading b) Absorption Loss c) Scattering and Reverberation d) Surface and Bottom Loss e) Duct Leakage	Reading Assignment Video Group Math Case Study T-Chart Diagram

Activity Steps

1) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 4 Foundations of Acoustics section 4.3 Propagation Loss subsections 4.3.1 Spreading Loss and all sub sections. Trainees may also read RP 33 Sections 2.2 (20 mins)
2) Instructor will show a video(s) on: "Spreading Loss" (10 mins)
3) Instructor will conduct a Group Math Case Study. The instructor, using an example, will demonstrate how to calculate the Spherical and Cylindrical spreading loss with the class. Then the class will be asked to determine the amount of Sound Spreading loss, using the whiteboard to draw / display their answers to the Group Math Case Study questions provided. (40 mins)
4) The instructor will conduct a T-Chart activity using a comparison between Absorption Loss vs. Scattering and Reverberation loss that affect the sound wave and how it travels. Trainees will be asked to give example of items that cause both: (15 mins)

Absorption Loss	Scattering Reverberation Loss
conversion to heat energy	significant waves will scatter sound
bottom composition (mud, silt, clay)	seabed roughness and composition
change in water temperature	bottom depth, shallow water causes

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	more reverberation on active sonar
higher frequencies absorb faster than lower frequencies	volume reverberation, the more items in the water (biological, sediments) the more reverberation
some biological life will absorb the sound	greater differences in density of layers can cause sound scattering / reverberation
Incident Angle of sound ray	Grazing Angle of sound ray

Trainees may give more examples, instructor will provide feedback on the answer provided by the trainees.

5) Instructor will give trainees a diagram of a Duct Leakage, the trainees will be asked to label the following: (10 mins)
   a) Depth Axis
   b) Sound Velocity (V) axis
   c) Sound source (submarine)
   d) Sonic Layer Depth (SLD)
   e) Sound Rays
   f) Wave Fronts
   g) Shadow zone
   h) Sound Leakage
6) Instructor may use Figure 4.19: Leakage of Sound from Refracted Paths into Shadow Zone from MILOC as reference and will take up the diagram up with the class and provide feedback. (5 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) RP 33 Manuals 6) Calculator per trainee 7) Duct Leakage diagram 8) Video “Spreading Loss” 9) Group Math Case Study 10) Group Math Case Study Answer Key 11) notepad per trainee	1) MILOC 2) RP 33

Notes to Instructor

Nil

EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.10 – Sources of Ambient Noise

Conditions		Sequence
Unclassified classroom with DWAN with projector		Instructed after EO 005.02.09
Time	**Classification**	**Instructor**
200 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Understanding causes of Ambient Noise in the Oceans to include: a) Shipping noise b) Sea State to include: i) Wind ii) Waves c) Precipitation Noise d) Ice Noise e) Biological Noise f) Seismic noise 2) Understanding how Biological life forms and Bioluminescence can affect ASW 3) Understanding the effects of Ocean Ambient noise has on ASW using the Wenz Curve.	Reading Assignment Best Summary Distributed Learning Module Video Interactive Lecture Math Quiz

Activity Steps	
	1) Instructor will have the class conduct a Best Summary Activity. Instructor may put trainees into groups of 1-2 and give each group one of the following topics to provide a Best Summary found in the MILOC Manual Chapter 6 section 6.4.2.2 Ambient Noise and RP 33 Chapter 3 Background Noise Sections 3.2 Ambient Noise and all sub sections. (40 mins) a) Shipping Noise b) Sea State to include Wind & Waves c) Precipitation noise d) Ice Noise e) Biological noise f) Seismic noise 2) Instructor will then have the trainees present their Best Summary to the class and allow trainees to discuss. (20 mins) 3) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Biological Oceanography. (40 mins) 4) Instructor will show a video(s) on: “Can you hear me?” (15 mins) 5) Instructor will hand out a Wenz Curve diagram to the trainees and display a large Wenz Curve diagram. Instructor will then explain how the Wenz

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	Curve diagram is used to determine Ambient Noise vs. a Frequency Level. (20 mins) 6) Instructor will conduct a Wenz curve Math Quiz, using the Wenz curve, trainees will be asked to provide the Ambient Noise level in Decibels for the Math questions provided. (20 mins) 7) The instructor will now place the trainees into groups of 2-3 trainees and each group will conduct a Case Study. (20mins) 8) Instructor will have each group present their Case Study, answer the questions and allow for group discussion. (25 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) RP 33 Manuals 6) Biological Oceanography DLN module 7) Video "Can you hear me?" 8) Biological Oceanography presentation 9) Wenz curve handouts 10) Wenz curve large diagram 11) Wenz curve Math Quiz 12) Calculator per trainee 13) notepad per trainee 14) Case Study Cards	1) MILOC 2) RP 33

Notes to Instructor

Nil

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.11 – Foundations of Acoustics & Sound Propagation Confirmation

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 005.02.10
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Confirmation of understanding Basic Sound Theory 2) Confirmation of understanding Sound Propagation 3) Confirmation of understanding Sound Propagation Loss 4) Confirmation of understanding Sources of Ambient Noise	Written Confirmation

Activity Steps

1) Instructor will conduct a confirmation check of the trainees comprehension of the subject material covered in EO 005.02.07 to 005.03.10. The confirmation check will comprise of questions and diagrams that the trainees will be required to perform on the following items: (50 mins)
   a) Label a Sound Wave diagram;
   b) Answer the math questions on Frequency;
   c) Label a Sound Velocity Profile diagram;
   d) Draw the sound Ray Paths for the following Sound Paths:
      i) Direct Path
      ii) Bottom Bounce
      iii) Surface Duct
      iv) Sound Channel
      v) Convergence Zone
   e) Sort the Sound Propagation Loss characteristics into Absorption Loss or Scattering Loss categories; and
   f) Using the Wenz Curve diagram, determine the Ambient Noise levels for the questions.
   g) Trainees may use the MILOC and RP 33 manuals for reference during this confirmation check.
2) Once the trainees have completed the confirmation check, they will pass it to the instructor who will mark them later and provide feedback to the class during EO 005.02.13, 15 mins of time has been allocated.

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Resources	References
1) Whiteboard with markers 2) MILOC Manuals 3) RP 33 Manual 4) Confirmation Check Questions & Diagrams 5) Confirmation Check Questions & Diagrams Instructor Answer Key 6) notepad per trainee	1) MILOC Manual 2) RP 33 Manual

Notes to Instructor
Nil

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.12 – Fronts & Eddies

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 005.02.11
Time	**Classification**	**Instructor**
350 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Identifying ocean fronts and currents to include: a) Temperature Dominated Fronts b) Salinity Dominated Fronts 2) Identifying dominant ocean currents 3) Identifying Eddies to include: a) How are they formed? b) How do they affect sound in water? c) How do Submarines exploit them? 4) Understanding the affects fronts, currents and eddies have on ASW.	Distributed Learning Reading Assignment Video Floating Flipchart T-Chart Map diagram 3 Star Activity Best Summary Diagram Case Study

Activity Steps	
	1) Trainees will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Regional Oceanography & Fronts and Eddies. (60 mins) 2) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 5 Range Dependent Oceanic Features section 5.1 Oceanic Fronts and RP 33 Section 5.3.1 Ocean Fronts. During the reading assignment, the trainees will be required to identify one Ocean front from the following oceans: (30 mins) a) Atlantic Ocean b) Pacific Ocean c) Indian Ocean d) Mediterranean Sea 3) Instructor will show a video(s) on: "NOAA Motion in the Ocean" and "TED Talk video on "How do Ocean Current Work". (15 mins) https://www.ted.com/talks/jennifer_verduin_how_do_ocean_currents_work 4) Instructor will conduct a Floating Flipchart activity, on each flipchart they will have an ocean label; Pacific Currents, Atlantic Currents, Indian Ocean & Arctic/Southern Currents. Trainees will be asked to provide at least one warm or cold current (use red markers for warm, blue or black markers for cold) on each of the oceans flipcharts. Trainees may use MILOC and RP 33 manuals for reference. (20 mins) 5) The instructor will then conduct a T-Chart activity, using the information from the previous Floating Flip Chart activity, the trainees will be asked to

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sort the ocean fronts into two categories; Temperature Dominated Fronts and Salinity Dominated Fronts. Some of the ocean fronts may apply to both categories. The goal of the exercise is to get the trainees to consider which fronts will have temperature and salinity considerations. (20 mins)

Temperature Dominated Front	Salinity Dominated Front
Gulf Stream vs. Labrador Current	Gulf Stream vs. Sargasso Sea
North Pacific Current vs. California Current	Bering Current vs. Alaskan Current

6) Instructor will give each trainee a world map and ask the trainees to identify which are the two most predominant Ocean Currents in the northern hemisphere and draw how they flow. (15 mins)
7) Instructor will then take up the previous diagram using a large scale world map and asks trainees to identify the 2 major currents and draw their direction of movement. (15 mins)
8) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 5 Range Dependent Oceanic Features section 5.2 Boundary Currents and Eddies and RP 33 Section 5.3.2 Eddies. During the reading assignment, the trainees will conduct a 3 Star Activity and be required to identify 3 characteristics from a Cold Core and Warm Core eddy. (30 mins)
9) Instructor will then have the trainees present their findings on the 3 Star activity and give feedback. (15 mins)
10) Instructor will show the trainees what a METOC Ocean Features Analysis looks like and where to obtain one on the METOC DWAN website. (15 mins)
11) Instructor will break the class into 3 groups. Each group will be given an Ocean Front Relative Strength category; Strong, Moderate, and Weak. Each group will provide the following characteristics for their category: (20 mins)
    a) Maximum Change in Sound Speed (m/s)
    b) Sonic Layer Depth (meters)
    c) Deep Sound Channel (meters)
    d) Persistence
    e) Trainees may use the MILOC & RP 33 manuals along with the Distributed Learning Module for reference.
12) Once they have gathered the information, the instructor will have each group present their information verbally and shown on whiteboard. (15 mins)

	13) Instructor will give the trainees a diagram of the formation of Warm and Cold Eddies. Trainees will label the following information, they are to use geographic names to label their diagram: (15 mins) a) Cold Water mass (i.e. slope or cold shelf water, Labrador current, etc.) b) Warm Water mass (i.e. Sargasso Sea, North Pacific Ocean, etc.) c) Current (i.e. Gulf Stream, Kurshio current) d) Warm Eddy e) Cold Eddy f) Label the direction of flow of currents and eddies. g) Instructor may use MILOC Figure 5.2 and RP 33 Figure 5-7 as reference. 14) The instructor will take up the Warm and Cold Eddies diagram with the class. (15 mins) 15) Instructor will break the class into 2 groups and conduct a Best Summary activity. One group will be given Warm Core Eddies, the other group the Cold Core Eddies, each group will provide a summary of their eddies characteristics. (15 mins) 16) The instructor will now conduct a Case Study, the instructor will break the groups and give each group a Case Study. (35 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) Distributed Learning Module on Regional Oceanography 5) Distributed Learning Module on Fronts & Eddies 6) MILOC Manuals 7) RP 33 Manuals 8) Video "NOAA Motion in the Ocean" 9) TED Talk video "How do Ocean Currents Work" https://www.ted.com/talks/jennifer_verduin_how_do_ocean_currents_work 10) Flipchart 11) Red & Blue markers 12) METOC Ocean Feature Analysis 13) Diagram of Current & Eddy 14) Large Scale World Map 15) notepad per trainee 16) Case Study Cards / Presentation	1) MILOC 2) RP 33

Notes to Instructor

Nil

EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.13 – Shallow Water Propagation

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 005.02.12
Time	**Classification**	**Instructor**
300 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Identifying Shallow Water characteristics to include: a) Dynamic Sound Velocity Profiles b) Bottom Topography c) Bottom Composition d) Ambient Noise e) Biological f) Sound Paths 2) Identifying how sound paths are affected by: a) Downslope Enhancement b) Upslope Enhancement 3) Identifying Topographic Shading 4) Applying Topographic Noise Stripping 5) Understanding the effects Shallow water and slope topography has on ASW.	Reading Assignment Team Quiz Floating Flipchart Best Summary Diagrams Group Case Study Class Case Study

Activity Steps	
	1) Instructor will give the trainees back their completed EO 005.02.11 Confirmation Check #1 and take up the questions with the class. (15 mins) 2) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 5 Range Dependent Oceanic Features section 5.3 Shallow Water Propagation and RP 33 Chapter 8 Environmental Effects on Sound Propagation in Shallow Water Sections 8.1 to 8.4. During the reading assignment, trainees will be asked to create 5 questions on Shallow Water Propagations. These questions will be used in a team quiz later. (30 mins) 3) Instructor will break the class into 2 teams and conduct a Team Quiz. Each team will ask questions, developed from the Shallow Water Propagation reading assignment, to the other team. Each trainee must ask at least 1 question each. Instructor will have the team's alternate questions and will act as the moderator of the quiz. (30 mins) 4) Instructor will conduct a Floating Flip Chart activity. On each flipchart will indicate a Shallow water characteristic of; Sound Velocity Profile, Bottom Topography, Bottom Composition, Ambient Noise, Biological and Sound Paths. Trainees will be asked to provide different factors that will affect

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these characteristics. Instructor and trainees may use MILOC and RP 33 manuals for reference. (20 mins)

5) Instructor will then ask 1 or 2 trainees per flip chart to present and explain how these factors affect the shallow water characteristics. Instructor and trainees may discuss additional factors during this presentation. (20 mins)
6) The instructor will break the class into 2 groups and the groups will conduct a Best Summary activity. One group will be given Upslope Enhancement, the other group Downslope Enhancement and they will read the MILOC Manual Chapter 5 Range Dependent Oceanic Features section 5.4 Slope Enhancement and RP 33 Chapter 8 Environmental Effects on Sound Propagation in Shallow Water Sections 8.5 subsection 8.5.1 Slope Enhancement focusing on their Slope enhancement. (20 mins)
7) The instructor will then ask each group to present their Best Summary Topic by drawing a diagram and verbally explaining their topic. (15 mins)
8) The instructor will break the class into 2 groups and the groups will conduct a Best Summary activity and draw/label a diagram. One group will be given Topographic Shading and read from the MILOC Manual Chapter 5 section 5.5 and RP 33 section 8.5.2 Topographic Shading. The other group will be given Topographic Noise Stripping and read from the MILOC Manual Chapter 5 section 5.6 and RP 33 section 8.5.3 Topographic Noise Stripping. (20 mins)
9) The instructor will then ask each group to present their Best Summary Topic by drawing a diagram and verbally explaining their topic. (15 mins)
10) Instructor will hand out a diagram of Topographic Noise Stripping Terms and Conditions, trainees will draw and label the following information on the diagram: (10 mins)
    a) Sound Velocity
    b) Sonic Layer Depth (SLD)
    c) Source Depth
    d) Conjugate depth
    e) Critical Depth
    f) Depth Excess
    g) Deep Sound Channel Axis (DSCA)
    h) Instructor may use MILOC Figure 5.7 as reference.
11) Instructor will hand out a diagram of Topographic shading, trainees will draw and label the following information on the diagram: (10 mins)
    a) Sonic Layer Depth (SLD)
    b) Conjugate depth
    c) Depth Excess
    d) Deep Sound Channel Axis (DSCA)
    e) Seamount or Ridge
    f) Draw a Deep Sound Channel (DSC) sound path from the submarine
    g) Draw a Convergence Zone (CZ) sound path from the submarine

h) Instructor may use RP 33 Figure 8-4 as reference.

12) Instructor will hand out a diagram of Topographic Noise Stripping, trainees will draw and label the following information on the diagram: (10 mins)
    a) Source depth
    b) Stripping Zone
    c) Draw a Deep Sound Channel (DSC) sound path from the submarine
    d) Draw a Convergence Zone (CZ) sound path from the shipping
    e) Instructor may use RP 33 Figure 8-7 as reference.
13) Instructor will take up the diagrams with the class and discuss how and why submarines and warships use these bottom topographic features to their advantage. (25 mins)
14) The instructor will now break the class into 2 groups. Each group will be given 2 scenarios and conduct a Case Study. (20 mins)
15) The instructor will ask the groups to present their Case Studies to the class, instructor and trainees will discuss the cases. (20 mins)
16) The instructor will now conduct a class Case Study.(20 mins)

Resources

1) Whiteboard with markers
2) DWAN terminals (1/trainee)
3) Projector & Speakers
4) MILOC Manuals
5) RP 33 Manuals
6) Diagram of Topographic Noise Stripping Terms and Conditions
7) Diagram of Topographic shading
8) Diagram of Topographic Noise Stripping
9) Upslope/Downslope Diagrams
10) notepad per trainee
11) Flip chart papers and markers
12) Case Study Cards / Presentation

References

1) MILOC
2) RP 33

Notes to Instructor

Nil

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EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.14 – SONAR Equations

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 005.02.13
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Development of the Active SONAR Equation to include: a) Transmission Loss or Signal Excess b) Source Level of Active SONAR c) Target Strength d) Noise Level or Propagation Loss e) Recognition Differential to include: i) Directivity Index ii) Detection Threshold f) Reverberation Loss 2) Development of the Passive SONAR Equation to include: a) Transmission Loss or Signal Excess b) Source Level of Active SONAR c) Noise Level or Propagation Loss d) Recognition Differential to include: i) Directivity Index ii) Detection Threshold 3) Understanding SONAR Equation Variables to include: a) Active Detection b) Active Counter Detection c) Passive Detection d) Passive Counter Detection	Reading Assignment Diagram Jigsaw Math Quiz Case Study

Activity Steps	
	1) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 6, The SONAR equations section 6.1 Active SONAR Equation and subsection 6.1.1 Development of the Active SONAR Equation and RP 33 Chapter 10 Active Sonar sections 10.1 to 10.3 and sub sections. (30 mins) 2) Instructor will show a diagram or draw out the Active SONAR equation on the whiteboard: 2TL= SL - NL + DI - DT +TS (see MILOC section 6.1 Active SONAR Equation). (10 mins) 3) The instructor will conduct a modified Jigsaw activity, they will assign one of the following portions of the Active SONAR equations to the trainees

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(instructor may pair them up) and ask the trainees to provide a description of their portion of the equation: (30 mins)

a) Source Level
b) Noise Level
c) Directivity Index
d) Detection Threshold
e) Target Strength

4) Instructor will conduct a Math Quiz with the trainees on the Active SONAR equation. Using the same groups from the previous activity, the instructor will give the trainees number(s)/values indicating the portion of their Active SONAR equation: Source Level, Noise Level, Directivity Index, Detection Threshold, and Target Strength. For each equation, the trainees/groups will write their portion of the equation on the whiteboard. The class will then calculate the answer using the formula and numbers. Instructor can help walk the class through the first calculation and allow the trainees to complete the remainder on their own. (30 mins)
5) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 6 the SONAR equations section 6.2 Passive SONAR Equation and subsection 6.2.1 Development of the Active SONAR Equation and RP 33 Chapter 9 Passive Sonar sections 9.1 and 9.2. (25 mins)
6) Instructor will show a diagram or draw out the Passive SONAR equation on the whiteboard: TL= SL - NL + DI - DT (see MILOC section 6.2 Passive SONAR Equation). (10 mins)
7) The instructor will conduct a Jigsaw activity, they will assign one of the following portions of the Passive SONAR equations to the trainees (instructor may pair them up) and ask the trainees to provide a description of their portion of the equation: (30 mins)

a) Source Level
b) Noise Level
c) Directivity Index
d) Detection Threshold

8) Instructor will now conduct a Math Quiz with the trainees on the Passive SONAR equation. Using the same groups from the previous activity, the instructor will give the trainees number(s)/values indicating the portion of their Passive SONAR equation: Source Level, Noise Level, Directivity Index and Detection Threshold. For each equation, the trainees/groups will write their portion of the equation on the whiteboard. The class will then calculate the answer using the formula and numbers. Instructor can help walk the class through the first calculation and allow the trainees to complete the remainder on their own. (30 mins)
9) Instructor will direct the trainees to complete a reading assignment found in the MILOC Manual Chapter 6 the SONAR equations section 6.4 SONAR Equation Variables and subsection 6.4.1 Source Level. (20 mins)

	10) The instructor will now break the class into 2 groups. Each group will be given 4 scenario cards labelled; Active Detection, Active Counter Detection, Passive Detection and Passive Counter Detection. The instructor will read the scenarios below and ask the groups to determine which of the 4 cards applies to the scenario. (30 mins)

Resources	References
1) Whiteboard with markers 2) DWAN terminals (1/trainee) 3) Projector & Speakers 4) MILOC Manuals 5) RP 33 Manuals 6) Diagram of Active SONAR equation 7) Diagram of Passive SONAR equation 8) Calculator per trainee 9) SONAR Equation Variable cards 10) Active Sonar Math Questions 11) Passive Sonar Math Questions 12) notepad per trainee	1) MILOC 2) RP 33

Notes to Instructor
Nil

EO 005.02 - Interpret Oceanographic & Environmental Conditions & Data

005.02.15 – Range Dependent Oceanographic Features & SONAR Equations Confirmation

Conditions		Sequence
Unclassified classroom with DWAN and Projector		Instructed after EO 005.02.14
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	RQ PO2 Instructor

Learning Objectives	Method
1) Confirmation of understanding Fronts & Eddies 2) Confirmation of understanding Shallow Water Propagation 3) Confirmation of understanding SONAR equations	Written Confirmation Math questions

Activity Steps	
	1) Instructor will conduct a confirmation check of the trainee's comprehension of the subject material covered in EO 005.02.12 to 005.03.14. The confirmation check will comprise of questions and diagrams that the trainees will be required to perform on the following items: (70 mins) a) Label an Ocean Features Analysis (OFA) diagram; b) Label and draw the sound path for an Upslope Enhancement diagram; c) Label and draw the sound path for a Downslope Enhancement diagram; d) Label a Sound Velocity Profile diagram to include; i) Sonic Layer Depth (SLD) ii) Conjugate depth iii) Depth Excess iv) Deep Sound Channel Axis (DSCA) e) Write out the Active SONAR equation; f) Write out the Passive SONAR equation; g) Complete the Active SONAR math equation with the numbers provided; and h) Complete the Passive SONAR math equation with the numbers provided. 2) Once the trainees have completed the confirmation check, the trainees will pass their answers to another trainee. The instructor will take up the answers with the class, trainees will correct their classmates work and return the answers to the trainees. The instructor will collect the final score and provide individual feedback to trainees prior to EC 05.02. (30 mins)

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Resources	References
1) Whiteboard with markers 2) MILOC Manuals 3) RP 33 Manual 4) Confirmation Check Questions & Diagrams 5) Confirmation Check Questions & Diagrams Instructor Answer Key 6) Calculator per trainee 7) notepad per trainee	1) MILOC Manual 2) RP 33 Manual

Notes to Instructor

Nil

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EC 005.02 - Interpret Oceanographic and Environmental Conditions

Conditions		Sequence
Unclassified classroom with DWAN and Projector		Conducted after EO 005.02.15
Time	**Classification**	**Instructor**
300 mins	UNCLASSIFIED	RQ PO2 Instructor

Scenario	Practical assessment based on content from EO 005.02
Assessment Steps	**Input:** Trainees are provided with an Oceanography package containing; an Ocean Features Analysis (OFA), a chart with Ocean Bottom Topography, a chart with variable seawater characteristics, a Sound Velocity Profile and a sound propagation loss Full Field display. **Performance:** Trainees will have 1 x 5 minutes to interpret the oceanographic data provided followed by 1x 15 minute question and answer period with the instructor(s) followed by 5 minute period instructor debrief, 25 mins total per trainee. **Output:** The trainee must achieve 21/28 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback provided via EC 005.02 checklist.

Resources	References
1) Whiteboard with markers 2) Oceanography package to include: a) Ocean Features Analysis (OFA) b) Chart with Ocean Bottom Topography c) Chart with variable seawater characteristics d) Sound Velocity Profile e) Sound Propagation loss Full Field display.	1) MILOC 2) RP 33

Notes to Instructor	
Nil	
Limitations	No use of publications or manuals
Impact of Failure	Retrain opportunities and reassess

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Item	EC 005.02 Interpret Oceanographic and Environmental Conditions	SAT	UNSAT	Remarks
1. Explain the Ocean Features Analysis (OFA)				
a.	Warm water masses	☐	☐	
b.	Cold water masses	☐	☐	
c.	Current / Fronts	☐	☐	
d.	Eddies (Warm/Cold)	☐	☐	
2. Describe the Bottom Topography and Composition				
a.	Bottom Composition and effects	☐	☐	
b.	Shallow Water	☐	☐	
c.	Slope Water	☐	☐	
d.	Deep Water	☐	☐	
e.	Pinnacle /Seamounts /Ridges /Trenches	☐	☐	
3. Explain the effects of Chemistry of Water in the region				
a.	Temperature	☐	☐	
b.	Salinity	☐	☐	
c.	Pressure	☐	☐	
d.	Biologics	☐	☐	
4. Explain the Sound Velocity Profile				
a.	Upper Mixed Layer	☐	☐	
b.	Sonic Layer Depth	☐	☐	
c.	Deep Sound Channel Axis	☐	☐	
d.	Conjugate Depth	☐	☐	
e.	Depth Access	☐	☐	

5. Identify the potential Sound Paths present			
a. Direct Path	☐	☐	
b. Bottom Bounce	☐	☐	
c. Deep Sound Channel	☐	☐	
d. Convergence Zone	☐	☐	
e. Surface Duct / Half Channel	☐	☐	
6. Using the Full Field display, identify the presence of the following sound characteristics:			
a. Bottom Bounce	☐	☐	
b. Convergence Zone	☐	☐	
c. Annulus	☐	☐	
d. Deep Sound Channel	☐	☐	
e. Shadow Zones	☐	☐	

Score: /28 The trainee must achieve 21/28 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 005.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 005.02 with the trainee.		**Trainee** I have been debriefed on the result of EC 005.02.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 005 – Display Intelligence via Underwater Warfare Brief

Scope	Conditions	Sequence
PO 005	Unclassified classroom with DWAN and Projector	Conducted after EC 005.02
Time	**Classification**	**Instructor**
250 min	UNCLASSIFIED	RQ PO2 Instructor

Assessment Steps	
	Input: Each trainee is given a platform to research and an oceanographic data pack. **Performance:** Trainees will be given 70 minutes to develop brief. Each trainee will then be given 10 minutes to brief class IAW PC 005 checklist. Instructor(s) will have 5 minutes per trainee for debrief. **Output:** The trainee must achieve 8/11 SATs to pass this PC. **Feedback:** Written and Verbal feedback to be provided IAW PC 005 checklist

Resources	References
1) Instructor DWAN with projector 2) UWW intelligence brief template 3) DWAN per trainee 4) Oceanography package to include: a) Ocean Features Analysis (OFA) b) Chart with Ocean Bottom Topography c) Chart with variable seawater characteristics d) Sound Velocity Profile e) Sound Propagation loss Full Field display.	1) MILOC Manual 2) RP 33 3) Open source websites

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

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Item	PC 005 DISPLAY INTELLIGENCE VIA UWW BRIEF	SAT	UNSAT	Remarks
1. Present Platform Brief				
a.	Relevant submarine information	☐	☐	
b.	Describe Historic data of platform	☐	☐	
c.	Identify Type of propulsion	☐	☐	
d.	Identify Type of weapons with images	☐	☐	
e.	Identify Type of sensors	☐	☐	
f.	List 3 platform Capabilities	☐	☐	
2. Present and Explain Oceanography Brief to include:				
a.	Ocean Feature Analysis	☐	☐	
b.	Sound Velocity Profile	☐	☐	
c.	Bottom Topography / Composition	☐	☐	
d.	Sound Paths present	☐	☐	
e.	Submarine Operating areas	☐	☐	

Score: /11 The trainee must achieve 8/11 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 005 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 005 with the trainee.	**Trainee** I have been debriefed on the result of PC 005.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 006 – OPERATE ACTIVE SONAR EQUIPMENT

1. **Performance**: Operate Active SONAR Equipment (C0015).

2. **Conditions**:

   a. Given:

      (1) Active SONAR equipment;
      (2) 510 Instructor's console;
      (3) Logs and records;
      (4) Internal communications;
      (5) Emission Control (EMCON) policy;
      (6) References;
      (7) Mentoring by minimum QL5A SONAR Op for the Embedded Training System (ETS); and
      (8) Supervision.

   b. Denied: Nil

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall operate active SONAR equipment (C0015), IAW refs, by:

   a. Ensuring HMS is operational IAW NCPM 600 and Operating Instructions Detecting-Ranging Set, Sonar, AN/SQS-510 by:
      (1) Performing pre sail checks on equipment (B0005) IAW refs;
      (2) Adhering to emission control (EMCON) restrictions (B0025);
      (3) Recording acoustic intelligence data (F0010) IAW NAVORD 3470-1;
      (4) Performing operating procedures IAW NCPM 600 and NAVORD 4003-6;
      (5) Performing initial contact procedures IAW NCPM 600;
      (6) Performing lost contact procedures IAW NCPM 600;
      (7) Performing bearing investigation IAW NCPM 600;
      (8) Tracking acoustic contacts IAW NCPM 600;
      (9) Performing torpedo procedures IAW ref NCPM 600;

(10) Performing mine investigation procedures IAW NCPM 600; and
(11) Maintaining operational logs (B0055) IAW NCPM 600 and A26.

4. **Training Limitations:** Nil

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EO 006.01 - Ensure HMS Is Operational

006.01.01 – Conduct HMS Flash Up

Conditions		Sequence
HMC Ship or SEACOT or Classified classroom		
Time	**Classification**	**Instructor**
350 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Introduction to 510 HMS 2) Flashing up and shutting down HMS to include: a) Operations room – data recorder b) SONAR operator console (SOC) 3) Perform Pre-sail checks	Ship Tour/Presentation EDI

Activity Steps	
	1) Instructor will either conduct an explanation of the parts of the 510 HMS on board ship (if available) or using the Power Point presentation in the classroom. (50 min) 2) Instructor will demonstrate how to flash up HMS using job sheet to include: Data recorder, and Sonar Operating Console. (30 min) 3) Instructor will demonstrate the pre-sail checks using the HMS job sheet to include operator checks and shut down procedures. (20 min) 4) Instructor will have trainees simulate flashing up HMS, conducting pre-sail checks and simulate shut down procedure. Instructor will ensure trainees have 2 opportunities to practice. Each trainee will be given 10 minutes per opportunity. (240 min) 5) Instructor will allow a question and answer period for feedback from trainees. (10 min)

Resources	References
1) HMC Ship or SEACOT 2) HMS with data recorder 3) Headset 4) Logs 5) HMS Flash up and pre-sail job sheet IAW EC Checklist	1) NCPM 600 2) Operating Instructions Detecting-Ranging Set, Sonar, AN/SQS-510 (V) 4

Notes to Instructor
Preferred option is ship tour and demonstration of flashing up HMS. Alternative option instructor to utilize pictures and/or video of HMS flash up procedures, if ship is unavailable. Instructor can conduct Log lesson on HMC ship or classroom.

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EC 006.01A - Ensure HMS Is Operational

Scope	Conditions	Sequence
EO 006.01	SEACOT Trainer or HMC Ship	Conducted after EO 006.01.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console, Data Recorder, Headset, and Logs. **Performance:** Trainees will be required to ensure HMS is operational IAW EC 006.01A checklist. 25 min/trainee is allocated for scheduling purposes. **Output:** The trainee must achieve 13/16 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback provided via EC 006.01A checklist.

Resources	References
1) HMC Ship or SEACOT with HMS console 2) HMS with data recorder 3) Headset 4) Logs	1) NCPM 600

Notes to Instructor
If HMC Ship is unavailable, instructor will have to simulate the Flashing up procedure process in SEACOT.

Limitations	If no ship available, instructor will have to simulate Data Recorder, Control Panel Remote
Impact of Failure	Feedback, time to make amendments and reassess

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Item	EC 006.01A ENSURE HMS IS OPERATIONAL	SAT	UNSAT	Remarks
1.	**Flash up HMS to include;**			
a.	Ensure power is available	☐	☐	
2.	**SONAR Operator Console (SOC)**			
a.	Ensure the CUD power ON/OFF switch to the ON position	☐	☐	
b.	Ensure the standby lamp illuminates amber on the CPR	☐	☐	
c.	Apply power to the SOC by pressing the power ON/OFF push button	☐	☐	
d.	Ensure the INHTX/ENTX key switch is set to INHTX position	☐	☐	
e.	Set the CPA power on button to the ON Position	☐	☐	
f.	Observe the following:			
	- Power ON indication illuminates green on the CPR;	☐	☐	
	- Power On indication illuminates green on the CPA;	☐	☐	
	- Power ON indication illuminates orange on the CUD;	☐	☐	
	- All lamps on the lower controls of the TVMs will flash ON;	☐	☐	
	- Test pattern will appear in the TVMs followed by the BIT test for approx. (one) minute.	☐	☐	
3.	**Perform pre sail checks**			
a.	Conduct operator checks as follows:			
	- Verify display formats on upper/lower TVMs;	☐	☐	
	- Verify joystick, trackball and keyboard operation; and	☐	☐	
	- Ensure INHTX/ENTX key switch is in the INHTX position;	☐	☐	
4.	**Shut Down Procedure**			

- Set CUD power ON/OFF switch to OFF position;	☐	☐	
- Set CPR power ON/OFF push button to OFF position; and	☐	☐	

Score: /16

The trainee must achieve 13/16 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 006.01A – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 006.01A with the trainee.	**Trainee** I have been debriefed on the result of EC 006.01A.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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006.01.02 – Conduct Closing Up

Conditions		Sequence
SEACOT Trainer or HMC Ship		Conducted after EC 006.01A
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Conducting closing up procedures 2) Applying ordered settings from SCS and report transmitting.	EDI

Activity Steps	
	1) Instructor will demonstrate how to conduct closing up procedures using HMS job sheet to include: (25 min) a) Set up Date/Time b) Set Up PSCAN display c) Select PIP Short d) Select ASCAN display e) Ensure List parameters is displayed in list area f) Set Up TX CONTROL menu in CSM area g) Conduct communications check and Closing Up report 2) Instructor will have trainees practice close up procedures using job sheet. (100 min) 3) Instructor will demonstrate how to apply ordered settings and proper reports to include: (25 min) a) Apply NORMAL/SHALLOW/MINE configuration b) Set OMNI/DTW/DTN mode c) Set power LOW/HIGH d) Set frequency e) Set FM/CW TX type f) Set Manual/AUTO sequence g) Set TIOW h) Set VOSIW i) Report “Carrying out standard active/ passive search procedures 4) Instructor will have trainees practice closing up and proper reporting using job sheet. (150 min)

Resources	References
1) SEACOT 1 per trainee, 1 instructor 2) Job Sheets for close up and settings IAW EC Checklist	1) NCPM 600

Notes to Instructor

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Nil.

EC 006.01B – Perform HMS Closing Up Procedures

Scope	Conditions	Sequence
EO 006.01	SEACOT Trainer or HMC Ship	Conducted after EO 006.01.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console, and settings for Closing Up. **Performance:** Trainees will be required to perform HMS closing up procedures IAW EC 006.01B checklist. 25 min/trainee have been allocated for scheduling purposes. **Output:** The trainee must achieve 12/15 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback provided via EC 006.01B checklist.

Resources	References
1) SEACOT HMS console 2) Headset 3) Logs 4) Sheet with Closing Up setting IAW EC Checklist	1) NCPM 600

Notes to Instructor
EC 006.01A and B can be conducted concurrently.

Limitations	Nil
Impact of Failure	Feedback, time to make amendments and reassess

Item	EC 006.01B PERFORM HMS CLOSING UP PROCEDURES	SAT	UNSAT	Remarks
1.	**Conduct closing up procedures**			
a.	Select system DATE/TIME CSB and enter the current date/time group	☐	☐	
b.	Select full PSCAN display format in upper TVM	☐	☐	
c.	Select PIP Short	☐	☐	
d.	Select full ASCAN display format in lower TVM	☐	☐	
e.	Ensure LIST PARAMETERS table is displayed in list area	☐	☐	
f.	Ensure LIST TRACKS table is displayed in list area	☐	☐	
g.	Select TX CONTROL menu in CSM area	☐	☐	
h.	Conduct communication checks	☐	☐	
2.	**Apply ordered settings from SCS and report transmitting**			
a.	NORMAL/SHALLOW/MINE configuration	☐	☐	
b.	OMNI/DTN/DTW mode	☐	☐	
c.	LOW/HIGH power	☐	☐	
d.	6.4/7.2/8.0 kHz	☐	☐	
e.	FM/OW TX type	☐	☐	
f.	MANUAL/AUTO sequence	☐	☐	
g.	TIOW ___ kyds	☐	☐	
h.	VOSIW ___ meters/sec	☐	☐	
i.	Conduct Closing Up Report	☐	☐	

Score: /17

The trainee must achieve 14/17 SATs to pass this EC.

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Positive Feedback		Improvement Feedback	
EC 006.01B – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 006.01B with the trainee.		**Trainee** I have been debriefed on the result of EC 006.01B.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 006.02 – Operate HMS

006.02.01 – HMS Logging and Marine Mammal Mitigation

Conditions		Sequence
Unclassified classroom with DWAN		Conducted after EC 006.01B
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Maintain HMS logs 2) Marine Mammal Mitigation	EDI Think Pair Share Practical Exercise

Activity Steps	
	1) Instructor will demonstrate how to fill out a HMS log IAW example and template (20 min) 2) Instructor will hand out blank hard copies of logs. Trainees will fill out logs based on instructor provided data. (30 min) 3) Instructor will take up answers with trainees for feedback. (15 min) 4) Trainees are provided opportunity to read NAVORD 4003-6 and trainees are to write down the 3 most important pieces of information they gained from the readings. (25 min) 5) Trainees are to share their responses with the class and discuss. (10 min). 6) Instructor will show SEMS section on Ramp Up procedures and explain the process. (20 min) 7) Instructor will pair up trainees and have them practice ramp up procedures using handouts. (50 min) 8) Instructor will run a practical exercise where trainees will perform and log Ramp up procedures. (30min)

Resources	References
1) DWAN with projector for instructor 2) HMS logs 3) NAVORD 4003-6 4) SEMS section EO5A 5) RAMP UP procedure sheet	1) NCPM 600 2) NAVORD 4003-6 3) SEMS section EO5A

Notes to Instructor
Instructor will either provide trainees with paper Log handouts or allow trainees to type them on their workstations.

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EO 006.02 – Operate HMS

006.02.02 – Operating HMS in Listen Mode

Conditions		Sequence
SEACOT Trainer or HMC Ship		Conducted after EO 006.02.01
Time	**Classification**	**Instructor**
100 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Operate in Listen mode to include: a) Broadband display b) Narrowband display	EDI

Activity Steps	
	1) Instructor will demonstrate how to operate HMS in listen mode to include: (30 mins) a) Broadband display b) Narrowband display 2) Instructor will have trainees practice how to operate HMS in Listen Mode. (70 mins)

Resources	References
1) SEACOT 1 per trainee and 1 for instructor 2) Listen mode job sheet IAW EC Checklist	1) NCPM 600

Notes to Instructor
Instructor will demonstrate all the passive display modes and provide explanation on why it is used.

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EO 006.03 – HMS Contact Procedures

006.03.01 – Initial, Lost Contact and Bearing Investigation

Conditions		Sequence
SEACOT Trainer or HMC Ship		Conducted after EO 006.02.02
Time	**Classification**	**Instructor**
1100 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Conducting initial active contact procedures 2) Conducting initial passive contact procedures. 3) Conducting Lost contact procedures 4) Conducting Bearing Investigation	EDI

Activity Steps	
	1) Instructor will demonstrate how to conduct initial active contact procedures IAW job sheet. (50 min) 2) Instructor will have trainees practice initial active contact procedures using job sheet. (100 min) 3) Instructor will demonstrate how to conduct initial active contact procedures IAW job sheet. (50 min) 4) Instructor will have trainees practice initial active contact procedures using job sheet. (100 min) 5) Instructor will demonstrate how to conduct lost contact procedures IAW job sheet. (50 min) 6) Instructor will have trainees practice lost contact procedures using job sheet. (100 min) 7) Instructor will demonstrate how to conduct bearing investigation procedures IAW job sheet. (50 min) 8) Instructor will have trainees practice bearing investigation procedures using job sheet. (100 min) 9) Instructor will conduct individual confirmation check on all procedures with each trainee, 40 min per trainee. Remaining trainees to continue practice on their own or in pairs.(500 min total)

Resources	References
1) SEACOT 1 per trainee and 1 instructor 2) Initial/ Lost contact/bearing investigation job sheet IAW EC Checklist	1) NCPM 600

Notes to Instructor
Trainees may pair up for practice.

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EC 006.03 – HMS Contact Procedures

Scope	Conditions	Sequence
EO 006.03	SEACOT Trainer or HMC Ship	Conducted after EO 006.03.01
Time	**Classification**	**Instructor**
400 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console a minimum of 1 x Underwater contact to conduct Initial, Lost Contact and Bearing Investigation procedures. **Performance:** Trainees will be required to Initial, Lost Contact and Bearing Investigation procedures IAW EC 006.03 checklist. 30 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 29/34 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback provided via EC 006.03 checklist.

Resources	References
1) SEACOT 2) HMS LOG	1) NCPM 600

Notes to Instructor
Trainee may have voice reports.

Limitations	Nil
Impact of Failure	Retrain and Retest

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Item	EC 006.03 HMS CONTACT PROCEDURES	SAT	UNSAT	Remarks
1.	**Conduct initial active contact procedures**			
a.	Monitor ASCAN for three consecutive echoes	☐	☐	
b.	Switch to ASCAN/BSCAN	☐	☐	
c.	Monitor PSCAN for broadband bearing correlation	☐	☐	
d.	Report: "New HMS contact, bearing ___, range ___, passive correlation bearing ___, (or no passive correlation)	☐	☐	
e.	CREATE TRACK CSB then REPORT TRACK CSB	☐	☐	
f.	Update symbol position on the next transmission in BSCAN (if necessary)	☐	☐	
g.	Select BSCAN/TSCAN display format in lower TVM (TSCAN Auto)	☐	☐	
h.	Perform second update of contact if necessary	☐	☐	
i.	Request and switch to CW for Doppler.	☐	☐	
j.	Provide AMPLIFYING REPORT to include: • Track# • Bearing • Range • Tracking • Indicating • Doppler • Classify	☐	☐	
k.	Request and switch to FM	☐	☐	
l.	Commence logging, report and update every three TX and monitor auto tracking for accuracy. Report both active and passive if held	☐	☐	
2.	**Conduct initial Passive Contact**			
a.	Select full PSCAN display format	☐	☐	

b. Position the ball tab on broadband bearing line	☐	☐	
c. Report "New passive contact, bearing ___ classified possible submarine/surface ship"	☐	☐	
d. Select CREATE TRACK CSB and continue to update and report bearing	☐	☐	
e. If in ACTIVE state, monitor ASCAN for active data correlation	☐	☐	
3. Conduct lost active contact procedures			
a. If no echoes after three (3) transmissions, report "No echoes, trk # ___, carrying out lost contact procedures"	☐	☐	
b. Select ASCAN/BSCAN display format in lower TVM	☐	☐	
c. Ball tab last known bearing in BSCAN	☐	☐	
d. Select DTW and conduct lost contact procedure IAW NCPM 600	☐	☐	
e. If regained prior to three (3) full transmissions report "Bearing drawing left/right/steady"	☐	☐	
f. If no echoes after three (3) full transmissions report "Lost contact procedures complete to include: • Track# • Last known bearing • Range • Tracking • Indicating • Doppler • Request new search order	☐	☐	
g. Assign LOST CONTACT symbol as directed by SCS	☐	☐	
4. Conduct lost active contact procedures			
a. Select DTN mode of transmission	☐	☐	
b. Select SPLIT ASCAN/BSCAN	☐	☐	

c. Line up the ball tab in BSCAN display to the ordered bearing and complete three (3) transmissions	☐	☐	
d. Step 20 degrees aft of the ordered bearing and transmit once if no contact is gained	☐	☐	
e. Step 10 degrees fwd towards the head bearing and transmit once	☐	☐	
f. Step back to initial bearing for one transmission	☐	☐	
g. Step 20 degrees forward of the head bearing and transmit once	☐	☐	
h. Step 10 degrees aft towards the head bearing and transmit once	☐	☐	
i. Step back to the initial bearing and transmit once	☐	☐	
j. If no contact is gained, reports "Bearing investigated, no echoes, reinvestigating." **Or** if contact is gained, Initiate initial contact procedures	☐	☐	

Score: /34

The trainee must achieve 29/34 SATs to pass this EC.

Positive Feedback	Improvement Feedback

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EC 006.03 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 006.03 with the trainee.		**Trainee** I have been debriefed on the result of EC 006.03.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 006.04 - Perform Torpedo Procedures

006.04.01 – Torpedo Procedures

Conditions		Sequence
SEACOT Trainer or HMC Ship		Conducted after EC 006.03
Time	**Classification**	**Instructor**
350 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Perform torpedo procedures IAW NCPM 600	EDI

Scenario	
Activity Steps	1) Instructor will demonstrate how to conduct torpedo procedures IAW job sheet. (30 min) 2) Instructor will have trainees practice torpedo procedures using job sheet. (80 min) 3) Instructor will conduct individual confirmation check on all procedures with each trainee, 20 min per trainee. Remaining trainees to continue practice on their own or in pairs.(240 min total)

Resources	References
1) SEACOT 1 per trainee, 1 instructor 2) Torpedo procedures job sheet IAW EC Checklist	1) NCPM 600

Notes to Instructor
Nil

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EC 006.04 - Perform Torpedo Procedures

Scope	Conditions	Sequence
EO 006.04	SEACOT Trainer or HMC Ship	Conducted after EO 006.04.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console and a minimum of 1 x torpedo contact to conduct Torpedo procedures. **Performance:** Trainees will be required to perform torpedo procedures IAW EC 006.06 checklist. 25 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 8/10 SATs to pass this EC and all critical(s). **Feedback:** Immediate verbal and written feedback provided via EC 006.04 checklist.

Resources	References
1) SEACOT 2) Voice procedures sheets	1) NCPM 600

Notes to Instructor
Nil

Limitations	Unable to simulate high bearing rate of torpedo
Impact of Failure	Retrain and Retest

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Item	EC 006.04 PERFORM TORPEDO PROCEDURES	SAT	UNSAT	Remarks
1. Perform Torpedo procedures				
a.	Select a full "PSCAN display format" in UPPER TVM	☐	☐	
b.	Ball tab torpedo HE and select ALERT QAB	☐	☐	
c.	Report "torpedo, torpedo, torpedo, (true) bearing ___"	☐	☐	***CRITICAL**
d.	Continue to update ABL and report true bearing	☐	☐	
e.	Give amplifying reports "torpedo drawing left/right/steady"	☐	☐	
f.	When torpedo HE fades, report "torpedo HE faded, bearing ___	☐	☐	
g.	Request permission to transmit and report "Investigating bearing"	☐	☐	
h.	Conduct bearing investigation	☐	☐	
i.	If contact is gained, conducts initial contact procedures, or if no contact gained, reports "Investigation complete, no echoes, reinvestigating"	☐	☐	
j.	Apply new search orders and settings as directed	☐	☐	

Score: /10 / 1 CRITICAL

The trainee must achieve 8/10 SATs on non-critical factors **AND** all critical factors to pass this EC.

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Positive Feedback		Improvement Feedback	
EC 006.04 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 006.04 with the trainee.		**Trainee** I have been debriefed on the result of EC 006.04.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 005.01 - Perform Mine Investigation Procedures

006.05.01 – Mine Investigation Procedures

Conditions		Sequence
SEACOT Trainer or HMC Ship		Conducted after EC 006.04
Time	**Classification**	**Instructor**
350 min	SEACOT	RQ-SONAROP-MS

Learning Objectives	Method
1) Perform mine investigation procedures	EDI

Scenario	
Activity Steps	1) Instructor will demonstrate how to conduct mine investigation procedures IAW job sheet. (30 min) 2) Instructor will have trainees conduct mine investigation procedures using job sheet. (80 min) 3) Instructor will conduct individual confirmation check on all procedures with each trainee, 20 min per trainee. Remaining trainees to continue practice on their own or in pairs.(240 min total)

Resources	References
1) SEACOT 1 per trainee, 1 instructor 2) Mine Investigation job sheet IAW EC Checklist	1) NCPM 600

Notes to Instructor
Nil

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EC 006.05 - Perform Mine Investigation Procedures

Scope	Conditions	Sequence
EO 006.05	SEACOT Trainer or HMC Ship	Conducted after EO 006.05.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console and a minimum of 3 x mines and/or mine like objects to conduct Mine Investigation procedures. **Performance:** Trainees will be required to perform mine investigation procedures IAW EC 006.07 checklist. 25 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 11/14 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback provided via EC 006.05 checklist.

Resources	References
1) SEACOT 2) Mine investigations voice report sheets IAW EC Checklist	1) NCPM 600

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and Retest

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122419

Item	EC 006.05 PERFORM MINE INVESTIGATION PROCEDURES	SAT	UNSAT	Remarks
1.	**Performs Mine Investigation Procedures**			
a.	Select MINE configuration mode; as directed	☐	☐	
b.	Select DTW	☐	☐	
c.	Select LOW power	☐	☐	
d.	Select 8.0 KHZ	☐	☐	
e.	Select FM TX type	☐	☐	
f.	Select AUTO TX sequence	☐	☐	
g.	Set TIOW 2.5 kyds as default (2 – 4 kyds)	☐	☐	
h.	Select ASCAN/BSCAN on lower TVM	☐	☐	
i.	Select full ASCAN on upper TVM	☐	☐	
j.	Set ship's head to auto: • Monitor ships head, bearing locks DTW to ships head	☐	☐	
k.	If contact is gained, report "New SONAR contact Bearing ___ range ___,"	☐	☐	
l.	Create track on contact: • Select Report Track CSB	☐	☐	
m.	Maintain and report priority contact (closest). ("Previously reported HMS contact # ___, Bearing (true) ___, Range ___ yds")	☐	☐	
n.	If contact is lost, report "By HMS, Lost Contact # ___, Bearing (true) ___, Range ___ yds. (Only if contact is lost ahead of the beam.)	☐	☐	

Score: /14	The trainee must achieve 11/14 SATs to pass this EC.

Positive Feedback	**Improvement Feedback**

EC 006.05 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 006.05 with the trainee.	**Trainee** I have been debriefed on the result of EC 006.05.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 006 – Operate Active SONAR Equipment

Scope	Conditions	Sequence
PO 006	SEACOT Trainer or HMC Ship	Conducted after EC 006.05
Time	**Classification**	**Instructor**
600 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees are provided the HMS console to conduct Set up/Closing Up procedures, a minimum of 1 x underwater contact to conduct Initial, Lost Contact and Bearing Investigation procedures, 1 x torpedo to conduct Torpedo procedures, 1 x mine contact to conduct Mine Investigation procedures and HMS logs to fill out correctly. **Performance:** Trainees will be required to operate active SONAR equipment and correctly maintain logs IAW PC 006 checklist. 50 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 61/76 SATs to pass this PC and all critical(s). **Feedback:** Immediate verbal and written feedback provided via PC 006 checklist.

Resources	References
1) SEACOT	1) NCPM 600

Notes to Instructor
Nil

Limitations	Unable to simulate high bearing rate of torpedo
Impact of Failure	IAW NPTORD 4500-54.

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Item	PC 006 OPERATE ACTIVE SONAR EQUIPMENT	SAT	UNSAT	Remarks
1.	**Operating procedures to include;**			
a.	Select system DATE/TIME CSB and enter the current date/time group	☐	☐	
b.	Select full PSCAN display format in upper TVM	☐	☐	
c.	Select PIP Short	☐	☐	
d.	Select full ASCAN display format in lower TVM	☐	☐	
e.	Ensure LIST PARAMETERS table is displayed in the list area.	☐	☐	
f.	Select TX CONTROL menu	☐	☐	
g.	Conduct communication checks	☐	☐	
h.	Select ASCAN/BSCAN Lower TVM.	☐	☐	
i.	NORMAL/SHALL/MINE configuration	☐	☐	
j.	OMNI/DTN/DTW mode.	☐	☐	
k.	Select LOW/HIGH power as directed	☐	☐	
l.	6.4/7.2/8.0 kHz	☐	☐	
m.	FM/CW TX type	☐	☐	
n.	MANUAL/AUTO sequence	☐	☐	
o.	TIOW_____ kyds	☐	☐	
p.	VOSIW_____ m/sec	☐	☐	
q.	Conduct Closing Up Report	☐	☐	
2.	**INITIAL CONTACT**			
a.	Monitor Lower TVM for three consecutive echoes (A/BSCAN)	☐	☐	
b.	Monitor PSCAN for broadband bearing correlation	☐	☐	
c.	Report: “New HMS contact, bearing ___, range ___, passive correlation bearing ___, (or no passive correlation)	☐	☐	

d. CREATE TRACK CSB then REPORT TRACK CSB	☐	☐	
e. Updates symbol position on the next transmission in BSCAN	☐	☐	
f. Select BSCAN/TSCAN display format in the lower TVM (TSCAN auto)	☐	☐	
g. Perform second update of CTC if necessary	☐	☐	
h. Request and switch to CW for Doppler	☐	☐	
i. Provides AMPLIFYING REPORT as follows: • Track# • Bearing • Range • Tracking • Indicating • Doppler • Classify	☐	☐	
j. Request and switch to FM	☐	☐	
k. Commence logging, report and update every three TX and monitor auto tracking for accuracy. Report both active and passive if held	☐	☐	
3. CONDUCT INITIAL PASSIVE CONTACT			
a. Selects full PSCAN display format	☐	☐	
b. Positions the ball tab on the broadband bearing line	☐	☐	
c. Reports "New passive contact, bearing ___ classified possible submarine/surface ship"	☐	☐	
d. Select CREATE TRACK and continue to update and report bearing	☐	☐	
e. If in ACTIVE state monitors ASCAN for active data correlation	☐	☐	
4. LOST CONTACT PROCEDURES			
a. If no echoes after 3 consecutive TX, reports "No echoes, HMS, Ctc # ____ carrying out lost contact procedures	☐	☐	

b. Select ASCAN/BSCAN display format in lower TVM	☐	☐	
c. Ball tab last known bearing in BSCAN	☐	☐	
d. Select DTW and conduct lost contact procedure IAW NCPM 600	☐	☐	
e. If regained prior to three (3) full transmissions report "Bearing drawing left/right/steady"	☐	☐	
f. If no echoes after three (3) full transmissions report "Lost contact procedures complete to include: • Track# • Last known bearing • Range • Tracking • Indicating • Doppler • Request new search order	☐	☐	
g. Assign LOST CONTACT symbol as directed by SCS	☐	☐	
h. If contact is regained the operator will conduct initial contact procedures	☐	☐	
5. PERFORM BEARING INVESTIGATION			
a. Select DTN mode of transmission	☐	☐	
b. Select SPLIT ASCAN/BSCAN	☐	☐	
c. Line up the ball tab in BSCAN display to the ordered bearing and complete three (3) transmissions	☐	☐	
d. Step 20 degrees aft of the ordered bearing and transmit once if no contact is gained	☐	☐	
e. Step 10 degrees fwd towards the head bearing and transmit once	☐	☐	
f. Step back to initial bearing for one transmission	☐	☐	
g. Step 20 degrees forward of the head bearing and transmit once	☐	☐	
h. Step 10 degrees aft towards the head bearing and transmit once	☐	☐	

i. Step back to the head bearing and transmit once	☐	☐	
j. If no contact is gained, reports "Bearing investigated, no echoes, reinvestigating." Or if contact is gained, Initiate initial contact procedures	☐	☐	
6. TORPEDO PROCEDURES (PSCAN) – ACTIVE STATE			
a. Select a full 'PSCAN display format' in the UPPER TVM	☐	☐	
b. Ball tab torpedo HE and select ALERT QAB.	☐	☐	
c. Report "By HMS, torpedo, torpedo, torpedo, bearing (true) ___,	☐	☐	**Critical**
d. Continue to update TBL and report true bearing	☐	☐	
e. Give amplifying report "torpedo drawing left/right/steady"	☐	☐	
f. When torpedo HE fade/abrupt out, report "torpedo HE fade/abrupt out, bearing ___	☐	☐	
g. Request permission to transmit and report "Investigating bearing_____"	☐	☐	
h. Conduct bearing investigation	☐	☐	
i. If contact is gained, conduct initial contact procedures, if no contact is gained, reports "investigation complete, no echoes, re-investigating	☐	☐	
j. Apply new search orders and settings as directed	☐	☐	
7. MINE INVESTIGATION PROCEDURES			
a. Selects MINE configuration mode	☐	☐	
b. Selects DTW	☐	☐	
c. LOW Power	☐	☐	
d. 8.0 KHZ (or directed)	☐	☐	
e. FM TX type	☐	☐	
f. AUTO TX sequence	☐	☐	

g. TIOW 2-4 kyds (2.5 kyds or as directed)	☐	☐	
h. Select ASCAN/BSCAN on lower TVM	☐	☐	
i. Select full ASCAN on upper TVM	☐	☐	
j. Set ship's head to auto: • Monitor ships head, bearing locks DTW to ships head	☐	☐	
k. If contact is gained, report "by HMS, contact bearing (true) ___ range ___ yds"	☐	☐	
l. Create new track on contact; • Selects Report Track CSB	☐	☐	
m. Maintains and reports priority contact (closest). "Previously reported HMS contact # ___ Bearing (true) ___, Range ___ yds	☐	☐	
n. If contact is lost report `by HMS, Lost contact # ___, Bearing (true) ___, Range ___ yds. (only if the contact is lost ahead of the beam.)	☐	☐	
8. Maintain Logs			
a. Fill out logs correctly	☐	☐	

Score: /76

The trainee must achieve 61/76 SATs including all critical items to pass this PC.

Positive Feedback	Improvement Feedback

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PC 006 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 006 with the trainee.		**Trainee** I have been debriefed on the result of PC 006.	
PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 007 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS

1. **Performance**: Perform Basic Passive Acoustic Analysis (F0035).

2. **Conditions**:

   a. Given:

      (1) Sound source;
      (2) Calculators;
      (3) Analysis sheets;
      (4) Passive acoustic display;
      (5) Logs and records;
      (6) Chronoscope;
      (7) References;
      (8) Assistance; and
      (9) Supervision.

   b. Denied: Nil

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall perform basic passive acoustic analysis (F0035), IAW refs, by:

   a. Interpreting LOFAR Gram Acoustic Characteristics (D0080) IAW CFCD 113 Vol 1 & 2, CFCD 106, and Principles of LOFAR Gram Analysis to include operating acoustic analysis system IAW refs;

   b. Determining mathematical relationships of signature sources IAW CFCD 113 Vol 1 & 2, CFCD 106, and Principles of LOFAR Gram Analysis;

   c. Performing Passive Aural Listening (PAL) (C0195) IAW refs CFCD 113 Vol 1 & 2, and Principles of LOFAR Gram Analysis;

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d. Applying the five steps to analysis IAW CFCD 113 Vol 1 & 2, CFCD 106, and Principles of LOFAR Gram Analysis;

e. Localizing passive acoustic data contacts by Doppler (C0085) IAW refs CFCD 106, CFCD 113 Vol 1 & 2, and ATP 28; and

f. Providing generic classification of passive contacts IAW refs A16 and A48 to include:

(1) Analyze passive acoustic data (C0065) IAW refs; and
(2) Classify passive acoustic data (C0080) IAW refs.

4. **Training Limitations:** Nil

EO 007.01 - Interpret LOFAR Gram Acoustic Characteristics

007.01.01 - LOFAR Grams

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		
Time	**Classification**	**Instructor**
450 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Introduction to the passive acoustic environment 2) Interpreting the principles of sound as applied to passive acoustics 3) Interpreting different types of noise 4) Examining a passive acoustic display 5) Identifying frequency bands 6) Interpreting fundamental frequencies and harmonics 7) Measuring fundamental frequencies and harmonics 8) Classifying signature characteristics	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator; 2) Osprey analysis software; 3) Headphones; 4) Calculators; 5) Stopwatch; 6) ARS; 7) Pencils and erasers; 8) Mathematical formula diagrams handouts;	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

Notes to Instructor
Nil

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EO 007.02 - Determine Mathematical Relationships of Signature Sources

007.02.01 – Engineering Relationships

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		Conducted after EO 007.01.01
Time	**Classification**	**Instructor**
1000 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Examining prime movers 2) Examining drive systems 3) Examining blade shaft components 4) Examining auxiliary systems 5) Examining resonant and modulation sources	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is Classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol. 1; 2) CFCD 113 Vol. 2

Notes to Instructor
Nil

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EC 007.02 - Determine Mathematical Relationships of Signature Sources

Scope	Conditions	Sequence
EO 007.02	Classified Classroom (ADAC for NFS(A))	Conducted after EO 007.02.01
Time	**Classification**	**Instructor**
50 min	SECRET	RQ-SONAROP-MS ADAC Staff

Assessment Steps	
	Input: Trainees are provided an Osprey Passive Acoustic simulator, an Acoustic Response sheet, and passive acoustic contacts with mathematical relationships to be determined. **Performance:** Trainees will be required to determine the mathematical relationships. **Output:** The trainee must achieve 70% to pass this EC IAW TRAINITY ADAC requirements. **Feedback:** Written and Verbal feedback to be provided IAW EC 007.02 Acoustic Response Answer Sheet.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	**1)** CFCD 113 Vol 1 **2)** CFCD 113 Vol 2

Notes to Instructor
Practical Assessment IAW TRINITY ADAC Requirements

Limitations	Nil
Impact of Failure	Retrain and Retest

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EO 007.03 - Perform Passive Aural Listening (PAL)

007.03.01 – PAL

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		
Time	**Classification**	**Instructor**
1150 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Recognizing propulsion and machinery sound sources; 2) Recognizing marine and biological sound sources; 3) Recognizing torpedo and miscellaneous sound sources; 4) Recognizing active acoustic sound sources.	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is Classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol.1 2) CFCD 113 Vol. 2

Notes to Instructor
Nil

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EC 007.03 - Perform PAL

Scope	Conditions	Sequence
EO 007.03	Classified Classroom (ADAC for NFS(A))	Conducted after EO 007.03.01
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-SONAROP-MS ADAC Staff

Assessment Steps	
	Input: Trainees are provided an Osprey Passive Acoustic simulator, an Acoustic Response sheet, and Passive Acoustic contacts to conduct Passive Aural Listening procedures. **Performance:** Trainees will be required to perform PAL **Output:** The trainee must achieve 50% to pass this EC IAW TRAINITY ADAC requirements **Feedback:** Written and Verbal feedback to be provided IAW EC 007.03 Acoustic Response Answer Sheet.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and Retest

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EO 007.04 - Apply the Five Steps of Analysis

007.04.01 – Perform Five Steps of Analysis

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		
Time	**Classification**	**Instructor**
500 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Measuring and recording frequencies 2) Determining line sources 3) Determining propulsion systems 4) Matching the data to the class of targets 5) Determining class of target (type specific)	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is Classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol.1 2) CFCD 113 Vol. 2

Notes to Instructor
Nil

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EC 007.04 - Apply the Five Steps to Analysis

Scope	Conditions	Sequence
EO 007.04	Classified Classroom (ADAC for NFS(A))	Conducted after EO 007.04.01
Time	**Classification**	**Instructor**
50 min	SECRET	RQ-SONAROP-MS ADAC Staff

Assessment Steps	
	Input: Trainees are provided an Osprey Passive Acoustic simulator, an Acoustic Response sheet, and Passive Acoustic contacts to apply the Five Steps of Analysis. **Performance:** Trainees will be required to apply the five steps to analysis **Output:** The trainee must achieve 60% to pass this EC IAW TRAINITY ADAC requirements. **Feedback:** Written and Verbal feedback to be provided IAW EC 007.04 Acoustic Response Answer Sheet.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and reassess

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EO 007.05 - Localize Passive Acoustic Data Contacts by Doppler

007.05.01 – Passive Acoustic Data by Doppler

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Recognizing interference patterns and influences; and 2) Determining Closest Point of Approach (CPA).	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is Classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol. 1 2) CFCD 113 Vol. 2

Notes to Instructor
Nil

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EO 007.06 - Provide Generic Classification of Passive Contacts

007.06.01 – Mixed Contact Labs

Conditions		Sequence
Classified Classroom (ADAC for NFS(A))		
Time	**Classification**	**Instructor**
6650 min	SECRET	RQ-SONAROP-MS ADAC Staff

Learning Objectives	Method
1) Identifying sound sources from friendly surface vessels 2) Identifying sound sources from Russian/Rest of World (ROW) 3) Identifying sound sources from friendly conventional submarines 4) Identifying sound sources from friendly nuclear submarines 5) Identifying sound sources from Russian/ROW conventional submarines 6) Identifying sound sources from Russian/ROW nuclear submarines 7) Identifying miscellaneous sound sources 8) Identifying Russian/ROW and friendly sound sources	EDI

Activity Steps	
	1) All Basic Passive Acoustic Analysis Master Lesson Guidance is Classified and will be provided by TRINITY ADAC. 2) TRINITY ADAC will provide training content and instructional support to both NFS(A & P) upon request. 3) Trainees will be granted appropriate practical time for each lesson.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol. 1 2) CFCD 113 Vol. 2

Notes to Instructor
Nil

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EC 007.06 – Provide Generic Classification of Passive Contacts

Scope	Conditions	Sequence
EO 007.06	Classified Classroom (ADAC for NFS(A))	Conducted after EO 007.06.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS ADAC Staff

Assessment Steps	
	Input: Trainees are provided an Osprey Passive Acoustic simulator, an Acoustic Response sheet, and 5 x mixed Passive Acoustic contacts to analyze. **Performance:** Trainees will be required to provide generic classification of passive contacts. They will have 4 x 50 min periods to complete the EC, followed by 2 x 50 min periods for instructor debrief. **Output:** The trainee must achieve 60% to pass this EC. **Feedback:** Written and Verbal feedback to be provided IAW EC 007.06 Acoustic Response Answer Sheet.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and reassess

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UNCLASSIFIED

Contact Number: _____ **Contact Location:** _____ **Measurement Date / Time:** _____

Sensor: _____ **Sensor Depth (ft):** _____ **SLD (ft):** _____ **AN:** _____

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPRM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry... ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2% tolerance	

Class Ind/ID Swath: 1 1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Drive Mode	RR
D/E	

Shaft and Engine Related Sources

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed

Class Indicators: 1 1 1

Modulation/Banding: 1 1 1

Others: 1

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1

Transients, Significant Events, Tactical Info

Time	Details
	1 pt for each *significant* event listed

Aural Characteristics

Time	Details
	1 pt for each significant aural characteristic listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1
	1/2 harmonics must be listed	
Augmentors	1	
for the frequency and calling it an augmentor		
Auxiliaries		
1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter		
Others	1	
1pt awarded to other significant sources		
Unknowns	0	

Classification

4 points total

1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

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PC 007 – Perform Basic Passive Acoustic Analysis

Scope	Conditions	Sequence
PO 007	Classified Classroom (ADAC for NFS(A))	Conducted after EC 007.06
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-SONAROP-MS ADAC Staff

Assessment Steps	
	Input: Trainees are provided an Osprey Passive Acoustic simulator, an Acoustic Response sheet, and 4 x mixed Passive Acoustic contacts to analyze. **Performance:** Trainees will be required to perform basic passive acoustic analysis. They will have 4 x 50 min periods to complete the PC, followed by 2 x 50 min periods for instructor debrief. **Output:** The trainee must achieve 60% to pass this EC. **Feedback:** Written and Verbal feedback to be provided IAW PC 007 Acoustic Response Answer Sheet.

Resources	References
1) Passive acoustic simulator 2) Osprey analysis software 3) Headphones 4) Calculators 5) Stopwatch 6) ARS 7) Pencils and erasers 8) Mathematical formula diagrams handouts	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54.

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UNCLASSIFIED

Contact Number: | **Contact Location:** | **Measurement Date / Time:**

Sensor: | **Sensor Depth (ft):** | **SLD (ft):** | **AN:**

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPRM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry... Ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2%	
tolerance	

Class Indf/D Swath

1 1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Drive Mode	RR
D/E	

Shaft and Engine Related Sources

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed

Class Indicators

1 1 1

Modulation/Blanding

1 1 1

Others

1

Transients, Significant Events, Tactical Info

Time	Details
	1 pt for each *significant* event listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1
	1/2 harmonics must be listed	
Augmentors	1	
for the frequency and calling it an augmentor		
Auxiliaries		
1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter		
Others	1	
1pt awarded to other significant sources		
Unknowns	0	

Aural Characteristics

Time	Details
	1 pt for each significant aural characteristic listed

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1

Classification

4 points total

1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

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PO 008 – OPERATE PASSIVE ACOUSTIC EQUIPMENT

1. **Performance**: Operate Passive Acoustic Equipment (C0020).

2. **Conditions**:

   a. Given:

      (1) Passive acoustic equipment;
      (2) Logs and records;
      (3) Search orders;
      (4) Operational status report;
      (5) References; and
      (6) Supervision.

   b. Denied: Assistance

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall operate passive acoustic equipment (C0020), IAW refs, by:

   a. Setting up Sonobuoy Processing System (SPS) IAW Sonobuoy Reference Guide, NCPM 600 and Acoustic Data Recorder Operation and Maintenance Manual (REACH);

   b. Operating CANTASS, IAW NCPM 600, Operators Manual, CANTASS Book 1 & 2 by:
      (1) Performing pre-sail checks on equipment (B0005);
      (2) Setting up operating parameters IAW;
      (3) Operating Command System (CS) (B0050);
      (4) Maintain operational logs (B0055);
      (5) Record acoustic intelligence data (F0010);
      (6) Amplifying track information (E0020; and
      (7) Tracking acoustic contacts (C0070).

4. **Training Limitations**: Nil.

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EO 008.01 - Set up Sonobuoy Processing System (SPS)

008.01.01 – Setup Sonobuoy Processing System

Conditions		Sequence
HMC Ship or SEACOT		Conducted after PC 007
Time	**Classification**	**Instructor**
400 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Identify major Components 2) SPS Flash up 3) SPS Set UP	EDI

Activity Steps	
	1) Instructor will demonstrate and explain the major components of SPS to include; (30 min) a) CMS 330 Console b) Time-Code Generator; and c) Digital Recorder 2) Trainees will practice individually or in groups to identify the major components of SPS (40 min) 3) Instructor will explain and demonstrate SPS flash up by; (30 min) a) Selecting RF channels b) Bit Test c) Power up Recorder 4) Trainees will practice individually or in groups practicing SPS flash up to include; (100 min) a) Selecting RF channels b) Bit Test c) Power up Recorder 5) Instructor will explain and demonstrate SPS Set up to include; (100 min) a) login as SPS role b) Selecting SPS video c) Perform SPS set up d) Edit system parameters e) Edit processing setting up data page f) Implement parameters g) Edit display set up page 6) Trainees will practice individually or in pairs SPS set up. (100 min)

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Resources	References
1) SEACOT Trainer; 2) 1 x SPS console per trainee 3) 1 x Instructor console	1) C-59-007-006/MB-001 Sonobuoy Reference Guide 2) NCPM 600

Notes to Instructor
Nil

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EC 008.01 - Setup Sonobuoy Processing System

Scope	Conditions	Sequence
EO 008.01	HMC Ship or SEACOT	Conducted after EO 008.01.01
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided an SPS console and Flash up / Set Up requirements. **Performance:** Trainees will be required to Flash up and Set up SPS individually IAW EC 008.01 checklist. 15 min has been allocated per trainee for scheduling purposes. **Output:** The trainee must achieve 10/13 SATs to pass this EC. **Feedback:** Verbal and written feedback IAW with EC 008.01 checklist

Resources	References
1) SEACOT SPS console	1) NCPM 600

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and Retest

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Item	EC 008.01 SET UP SPS	SAT	UNSAT	Remarks
1.	**Identify major components**			
a.	CMS Console	☐	☐	
b.	Time-Code generator	☐	☐	
c.	Digital data recorder	☐	☐	
2.	**Perform Sonobuoy Processing System flash-up**			
a.	Select RF channels	☐	☐	
b.	Bit test	☐	☐	
c.	Power up recorder	☐	☐	
3.	**Select SPS through CMS**			
a.	Login as SPS role	☐	☐	
b.	Select SPS video	☐	☐	
c.	Perform Sonobuoy Processing System setup	☐	☐	
d.	Edit system parameters	☐	☐	
e.	Edit processing setting up data page	☐	☐	
f.	Implement parameters	☐	☐	
g.	Edit display set up page	☐	☐	

Score: **/13** The trainee must achieve 10/13 SATs to pass this EC.

Positive Feedback	Improvement Feedback

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EC 008.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 008.01 with the trainee.		**Trainee** I have been debriefed on the result of EC 008.01.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 008.02 – Ensure CANTASS is Operational

008.02.01 – CANTASS Flash-up/Function Check

Conditions		Sequence
CANTASS Mission Simulator or SEACOT		Conducted after EC 008.01
Time	**Classification**	**Instructor**
350 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Flashing up and shutting down CANTASS 2) Performing function checks on equipment	EDI

Activity Steps	
	1) Instructor will demonstrate how to properly flash-up and shut–down CANTASS while explaining each major components IAW NCPM 600. (50 min) 2) Trainees will then practice the flash-up procedures by themselves including naming each components used IAW NCPM 600. (100 min) 3) Instructor will demonstrate how to perform function checks on the system IAW NCPM 600 to include: (50 min) a) Check System test Pattern displayed b) Ensure SMHT selected in Tabular Area of DCA; and c) Check Function Controls of VFKs and QABs on each DCA. 4) Trainees will then practice on their own how to shut-down CANTASS IAW NCPM 600. (150 min)

Resources	References
1) CANTASS Mission Simulator or SEACOT 2) EC 008.02 Checklist	1) NCPM 600 2) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II

Notes to Instructor
Nil

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EC 008.02 - Ensure CANTASS is Operational

Scope	Conditions	Sequence
EO 008.02	CANTASS Mission Simulator Trainer or SEACOT	Conducted after EO 008.02.01
Time	**Classification**	**Instructor**
400 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a CANTASS console and Flash up requirements. **Performance:** Trainees will be required to ensure CANTASS is operational IAW EC 008.02 checklist. 30 min/trainee has been allocated for scheduling purposes **Output:** The trainee must achieve 20/24 SATs to pass this EC. **Feedback:** Verbal and written feedback IAW with EC 008.02 checklist

Resources	References
1) SEACOT	1) NCPM 600 2) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II

Notes to Instructor
Nil

Limitations	If SEACOT not available can conduct onboard HMC Ship
Impact of Failure	Retrain and retest

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Item	EC 008.02 ENSURE CANTASS IS OPERATIONAL	SAT	UNSAT	Remarks
1. Flash Up CANTASS to include;				
a) Ensure Mode Switch is set to "NORMAL"		☐	☐	
b) Ensure Battle Short is "OFF"		☐	☐	
c) Set MTM Power "ON/OFF" switch to "ON"		☐	☐	
d) Set Power "ON/OFF" switch to "ON"		☐	☐	
e) Set Internal and External Volumes to "MINIMUM"		☐	☐	
f) Select CPA Power "ON" push button		☐	☐	
g) Check Internal and External Power Lights are "ON"		☐	☐	
h) Check 115V Circuit Breaker is "ON"		☐	☐	
i) Check 115V AC Power Available light is "ON"		☐	☐	
j) Ensure Mode Switch is set to "MASTER"		☐	☐	
k) Ensure System Power Switch is set to "OFF"		☐	☐	
l) Set System Power Switch to "ON RELOAD"		☐	☐	
m) Set VGR Power to "ON		☐	☐	
2. Function Checks to include;				
a) Check System test Pattern displayed		☐	☐	
b) Ensure SMHT selected in Tabular Area of DCA		☐	☐	
c) Check Function Controls of VFKs and QABs on each DCA		☐	☐	
d) Verify display formats on upper/lower MTM's				
e) Verify joystick, trackball and keyboard operation				
3. Shut down CANTASS to include;				
a) Set VGR Power to "OFF"		☐	☐	

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b) Set System Power switch to “OFF” on CPR	☐	☐	
c) Set Power push button to “OFF” on CPA	☐	☐	
d) Set Power switch to “OFF” on MTMs	☐	☐	
e) Set Power push button to “OFF” on CPA	☐	☐	
f) Set Power ON/OFF switch to “OFF” on CUD	☐	☐	

Score: /24

The trainee must achieve 20/24 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 008.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 008.02 with the trainee.	**Trainee** I have been debriefed on the result of EC 008.02.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 008.03 - Set up CANTASS Operating Parameters

008.03.01 – CANTASS Operating Parameters and Search Procedures

Conditions		Sequence
CANTASS Mission Simulator Trainer or SEACOT		Conducted after EC 008.02
Time	**Classification**	**Instructor**
1350 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Setting up CANTASS parameters 2) Conducting search procedure IAW refs	EDI

Activity Steps	
	1) Instructor will demonstrate setting up CANTASS parameters in accordance with TASSUP directions to include: (50 min) a) Set up Vernier's b) Set up Narrowband Integration c) Set up demon Integration d) Update Salinity e) Update Magnetic Declination f) Update Array Heading 2) Trainees will practice individually or in pairs setting up CANTASS parameters, while the instructor is directing the parameters to include; (300 min) a) Set up Vernier's b) Set up Narrowband Integration c) Set up demon Integration d) Update Salinity e) Update Magnetic Declination f) Update Array Heading 3) Instructor will demonstrate CANTASS set up IAW with 008.03 checklist. (50 min) 4) Trainees will practice CANTASS set up individually or in pairs IAW 008.03 checklist. (300 min) 5) Instructor will demonstrate Search procedures IAW 008.03 checklist to include; (50 min) a) Selecting beam scroll number b) Select TBB format on lower TVM c) Selecting SS format on upper TVM d) Perform correct beam scrolling e) Adjust CANTASS parameters as directed to include; f) Integration Period

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	g) Vernier's h) Windows Spectrum (A/B) i) Implement correct bearing investigations as directed. 6) Trainees will practice Search procedures individually or in pairs IAW with 008.03 checklist. (300 min) 7) Trainees will practice CANTASS set up and search procedures together. (300 min)

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) EC 008.03 checklist	1) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II 2) NCPM 600

Notes to Instructor
Nil

EC 008.03 - Set up CANTASS Operating Parameters

Scope	Conditions	Sequence
EO 008.03	CANTASS Mission Simulator Trainer or SEACOT	Conducted after EO 008.03.01
Time	**Classification**	**Instructor**
600 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a CANTASS console and Set up requirements to input operating parameters for Search procedures. **Performance:** Trainees will be required to set up CANTASS operating parameters and search procedures IAW EC 008.03 checklist. 50 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 16/19 SATs to pass this EC. **Feedback:** Written and verbal feedback IAW with EC 008.03 checklist.

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) Closing up voice procedure handout	1) NCPM 600 2) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Retrain and Retest

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Item	EC 008.03 SET UP PARAMETERS AND OPERATE CANTASS	SAT	UNSAT	Remarks
1. Input system setup parameter to include;				
a) Set up in accordance with TASSUP to include; i. Set up Vernier's ii. Set up Narrowband Integration iii. Set up Demon Integration iv. Update Salinity v. Update magnetic Declination vi. Update array heading		☐	☐	
b) Select array receiver mode		☐	☐	
c) Select array gain		☐	☐	
d) Update operators array gain		☐	☐	
e) Select auto zero mode		☐	☐	
f) Specify array channel control		☐	☐	
g) Ensure CAL signals OFF		☐	☐	
h) Select system array heading source		☐	☐	
i) Select system time source		☐	☐	
j) Update operators time		☐	☐	
k) Select HDDR mode		☐	☐	
l) Sanitize array		☐	☐	
m) Verbally state closing up report		☐	☐	
2. Search Procedures				
a) Select beam scroll		☐	☐	
b) Select TBB format lower TVM		☐	☐	
c) Select SS format upper TVM		☐	☐	
d) Perform beam scroll		☐	☐	
e) Adjust CANTASS parameters as directed to include; i. Integration Period ii. Vernier's iii. Windows spectrum format (A/B)		☐	☐	

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f) Implement correct bearing investigations as directed	☐	☐	

Score: /19

The trainee must achieve 16/19 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 008.03 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 008.03 with the trainee.	**Trainee** I have been debriefed on the result of EC 008.03.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 008.04 – Maintain Passive Acoustic Equipment Log

008.04.01 – CANTASS Log

Conditions		Sequence
CANTASS Mission Simulator Trainer or SEACOT		Conducted after EC 008.03
Time	**Classification**	**Instructor**
50 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Maintain passive acoustic equipment log	EDI

Activity Steps	
	1) Instructor will demonstrate maintaining CANTASS log IAW EC 008.05 check list. (10 min) 2) Instructor will provide trainees a scenario for trainees to practice (40 min)

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) CANTASS Log handout	1) NCPM 600

Notes to Instructor
1) CANTASS Log handouts to be distributed in class 2) Will be tested in conjunction with EC 008.05.

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EO 008.05 – Tracking Contacts and Amplify Track Information

008.05.01 – Track & Amplify Acoustic Contacts

Conditions		Sequence
CANTASS Mission Simulator Trainer or SEACOT		Conducted after EC 008.03
Time	**Classification**	**Instructor**
650 min	SECRET	RQ-SONAROP-MS

Learning Objectives	Method
1) Create Contact 2) Resolving ambiguity 3) Carrying out tracking procedures 4) Correlating NB/BB data 5) Amplifying/updating TD/CCD data 6) Carrying out lost/regain contact procedures 7) Performing torpedo detection procedures	EDI

Activity Steps	
	1) Instructor will demonstrate how to track and update acoustic contact to include; (50 min) a) Create tracks; b) Resolving for ambiguity; c) Carrying out tracking procedures; d) Correlating NB/BB data; e) Amplifying/updating TD/CCD data; and f) Carrying out lost/regain contact procedures. 2) Trainee will then practice all previously learned techniques by themselves or in group; (450 min). 3) Instructor will demonstrate how to perform torpedo detection procedures; (50 min) 4) Trainees will practice by themselves or in group torpedo detection procedures; (100 min)

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) CANTASS operator voice reports handouts	1) NCPM 600

Notes to Instructor
Nil

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EC 008.05 – Tracking Contacts and Amplify Track Information

Scope	Conditions	Sequence
EO 008.04 EO 008.05	CANTASS Mission Simulator Trainer or SEACOT	Conducted after EO 008.05.02
Time	**Classification**	**Instructor**
600 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a CANTASS console, CANTASS Logs, a minimum of 1 x underwater contact to track and provide amplifying track information, and 1 x torpedo contact to conduct torpedo detection procedures. **Performance:** Trainees will be required to track an acoustics contact, provide amplifying track information, conduct torpedo detection procedures and maintain a CANTASS Log IAW EC 008.05 checklist. 50 min/trainee has been allocated for scheduling purposes. **Output:** The trainee must achieve 20/25 SATs to pass this EC. **Feedback:** Written and verbal feedback IAW with EC 008.05 checklist.

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) CANTASS Logs	1) NCPM 600 2) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II.

Notes to Instructor
Nil

Limitations	Trainer availability
Impact of Failure	Retrain and Retest

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Item	EC 008.05 TRACK CONTACTS AND AMPLIFY TRACK INFORMATION	SAT	UNSAT	Remarks
1. Track acoustic contact by conducting the following:				
a) Identify and create contact		☐	☐	
b) Resolve ambiguity		☐	☐	
c) Carry out tracking procedures		☐	☐	
d) Correlate NB/BB data		☐	☐	
e) Amplify/update TD/CCD data		☐	☐	
f) Carry out lost/ regain contact procedures		☐	☐	
2. Provide CCS track information to include;				
a) Frequency		☐	☐	
b) Bearing		☐	☐	
c) Time		☐	☐	
d) Information		☐	☐	
e) Input/update		☐	☐	
3. Predict target movement, to include;				
a) Broadband/search summary		☐	☐	
b) TBB component intensity		☐	☐	
c) Tracker assist		☐	☐	
4. Lost contact procedures to include;				
a) Verbally states Lost contact report		☐	☐	
b) Implement Lost contact procedures by; i. Scrolling fwd/aft; and ii. Implementing lost contact		☐	☐	
5. Torpedo Detection to include;				
a) **Identify Torpedo ***		☐	☐	***Critical**
b) **Torpedo voice report**		☐	☐	***Critical**
6. Maintain passive acoustic equipment logs to include;				
a. Recording in log the time (Zulu)		☐	☐	
b. Recording in log tape number		☐	☐	

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Item	EC 008.05 TRACK CONTACTS AND AMPLIFY TRACK INFORMATION	SAT	UNSAT	Remarks
7. Recording contact data, to include;				
a. Beam		☐	☐	
b. Bearing (true/amb)		☐	☐	
c. Sources holding		☐	☐	
d. Frequency		☐	☐	
e. Recording in log any extra information i.e. changes in set up, contact information		☐	☐	

Score: /25

The trainee must achieve 20/25 SATs and 2/2 critical to pass this EC.

Positive Feedback	Improvement Feedback

EC 008.05 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 008.05 with the trainee.	**Trainee** I have been debriefed on the result of EC 008.05.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 008 – Operate Passive Acoustic Equipment

Scope	Conditions	Sequence
PO 008	CANTASS Mission Simulator Trainer or SEACOT	Conducted after EC 008.05
Time	**Classification**	**Instructor**
600 min	SECRET	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a CANTASS console to flash and set up, a minimum of 1 x underwater contact to search, track and provide amplifying track information, lost contact, 1 x torpedo contact to conduct torpedo detection procedures, and CANTASS Logs to maintain. **Performance:** Trainees will be required to operate passive acoustic equipment IAW PC 008 checklist. Logs will be updated by trainee during assessment. 50 min/trainee has been allocated for scheduling purposes **Output:** The trainee must achieve 38/47 SATs to pass this EC. **Feedback:** Written and verbal feedback IAW with PC 008 checklist.

Resources	References
1) CANTASS Mission Simulator Trainer or SEACOT 2) CANTASS Logs	1) NCPM 600 2) C-69-748-000/MB-001, Operators Manual, CANTASS Book I and II.

Notes to Instructor
1) Logs will be updated by trainee during assessment

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54.

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Item	PC 008 OPERATE PASSIVE ACOUSTIC EQUIPMENT	SAT	UNSAT	Remarks
1. CLOSING –UP PROCEDURES to include;				
a)	Set up in accordance with TASSUP to include;	☐	☐	
	i. Set up Vernier's ii. Set up Narrowband Integration iii. Set up Demon Integration iv. Update Salinity v. Update magnetic Declination vi. Update array heading	☐	☐	
b)	Select array receiver mode	☐	☐	
c)	Select array gain source	☐	☐	
d)	Update operator's array gain	☐	☐	
e)	Select auto zero mode	☐	☐	
f)	Specify array channel control	☐	☐	
g)	Ensure CAL signals OFF	☐	☐	
h)	Select system array heading source	☐	☐	
i)	Select system time source	☐	☐	
j)	Update operator's time	☐	☐	
k)	Select HDDR mode	☐	☐	
l)	Sanitize Array			
m)	Verbally state closing up report	☐	☐	
2. SEARCH PROCEDURES				
a)	Select beam scroll number	☐	☐	
b)	Select TBB format lower monitor	☐	☐	
c)	Select SS format for upper monitor	☐	☐	
d)	Perform beam scroll	☐	☐	
e)	Adjust CANTASS parameters as directed to include; i. Integration Period ii. Vernier's iii. Windows spectrum format (A/B)	☐	☐	

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f) Implement correct bearing investigations as directed	☐	☐	
3. TRACKING AND AMPLIFYING PROCEDURES			
a) Operator identifies contact	☐	☐	
b) Create target	☐	☐	
c) Create OIT	☐	☐	
d) Verbally states Initial Contact Report	☐	☐	
e) Create Contact	☐	☐	
f) Setup display to resolve ambiguity	☐	☐	
g) Resolve ambiguity	☐	☐	
h) Setup display for Analysis/tracking	☐	☐	
i) Correlate narrowband/ broadband info	☐	☐	
j) Run Fine Bearing Estimates	☐	☐	
k) Predict beam-to-beam tracking	☐	☐	
l) Associate all target components to target detail	☐	☐	
m) Complete target detail format	☐	☐	
n) Verbally state Amplifying Report	☐	☐	
o) Update OIT			
4. LOST CONTACT PROCEDURES			
a) Verbally states Lost Contact report	☐	☐	
b) Implement Lost contact procedures by; i. Scrolling fwd/aft ii. Implementing lost contact symbology	☐	☐	
5. TORPEDO DETECTION			
a) Identify Torpedo	☐	☐	***Critical**
b) Torpedo Voice report	☐	☐	***Critical**
6. RECORDING IN LOG, to include			
a) Recording in log the time (Zulu)	☐	☐	
b) Recording in log tape number	☐	☐	
c) Beam	☐	☐	
d) Bearing (true/amb)	☐	☐	
e) Sources holding	☐	☐	
f) Frequency of contact	☐	☐	

g) Recording in log any extra information i.e. changes in set up, contact information	☐	☐	

Score: /47

The trainee must achieve 38/47 SATs and 2/2 critical to pass this PC.

Positive Feedback	Improvement Feedback

PC 008 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 008 with the trainee.	**Trainee** I have been debriefed on the result of PC 008.

PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PO 009 – OPERATE SURFACE VESSEL TORPEDO TUBES (SVTT) WEAPON SYSTEMS

1. **Performance**: Operate SVTT Weapon Systems.

2. **Conditions**:

   a. Given:

      (1) SVTT weapon system;
      (2) HOT TORP or torpedo;
      (3) Stowage Chocks;
      (4) Low Pressure air system;
      (5) Loading tray;
      (6) Safety equipment;
      (7) Internal communications equipment;
      (8) References;
      (9) Supervision; and
      (10) Assistance.

   b. Denied: Nil

   c. Environment:

      (1) Any sea state;
      (2) Any noise level;
      (3) Any depth;
      (4) Any weather;
      (5) Day or night; and
      (6) Any operational state.

3. **Standard**: The Jr. SONAR Operator shall operate SVTT Weapon Systems, IAW NCPM 1001, NAVORD 4995-0 and NAVORD 4995-2, by:

   a. Conducting Torpedo Safety, States of Preparedness, Degrees of Readiness and Voice Procedures;
   b. Assisting in loading/unloading/handling torpedoes (C0170) by:
      (1) Charging torpedo tubes flask (C0185); and
      (2) Performing UWW weapon systems serviceability checks (air-shots) (D0010).
   c. Conducting weapon systems firing drills and procedures, by operating Torpedo Setting Panel (TSP) (C0175);

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d. Performing UWW weapon firings (D0025);
e. Perform weapon system misfire procedures; and
f. Fire torpedoes in emergency mode (C0180).

4. **Training Limitations**: Nil

EO 009.01 – Torpedo Safety and Preparedness

009.01.01 –Torpedo Safety, States of Preparedness, Degrees of Readiness and Voice Procedures

Conditions		Sequence
Unclassified Classroom with DWAN with projector		
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Torpedo Firing Safety 2) Identifying states of preparedness/degrees of readiness 3) Torpedo Voice Reports	Lecture Reading assignment EDI Video

Activity Steps	
	1) Instructor will give a PowerPoint presentation on torpedo safety. (50 min) 2) Instructor will hand out a copy of NCPM 1001 Ch. 2 to the trainees. Trainees will conduct a reading assignment. (15 min) 3) Instructor will select 1 trainee to explain the following states of preparedness: (15 min) a) Standing to b) Relaxed c) Stood Down d) Secured 4) Instructor will select 1 trainee to explain the term “Mechanically Safe” as it applies to the SVTT (5 min) 5) Instructor will select 1 trainee to explain the term “Electrically safe” as it applies to the SVTT (5 min) 6) Instructor will ask trainees to explain the crewing requirements for the following degrees of readiness: (10 min) a) First degree b) Second degree c) Third/Fourth degree 7) Instructor will hand out NCPM 1001 Ch.5 – Torpedo firing voice reports. Instructor will then verbally demonstrate each of the torpedo firing mode voice reports to include: (10 min) a) Closing up b) Primary c) Secondary d) Emergency 8) Instructor will now pair up the trainees and have them practice their voice reports. (40 min)

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	9) Instructor will now conduct random confirmation by choosing a trainee and having them conduct a report of the instructor's choice. Each trainee is to be chosen once. (30 min) 10) Instructor will now present the Sea Training Group torpedo loading video (20 min)

Resources	References
1) DWAN With Projector and speakers 2) NAVORD 4995-0 3) Sea Training Group torpedo loading video 4) NCPM 1001 Ch.2 Handout 5) NCPM 1001 Ch.5 Handout	1) NCPM 1001; 2) NAVORD 4995-0, Safety Firings Orders for Ships

Notes to Instructor

Torpedo tubes states of preparedness/degrees of readiness and firing modes procedures and voice reports checklist handouts to be distributed in class
http://halifax.mil.ca/SEA_TRG/videos/video.html?vid=../videos/Torpedo_Loading.mp4

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EO 009.02 - Assist in Loading/Unloading Torpedoes

009.02.01 – Assist in Loading/Unloading Torpedoes

Conditions		Sequence
HMC Ship or Torpedo Tubes Trainer		
Time	**Classification**	**Instructor**
400 min	UNCLASSIFIED	RQ-SONAROP-MS RQ-WENG-PO2 (Supervisor) RQ-WENG-S1

Learning Objectives	Method
1) Using components and functions of SVTT weapon system; 2) Conducting torpedo handling procedures; and 3) Conducting torpedo loading/unloading procedures.	EDI

Activity Steps	
	1) WENG PO2 Supervisor will conduct a safety brief on the SVTT. (5 min) 2) Instructor will conduct an explanation and demonstration of the components of the Surface Vessel Torpedo Tube Weapon system to include: (40 min) a) Stowage Chocks b) Charging Torpedo Flasks c) Operating Mk 32 Torpedo tubes d) Operating Torpedo Handling System e) HOTTORP 3) Instructor will explain the differences between the torpedo types to include: (5 min) a) HOTTORP b) EXTORP c) Warshot 4) Instructor and WENG Staff will explain and demonstrate and allow for trainees to practice how to safely conduct torpedo handling procedures to include: (100 min) a) Operating/Safety/environmental procedures b) Operation of chocks c) Operation of the crane d) Operation of the loading tray 5) Instructor and WENG Staff will explain and demonstrate how to conduct torpedo loading/unloading procedures to include: (100 mins) a) Operating/Safety/Environmental procedures b) Loading torpedo from the loading tray c) Connecting torpedo tube connections d) Charging/Installing flasks e) Unloading

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	6) Instructor and WENG Staff will have trainees practice conducting a torpedo load/unload. (150 mins)

Resources	References
1) HMC ship with SVTT weapon system; 2) SVTT weapon system Virtual Trainer (When available); 3) HOTTORP	1) NCPM 1001 2) Launch Accessories, Torpedo MK 46 Mod 5

Notes to Instructor

1) Rotate Trainee though to allow them Maximum participation
2) No EC 009.02, will be conducted as part of RQ S2 OJPR PC 009.

EO 009.03 - Operate Torpedo Setting Panel (TSP)

009.03.01 – Torpedo Setting Panel

Conditions		Sequence
HMC Ship / SEACOT / S82 Pullen Building (NFS(A) Only)		Conducted after EO 009.01.01 and EO 009.02.01
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Identifying components of the TSP 2) Operating TSP	EDI

Activity Steps	
	1) Instructor will explain and demonstrate the different components of the Mk 331 Torpedo Setting Panel IAW NCPM 1001. (15 mins) 2) Instructor will now explain and demonstrate how to conduct torpedo firing procedure IAW EC 009.02 Checklist. (35 mins) 3) Instructor will have each trainee practice conducting torpedo firing procedure IAW EC 009.02 Checklist. Instructor will ensure a minimum 2 runs per trainee, 5 min per run. (150 min)

Resources	References
1) HMC ship with Torpedo Settings Panel or 2) Torpedo Settings Panel Simulator program 3) SEACOT (If Ship unavailable)	1) NCPM 1001

Notes to Instructor
Nil

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EC 009.03 - Operate Torpedo Setting Panel (TSP)

Scope	Conditions	Sequence
EO 009.02	HMC Ship / SEACOT / S82 Pullen Building (NFS(A) Only)	Conducted after EO 009.03.01
Time	**Classification**	**Instructor**
200 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a HMC Ship with Torpedo Settings Panel or TSP emulation and torpedo settings to demonstrate operating the TSP. **Performance:** Trainee will operate TSP IAW EC checklist. 15 min per trainee has been allocated for scheduling **Output:** The trainee must achieve 17/21 SATs to pass this EC. **Feedback:** Written and Verbal feedback to be provided IAW EC 009.03 checklist.

Resources	References
1) HMC ship with Torpedo Settings Panel or 2) Torpedo Settings Panel Simulator program 3) SEACOT (If Ship unavailable)	1) NCPM 1001

Notes to Instructor
Nil

Limitations	If no ship available, must simulate TSP.
Impact of Failure	Retrain and reassess

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Item	EC 009.03 OPERATE TORPEDO SETTING PANEL	SAT	UNSAT	Remarks
1.	**Power up MK 331 TSP**			
a)	Turn on AC Power	☐	☐	
b)	Confirm 28V DC indicator is Green	☐	☐	
c)	Confirm 115V AC indicator is Green	☐	☐	
d)	Confirm Torpedo Setting Panel is ON	☐	☐	
2.	**Conduct System Setup**			
a)	Switch TSP to LOCAL	☐	☐	
b)	Select SETUP Soft key	☐	☐	
c)	Select TRAINING Soft key	☐	☐	
d)	Select Torpedo Inventory Soft key	☐	☐	
e)	Setup all 4 Tubes with MK46 5A(SW) torpedoes	☐	☐	
3.	**Set Up Torpedo Firing Solution**			
a)	Select TRAINING Soft key	☐	☐	
b)	Select Torpedo Tube	☐	☐	
c)	Select RUNOUT Mode	☐	☐	
d)	Select Initial Search Depth (ISD)	☐	☐	
e)	Select Tactical Floor	☐	☐	
f)	Select Gyro Angle	☐	☐	
4.	**Fire Torpedo from Local**			
a)	Select STANDBY	☐	☐	
b)	Confirm LAUNCH button is lit	☐	☐	
c)	Select LAUNCH Button	☐	☐	
d)	Confirm Status of Weapon	☐	☐	
5.	**Shut Down MK 331 TSP**			
a)	Switch TSP to REMOTE	☐	☐	
b)	Turn off AC Power	☐	☐	

Score: /21

The trainee must achieve 17/21 SATs to pass this EC.

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Positive Feedback	Improvement Feedback

EC 009.03 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 009.03 with the trainee.	**Trainee** I have been debriefed on the result of EC 009.03.

PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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EO 009.04 - Perform UWW Weapon Firings

009.04.01 – UWW Weapons Firings

Conditions		Sequence
HMC Ship / SEACOT / S82 Pullen Building (NFS(A) Only)		Conducted after EC 009.03
Time	**Classification**	**Instructor**
550 min	UNCLASSIFIED	RQ-SONAROP-MS

Learning Objectives	Method
1) Conducting firing modes and procedures 2) Performing weapon system misfire procedures 3) Conducting torpedo firing in emergency mode	EDI

Activity Steps	
	1) Instructor will explain and demonstrate weapon firing procedures IAW EC/PC checklist to include: (50 min) a) Primary mode b) Secondary mode 2) Instructor will explain and demonstrate procedure for when weapon misfire occurs and the emergency firing procedure. (20 min) 3) Instructor will have the trainees conduct all the firing procedures and weapon misfire drill IAW EC/PC Checklist. Each trainee will be given a minimum of 2 runs, each run will consist of primary, secondary, and emergency firing. (20 minutes/trainee for each run). (480 mins).

Resources	References
1) HMC ship with Torpedo Settings Panel or 2) Torpedo Settings Panel Simulator program 3) SEACOT (If Ship unavailable) 4) NCPM 1001 Ch.2 Handout 5) NCPM 1001 Ch.5 Handout	1) NCPM 1001

Notes to Instructor
Nil

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EC 009.04 - Perform UWW Weapon Firings

Scope	Conditions	Sequence
EO 009.04	HMC Ship / SEACOT / S82 Pullen Building (NFS(A) Only)	Conducted after EO 009.04.01
Time	**Classification**	**Instructor**
300 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a HMC Ship with Torpedo Settings Panel or TSP emulation, torpedo voice reports and torpedo settings to perform UWW Weapons firing. **Performance:** Trainees will be required to perform UWW weapon firings in Primary, Secondary and Emergency modes IAW EC checklist. 25 min per trainee has been allocated for scheduling **Output:** The trainee must achieve 37/41 SATs and all critical items to pass this EC. **Feedback:** Written and Verbal feedback to be provided IAW EC 009.04 checklist

Resources	References
1) HMC ship with Torpedo Settings Panel or 2) Torpedo Settings Panel Simulator program 3) SEACOT (If Ship unavailable) 4) NCPM 1001 Ch.2 Handout 5) NCPM 1001 Ch.5 Handout	1) NCPM 1001

Notes to Instructor
Nil

Limitations	If ship is unavailable, instructor will simulate.
Impact of Failure	Retrain and reassess

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Item	EC 009.04 PERFORM UWW WEAPON FIRING	SAT	UNSAT	Remarks
1. PERSONNEL & EQUIPMENT SAFETY MAINTAINED THROUGHOUT:				
a) Request possession of SVTT padlock keys		☐	☐	
b) Ensure power is available.		☐	☐	
c) Confirm safety equipment is in place to include: Out of bounds signs.		☐	☐	
d) Conduct Communication checks		☐	☐	
2. ASSUMING "STANDING TO STATE" FROM "SECURED" STATE				
a) Check all breech ward access opening covers are secured		☐	☐	
b) Ensure flex hoses are connected		☐	☐	
c) Check breech mechanism air pressures are 1600 psi (+ 0 / - 50 psi)		☐	☐	
d) Check manifold air valve is open (lower barrel) and ship air supply is connected to the tube pneumatic system (manifold gauge indicates between 100 - 150 psi).		☐	☐	
e) Ensure muzzle door heater switch is turned "ON" when ambient temperature drops below 4.4 degrees Celsius		☐	☐	
f) Operate manual muzzle door open switches on all tubes. Confirm doors are open and valve operating mechanism move to breech ward end.		☐	☐	
g) Close muzzle doors —ensure control valve operating mechanism moves to muzzle ward position		☐	☐	
h) Unlock padlocks on emergency fire levers and leave padlocks in place		☐	☐	
i) Place tube ready actuator for all tubes to the ON (remote) position		☐	☐	

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j) Ensure AC and DC circuit breakers are ON	☐	☐	
k) Switch TSP to REMOTE	☐	☐	
l) Conduct Closing up report to UWWD IAW NCPM 1001	☐	☐	
3. Conduct Weapon Firing in Primary Mode			
a) Confirms weapon ready to fire from TSP	☐	☐	
b) Confirms weapon status on TSP	☐	☐	
c) Conducts Primary Firing Drill voice reports IAW NCPM 1001	☐	☐	
4. Conduct Weapon Firing in Secondary Mode			
Trainee must complete the following steps when directed:			
a) Switch TSP to LOCAL	☐	☐	
b) Selects Tube #	☐	☐	
c) Sets mode (RUNOUT / NO RUNOUT)	☐	☐	
d) Sets Initial Search Depth	☐	☐	
e) Sets Tactical Floor	☐	☐	
f) Sets Gyro Angle	☐	☐	
Trainee will be directed to make tubes "READY":			
g) Place actuator switch to Open	☐	☐	
h) Selects STANDBY	☐	☐	
i) Confirms weapon ready to fire from TSP	☐	☐	
Trainee will be directed to "FIRE"			
j) Selects LAUNCH button	☐	☐	
k) Confirms weapon status on TSP	☐	☐	
l) Conducts Secondary Firing Drill voice reports IAW NCPM 1001	☐	☐	
5. TORPEDO MISFIRE PROCEDURES "FIRE IN EMERGENCY"			
Misfire is reported in secondary mode, directed to "FIRE IN EMERGENCY"			
a) Removes Padlock from Emergency Fire Lever	☐	☐	
b) Pulls Firing Lever	☐	☐	
c) Confirms weapon status from tube	☐	☐	
d) Conducts Emergency Firing Drill voice reports IAW NCPM 1001	☐	☐	

6. SECURE TORPEDO TUBES			
When Ordered, make a loaded torpedo tube mechanically safe:			
a) Lock padlocks on the EMERGENCY FIRING LEVERS	☐	☐	
b) Ensure control valve operating lever is in the muzzle ward position and muzzle door indicator light is out	☐	☐	
c) Confirm Flex hose is disconnected	☐	☐	
d) Switch TSP to LOCAL, and turn the AC Power switch to OFF	☐	☐	
e) Conducts "SECURED" state voice report IAW NCPM 1001	☐	☐	
f) Return keys on completion	☐	☐	

Score: **/41** The trainee must achieve 37/41 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 009 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 009 with the trainee.	**Trainee** I have been debriefed on the result of EC 009.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PC 009 – Operate Surface Vessel Torpedo Tubes (SVTT) Weapon Systems

Scope	Conditions	Sequence
PO 009	HMC Ship / SEACOT / S82 Pullen Building (NFS(A) Only)	Conducted after EC 009.04
Time	**Classification**	**Instructor**
300 min	UNCLASSIFIED	RQ-SONAROP-MS

Assessment Steps	
	Input: Trainees will be provided a HMC Ship with Torpedo Settings Panel or TSP emulation, torpedo voice reports and torpedo settings to perform UWW Weapons firing. **Performance:** Trainees will be required to operate SVTT weapon systems and perform UWW weapon firings in Primary, Secondary and Emergency modes IAW EC checklist. Each trainee has been allocated 25 min for scheduling purposes **Output:** Trainees must achieve 33/41 SATs to pass this PC **Feedback:** Verbal and written feedback IAW PC 009 checklist

Resources	References
1) HMC ship with Torpedo Settings Panel or 2) Torpedo Settings Panel Simulator program 3) SEACOT (If Ship unavailable)	1) NCPM 1001

Notes to Instructor
Nil

Limitations	If HMC Ship is unavailable, instructor will simulate
Impact of Failure	IAW NPTORD 4500-54.

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Item	PC 009 OPERATE SVTT WEAPON SYSTEMS	SAT	UNSAT	Remarks
1. PERSONNEL & EQUIPMENT SAFETY MAINTAINED THROUGHOUT:				
a) Request possession of SVTT padlock keys		☐	☐	
b) Ensure power is available.		☐	☐	
c) Confirm safety equipment is in place to include: Out of bounds signs.		☐	☐	
d) Conduct Communication checks		☐	☐	
2. ASSUMING "STANDING TO STATE" FROM "SECURED" STATE				
a) Check all breech ward access opening covers are secured		☐	☐	
b) Ensure flex hoses are connected		☐	☐	
c) Check breech mechanism air pressures are 1600 psi (+ 0 / - 50 psi)		☐	☐	
d) Check manifold air valve is open (lower barrel) and ship air supply is connected to the tube pneumatic system (manifold gauge indicates between 100 - 150 psi).		☐	☐	
e) Ensure muzzle door heater switch is turned "ON" when ambient temperature drops below 4.4 degrees Celsius		☐	☐	
f) Operate manual muzzle door open switches on all tubes. Confirm doors are open and valve operating mechanism move to breech ward end.		☐	☐	
g) Close muzzle doors —ensure control valve operating mechanism moves to muzzle ward position		☐	☐	
h) Unlock padlocks on emergency fire levers and leave padlocks in place		☐	☐	
i) Place tube ready actuator for all tubes to the ON (remote) position		☐	☐	

j) Ensure AC and DC circuit breakers are ON	☐	☐	
k) Switch TSP to REMOTE	☐	☐	
l) Conduct Closing up report to UWWD IAW NCPM 1001	☐	☐	
3. Conduct Weapon Firing in Primary Mode			
a) Confirms weapon ready to fire from TSP	☐	☐	
b) Confirms weapon status on TSP	☐	☐	
c) Conducts Primary Firing Drill voice reports IAW NCPM 1001	☐	☐	
4. Conduct Weapon Firing in Secondary Mode			
Trainee must complete the following steps when directed:			
a) Switch TSP to LOCAL	☐	☐	
b) Selects Tube #	☐	☐	
c) Sets mode (RUNOUT / NO RUNOUT)	☐	☐	
d) Sets Initial Search Depth	☐	☐	
e) Sets Tactical Floor	☐	☐	
f) Sets Gyro Angle	☐	☐	
Trainee will be directed to make tubes "READY":			
g) Place actuator switch to Open	☐	☐	
h) Selects STANDBY	☐	☐	
i) Confirms weapon ready to fire from TSP	☐	☐	
Trainee will be directed to "FIRE"			
j) Selects LAUNCH button	☐	☐	
k) Confirms weapon status on TSP	☐	☐	
l) Conducts Secondary Firing Drill voice reports IAW NCPM 1001	☐	☐	
5. TORPEDO MISFIRE PROCEDURES "FIRE IN EMERGENCY"			
Misfire is reported in secondary mode, directed to "FIRE IN EMERGENCY"			
a) Removes Padlock from Emergency Fire Lever	☐	☐	
b) Pulls Firing Lever	☐	☐	
c) Confirms weapon status from tube	☐	☐	
d) Conducts Emergency Firing Drill voice reports IAW NCPM 1001	☐	☐	

6. SECURE TORPEDO TUBES			
When Ordered, make a loaded torpedo tube mechanically safe:			
a) Lock padlocks on the EMERGENCY FIRING LEVERS	☐	☐	
b) Ensure control valve operating lever is in the muzzle ward position and muzzle door indicator light is out	☐	☐	
c) Confirm Flex hose is disconnected	☐	☐	
d) Switch TSP to LOCAL, and turn the AC Power switch to OFF	☐	☐	
e) Conducts "SECURED" state voice report IAW NCPM 1001	☐	☐	
f) Return keys on completion	☐	☐	

Score: **/41** The trainee must achieve 33/41 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 009 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 009 with the trainee.		**Trainee** I have been debriefed on the result of PC 009.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

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CHAPTER 3: ASSESSMENT OF TRAINEE

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GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs) are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. EC results do not factor into a trainee's successful completion of the PO.

3. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 – Conduct of Instructional Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimun to prepare for a re-test.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 – Training Review Board Process.

COURSE REPORTS

10. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

11. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

12. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

13. A record of trainee counselling shall be maintained by the TE.

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ANNEX A - TASK LIST

Number	Task Statement	T-N/T	PO
	Duty Area A - COMMUNICATIONS		
A0005	Perform communications circuit functional checks (comm checks)	N/T	
A0010	Operate communications equipment	N/T	
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	T	003
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	N/T	
	Duty Area B - OPERATIONS		
B0005	Perform pre-sail checks on equipment	T	003, 006, 008
B0015	Maintain stateboards	N/T	
B0025	Adhere to Emission control (EMCON) restrictions	T	006
B0030	Contribute to intelligence gathering Ops	T	005
B0035	Display intelligence via UWW brief	T	005
B0045	Identify Capacities and limitations of Platforms	T	005
B0050	Operate CS	T	008
B0055	Maintain operational logs	T	002, 003, 004, 006, 008
	Duty Area C - ACOUSTIC OPERATIONS		
C0012	Setup Oceanographic Equipment	T	003
C0015	Operate active SONAR equipment	T	006
C0020	Operate passive acoustic equipment	T	008
C0025	Operate ancillary acoustic equipment	T	003
C0030	Operate oceanographic equipment	T	003
C0035	Launch underwater expendables	T	004
C0040	Analyzing XBT/XSV Forms	T	003
C0045	Complete standard Bathymetric forms	T	003
C0050	Prepare underwater expendables for launch	T	004
C0055	Verify underwater expendables inventories	T	004
C0065	Analyze passive acoustic data	T	007
C0070	Track acoustic contacts	T	006, 008
C0075	Annotate recording media	N/T	
C0080	Classify passive acoustic data	T	007
C0085	Localize passive acoustic contacts by DOPPLER	T	007
C0090	Plot Target Motion Analysis (TMA) data	N/T	
C0105	Correlate active/passive/non-acoustic information	N/T	
C0115	Calculate acoustic sensor range information	N/T	
C0135	Perform noise rounds	T	002

C0170	Assist with loading/handling of torpedoes	T	009
C0175	Operate Torpedo Setting Panel (TSP)	T	009
C0180	Fire torpedoes in emergency mode	T	009
C0185	Charge torpedo tubes/flask	T	009
C0195	Perform Passive Aural Listening (PAL)	T	007
	Duty Area D - UNDER WATER WARFARE		
D0010	Perform UWW weapon systems serviceability checks (i.e. airshot)	T	009
D0025	Perform UWW weapon firings	T	009
D0030	Perform weapon system misfire procedures	T	009
D0035	Conduct weapon system misfire procedures	N/T	
D0080	Interpret LOFAR Gram Acoustic Characteristics	T	007
	Duty Area E - COMBAT INFORMATION ORGANIZATION (CIO)		
E0020	Amplify track information	T	008
	Duty Area F - ACOUSTIC DATA COLLECTION AND ANALYSIS		
F0010	Record acoustic intelligence data	T	006, 008
F0035	Perform basic passive acoustic analysis	T	007
F0045	Interpret oceanographic data	T	005

ANNEX B - RECORD OF DECISIONS

Dates: May – September 2020

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Jawornicki	B	SLt	00211	TDC(A) – LCS Div TDO
Morris	K	Lt	00211	TDC(A) – LCS Div TDO
Tallman	JR	CPO2	00324	TDC(A) – UWW Div
Rogers	TS	PO2	00324	HMCS VILLE DE QUEBEC
Deneau	J	MS	00324	TDC(A) – UWW Div
Grandmaison	MDE	MS	00324	TDC(A) – UWW Div
Blonde	A	MS	00324	TDC(A) – UWW Div
Payette	MJ	MS	00324	TDC(A) – UWW Div
Craig	B	MS	00324	NFS(A) – UWW Instructor
Walentynski		MS	00324	NFS(A) – UWW Instructor

Working Group Objectives:

1. Update and Amend QSP based on the outcomes of Post Course After Action Reports and Instructor feedback;

2. Update QSP to 3 Chapter QSP format;

3. Improve Oceanography instruction and training content;

4. Update and improve trainee / instructor training material;

5. Review timings based on Course After Action Reports (CAAR);

6. Review training methodologies; and

7. Recommend EOs for Technology Enabled Learning (TEL).

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communications circuit functional checks (comm checks)	Learn on NETP shipboard training
A0010	Operate communications equipment	Learn on NETP shipboard training

A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	Instructed on RQ-PO2 level, above current trade knowledge level.
B0015	Maintain stateboards	Instructed at the RQ-S1 level as stateboards are now incorporated with CMS 330.
B0015	Maintain stateboards	Instructed at the RQ-S1 level as stateboards are now incorporated with CMS 330.
C0075	Annotate recording media	No recording media available to annotate, task same as Maintaining Logs
C0090	Plot Target Motion Analysis (TMA) data	TMA plotting is now conducted on CMS 330, training in TMA plotting is now conducted at RQ-S1 level.
C0105	Correlate active/passive/non-acoustic information	Instructed on RQ-PO2 level, above current trade knowledge level.
C0115	Calculate acoustic sensor range information	Instructed on RQ-S1 level, above current trade knowledge level.
D0035	Conduct weapon system misfire procedures	RQ S3 level Performs task D0030, task D0035 should be for RQ PO1 / UWWD level.

Discussions / Decisions:

8. The following Performance and Enabling Objectives were the focus of the WG:

    a. PO 003 – Operate Ancillary Acoustic Equipment to include the development of:
        i. Inclusion of NITES2R Flash Up/Set Up checklists;
        ii. Use of MK-21 Bathythermograph software;
        iii. Underwater Telephone Standard Operating Procedures; and
        iv. Video / lecture on ESUS codes.
    b. PO 005 – Display Intelligence Via Underwater Brief to include:
        i. EO 005.01 – development of Submarine Recognition cards as training aids; and
        ii. EO 005.02 – improvements to Oceanography instruction that includes varied instructional methodologies and more interactive activities to enable more trainee participation and learning interest.
    c. PO 006 – Operate Active SONAR Equipment – restructured EO sequential format to make instruction in line with performance;
    d. PO 008 – Operate Passive SONAR Equipment – restructured EO sequential format to make instruction in line with performance; and
    e. PO 009 – Operate Surface Vessel Torpedo Tubes to include instruction on:
        i. Torpedo Safety procedures;
        ii. States of Preparedness;
        iii. Weapon Readiness Level: and

iv. Torpedo Voice procedures.

9. An initial WG was stood up by NTDC(A) UWW staff to rectify the Oceanography (PO 005 EO 005.02) training. Lessons were developed and included multiple instructional methodologies to provide a varied approach to instructing. These included: Reading Assignments, Lectures, Videos, TEL (to be developed) and AIT.

10. The QSP was in Legacy format, the MLG was poorly formatted, outdated and missing references, much of the training content was outdated or incorrect and that the Oceanography (PO 005 EO 005.02) training was insufficient and "boring" for RQ S3 trainees to comprehend.

11. From the 27-31 July 2020, NTDC(A) UWW section, along with instructors from NFS(A) and one member from the fleet conducted an onsite WG meeting to review the converted QSP and update all the lessons within the document. This review included the identification of various training content and material that was required for instruction.

12. Upon completion of the July Working Group, NTDC(A) UWW section commenced a review of the updated QSP for submission and began developing the training content.

13. This QSP is to be considered a bridge between the current ASW equipment, next phase of training development should focus on the development of Underwater Warfare Suite Upgrade (UWSU) equipment content. Follow-on WGs should attempt to address the misalignment between the PS and LG, EC, and PCs within each PO along with consistent use of terminology throughout.

Dates: July – September 2019

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Jawornicki	B	SLt	00211	TDC(A) – LCS Div TDO
Tallman	JR	CPO2	00324	TDC(A) – UWW Div
Grandmaison	MDE	MS	00324	TDC(A) – UWW Div
Hamilton	CR	PO2	00324	TDC(A) – UWW Div
Payette	MJ	MS	00324	TDC(A) – UWW Div

Working Group Objectives:

1. Update and Amend QSP based on the outcomes of the Combat Training Review (CTR);

2. Update QS to reflect current job requirements;

3. Review timings based on Course After Action Reports (CAAR) and CTR decisions;

4. Review training methodologies;

5. Develop Master Lesson Guidance in FIREFLY format to support QSP; and

6. Recommend EOs for Technology Enabled Learning (TEL).

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communications circuit functional checks (comm checks)	Learn on NETP shipboard training
A0010	Operate communications equipment	Learn on NETP shipboard training
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	Instructed on RQ-PO2 level, above current trade knowledge level.
B0015	Maintain stateboards	Instructed at the RQ-S1 level as stateboards are now incorporated with CMS 330.
B0015	Maintain stateboards	Instructed at the RQ-S1 level as stateboards are now incorporated with CMS 330.
C0075	Annotate recording media	No recording media available to annotate, task same as Maintaining Logs
C0090	Plot Target Motion Analysis (TMA) data	TMA plotting is now conducted on CMS 330, training in TMA plotting is now conducted at RQ-S1 level.

C0105	Correlate active/passive/non-acoustic information	Instructed on RQ-PO2 level, above current trade knowledge level.
C0115	Calculate acoustic sensor range information	Instructed on RQ-S1 level, above current trade knowledge level.
D0035	Conduct weapon system misfire procedures	RQ-S3 level Performs task D0030, task D0035 should be for RQ PO1 / UWWD level.

Discussions / Decisions:

7. The previous version of the QSP was completed in January 2016 but was not approved by NPTG HQ until March 2019. During the Combat Training Review in Jan – Feb 2019, it was determined that the QSP required amendments and updates to timings and instructional methodology based on Course After Action Reports, Post Course Dialogues and Course Critiques. It was also determined by Naval Fleet Schools (Atlantic & Pacific) that there required development of Master Lesson Guidance for use by instructors.

8. NTDC(A) along with NFS(A), NFS(P) and TRINITY ADAC determined during the CTR in February 2019 the requirements for the RQ S3 course content and training times required.

9. NTDC(A) staff stood up a Working Group in July 2019 to update and amend the QSP for formatting, content and timings. NTDC(A) staff also developed Master Lesson Guidance to supplement the QSP.

10. Based on the changes to the RQ S3 QSP, the number of training days was reduced from 101 training days to 98 training days with the largest reduction being in the amount of Active SONAR Operation training and the Passive Acoustic Analysis phase being condensed to 6 weeks in order to align with ADAC's Basic Passive Acoustic Analysis Course (BPAAC). These reductions were assessed as low risk to the trainees based on the outcomes of 3 x PILOT CAARs.

11. The implementation of spirals 2 and 3 TEL options will contribute to reduce training time as well and make training more effective by focussing on performance-based approach such as Explain – Demonstrate – Imitate.

4	Update checklists and teaching point tables to add more details when references are developed.	CPO2 Skinner	TBD

Dates: September – December 2016

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Cornfield	SL	PO2	00324	TDC(A) – UWW Div
Rushton	JO	MS	00324	TDC(A) – UWW Div
Plourde	AP	MS	00324	TDC(A) – UWW Div
Devine	SJ	Lt(N)	00207	TDC(A) – UWW Div
Skinner	DA	CPO2	00324	TDC(A) – UWW Div
Fisher	WJ	PO2	00324	TDC(A) – UWW Div
Bonneau	MJR	Lt	00211	TDC(A) – LSC Div
Matte	KML	SLt	00211	TDC(A) – LSC Div

Working Group Objectives:

1. Align QSP with Future Naval Training Strategy (FNTS) intent;
2. Update QS to reflect current job requirements;
3. Review timings to reduce training time;
4. Review training methodologies;
5. Modularize training; and
6. Recommend EOs for Technology Enabled Learning (TEL).

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communications circuit functional checks (comm checks)	Learn on NETP shipboard training
A0010	Operate communications equipment	Learn on NETP shipboard training
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	Learn on the job
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	Instructed on RQ-PO2 level, above current trade knowledge level.
B0015	Maintain stateboards	Instructed at the RQ-S1 level as stateboards are now incorporated with CMS 330.
C0075	Annotate recording media	No recording media available to annotate, task same as Maintaining Logs

C0105	Correlate active/passive/non-acoustic information	Instructed on RQ-PO2 level, above current trade knowledge level.
C0115	Calculate acoustic sensor range information	Instructed on RQ-S1 level, above current trade knowledge level.
D0035	Conduct weapon system misfire procedures	RQ S3 level Performs task D0030, task D0035 should be for RQ PO1 / UWWD level.

Discussions / Decisions:

7. The QS was updated to reflect the current job requirements and the instructional methods and timings were reviewed. These changes enabled the course duration to be reduced from 143 to 118 training days.

8. The implementation of spirals 2 and 3 TEL options will contribute to reduce training time as well and make training more effective by focussing on performance-based approach such as Explain – Demonstrate – Imitate.

9. The assessment plan was reviewed to reduce the amount of knowledge-based tests and enhance both formative and summative assessments of trainees.

10. The QL3 and QL4 have been merged together since the old QL4 QSP was a complete repetition of the QL3 QSP. The standards presented in both Course Control Documents (CCDs) were the exact same. Also, no extra training was required for any task of the QL4 QSP. Working group members mentioned that all that is necessary for trainees to be ready for their job after the in-house training is more expertise at sea. Therefore, to help them transfer their knowledge in real life situation before granting the final seal of approval a QL3 OJT is needed. By linking the OJPR to the QL3 instead of being on its own, the training progression will remain concretely unchanged for the members; the only difference will be in the CCD. The agreed upon modularization of the merge SONAR Op RQ S3/S2 QSP is as follow:

a. An in-house section (Module 1 to 9) that used to be the old QL3; and

b. An OJPR (Module 10), that used to be the old QL4.

11. The modules will keep the same MITE codes, ensuring that the new merged QSP doesn't affect the SONAR Op rank progression.

12. PO 001 has the potential to be developed as a DL course offered to various occupations as part of their QL3.

13. All EOs have potential for DL. In a near future, development of e-learning modules with low level of interactivity could be viable option to reduce in-house portion of the course. These modules could focus on knowledge-based components related to either buttonology, capabilities and limitations of equipment. There is also a DL potential for

classification and analysis portion. In-house part of the course could focus on demonstration and practice.

14. Video production was also identified as a viable option to show best practices and support classroom instruction.

15. References need to be developed for the following:

a. PO 004, Par 3 d. and e.;

b. PO 008, Par 3 c.;

c. EO 003.03, Par 3 d. ref for Bell Hop has not enough details;

d. EO 003.05, no references for Bathy and Under Water Telephone logs;

e. EO 004.01, Par 3 d. and e.;

f. EO 004.02, Par 3 d. and e.;

g. EO 004.03, Par 3 d. and e.;

h. EO 004.04, Par 3 d. and e.; and

i. EO 004.05, Par 3 d. and e.

16. IAW Par 14, checklists and teaching point tables need to be updated. Since a lot of references were non-existent, it was impossible for the working group to give further details when creating the checklists or defining teaching points. This is a critical consideration in order to make sure future Jr SONAR Operators are trained to expected level. Working group members stated that, even if they lack some details, checklists are the exact representations of job requirements.

Lessons Learned:

17. Most of the references were either non-accessible or non-existent. Pre-working group activities must allocate time or resources to gather references onto a laptop or secret hard drive.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	MLPs to be developed	CPO2 Skinner	TBD
2	Courseware material to be developed	CPO2 Skinner	TBD

3	References to be developed	CPO2 Skinner	TBD
4	Update checklists and teaching point tables to add more details when references are developed.	CPO2 Skinner	TBD

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ANNEX C - COSTING INFORMATION

Based on 12 trainees total, with 90 training days (plus weekends 126 calendar days)

Course: RQ-SONAROP-S3	Serial:					
TD	No. Of Trainees / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total (2 serial per yr)
Airfare (Return)	0	$1 000,00		$0	$0	$0,00
R&Q	0	$36,00	0	$0	$0	$0,00
Incidentals	0	$17,50	0	$0	$0	$0,00
Subtotal				**$0,00**	**$0,00**	**$0,00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	12	$50,00	1	$600	$600	$6 000.00
Classroom (square meters)	30	$13.00	92	$35,880	$35,880	$351,880.00
Computer requirements	12	$1 000,00	1	$12 000	$0	$12 000.00
IT Equip (Proj, Smart board etc)	1	$6 000,00	1	$6 000	$0	$6 000.00
DWAN connection (Set up)	1	$5 000,00	1	$5 000	$0	$5 000.00
Link Toolkit (on DVD)	15	$1,00	1	$15	$15	$150.00
Instructional Materials (Printed / Handouts etc)	15	$30,00	1	$450	$450	$4 500.00
Subtotal				**$59 945**	**$36 945**	**$385 430.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)5	0	$206,00	0	$0	$0	$0,00
Instructor (Lt(N)) (incremental)5	0	$235,00	0	$0	$0	$0,00
Incremental Civilian Instructor	0	0	0	$0	$0	$0,00
		subtotal		**$0,00**	**$0,00**	**$0,00**
	Total Estimated Costs			**$59 945**	**$36 945**	**$385 430.00**

ANNEX D - ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess trainee's ability to: - Maintain physical security.	Practical	5/5 SATs
PC 001	PO 001	To assess trainee's ability to: - Comply with security.	Practical	8/10 SATs
EC 002.01	EO 002.01	To assess trainee's ability to: - Perform noise rounds.	Practical	7/9 SATs
PC 002	PO 002	To assess trainee's ability to: - Perform noise rounds.	Practical	7/9 SATs
EC 003.01	EO 003.01	To assess trainee's ability to: - initialize oceanographic equipment; - launch Bathythermograph (XBT/XSV); and - shut down oceanographic equipment.	Practical	11/14 SATs
EC 003.02	EO 003.02	To assess trainee's ability to: -	Practical	10/13 SATs
PC 003A	EO 003.01 EO 003.02	To assess trainee's ability to: - initialize ancillary acoustic equipment; - launch Bathythermograph (XBT/XSV); - analyze XBT/XSV forms; and - Analyze XBT/XSV forms.	Practical	15/19 SATs
EC 003.03	EO 003.03	To assess trainee's ability to: - Communicate externally on Underwater Telephone.	Practical	25/33 SATs
PC 003B	EO 003.03	To assess trainee's ability to: - Communicate externally on Underwater Telephone.	Practical	22/29 SATs
EC 004.01A	EO 004.01.01 EO 004.01.02	To assess trainee's ability to: - Launch Sonobuoy.	Practical	5/6 SATs

EC 004.01B	EO 004.01.01 EO 004.01.03	To assess trainee's ability to: - Launch ADCs.	Practical	6/7 SATs
EC 004.01C	EO 004.01.01 EO 004.01.04	To assess trainee's ability to: - Launch NAEBs.	Practical	5/6 SATs
EC 004.01C	EO 004.01.01 EO 004.01.05	To assess trainee's ability to: - Launch ESUS.	Practical	4/5 SATs
PC 004	PO 004	To assess trainee's ability to: - Launch underwater expendables.	Practical	4/4 SATs
EC 005.01	EO 005.01	To assess trainee's ability to: - Deliver a platform brief.	Practical	7/9 SATs
EC 005.02	EO 005.02	To assess trainee's ability to: - Interpret oceanographic and environmental conditions.	Practical	21/28 SATs
PC 005	PO 005	To assess trainee's ability to: - Display intelligence via UWW brief.	Practical	8/11 SATs
EC 006.01A	EO 006.01	To assess trainee's ability to: - Ensure HMS is operational.	Practical	13/16 SATs
EC 006.01B	EO 006.01	To assess trainee's ability to: - Perform HMS operating procedures.	Practical	12/15 SATs
EC 006.03	EO 006.03	To assess trainee's ability to: - perform initial contact procedures; - perform lost contact procedures; and - Perform bearing investigation.	Practical	29/34 SATs
EC 006.04	EO 006.04	To assess trainee's ability to: - perform torpedo procedures	Practical	8/10 SATs and Critical(s)
EC 006.05	EO 006.05	To assess trainee's ability to: - perform mine investigation procedures	Practical	11/14 SATs
PC 006	PO 006	To assess trainee's ability to: - Operate active SONAR equipment.	Practical	61/76 SATs and Critical(s)
EC 007.02	EO 007.01 EO 007.02	To assess trainee's ability to: - interpret LOFAR Gram acoustic characteristics; and	Practical	70%

		- Determine mathematical relationship of signature sources.		
EC 007.03	EO 007.03	To assess trainee's ability to: - Perform PAL.	Practical	50%
EC 007.04	EO 007.04	To assess trainee's ability to: - Apply the five steps to analysis.	Practical	60%
EC 007.06	EO 007.05 EO 007.06	To assess trainee's ability to: - localize passive acoustic data contacts by Doppler; and - Provide generic classification of passive contacts.	Practical	60%
PC 007	PO 007	To assess trainee's ability to: - Perform basic passive acoustic analysis.	Practical	60%
EC 008.01	EO 008.01	To assess trainee's ability to: - Set up SPS.	Practical	10/13 SATs
EC 008.02	EO 008.02	To assess trainee's ability to: - Ensure CANTASS is operational.	Practical	20/24 SATs
EC 008.03	EO 008.03	To assess trainee's ability to: - Set up CANTASS operating parameters.	Practical	16/19 SATs
EC 008.05	EO 008.04 EO 008.05	To assess trainee's ability to: - track acoustic contacts; - amplify track information; - conduct torpedo detection procedures; and - Maintain logs.	Practical	20/25 SATs
PC 008	EO 008.02 EO 008.03 EO 008.04 EO 008.05	To assess trainee's ability to: - Operate passive acoustic equipment.	Practical	38/47 SATs

EC 009.03	EO 009.01 EO 009.03	To assess trainee's ability to: - Operate Torpedo Settings Panel.	Practical	17/21 SATs
EC 009.04	EO 009.04	To assess trainee's ability to: - perform UWW weapon firing in primary and secondary mode; and - Perform UWW weapon firings in emergency mode.	Practical	37/41 SATs
PC 009	PO 009	To assess trainee's ability to: - Operate SVTT weapon systems.	Practical	33/41 SATs

RQ-SONAR OP-S2 OJPR ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
PC 002	PO 002	To assess trainee's ability to: - Perform noise rounds.	Practical	19/19 SATs
PC 003A	EO 003.01 EO 003.02	To assess trainee's ability to: - initialize ancillary acoustic equipment; - launch Bathythermograph (XBT/XSV); - analyze XBT/XSV forms; and - Analyze XBT/XSV forms.	Practical	18/23 SATs
PC 003B	EO 003.03	To assess trainee's ability to: - Communicate externally on Underwater Telephone.	Practical	30/30 SATs
PC 004	PO 004	To assess trainee's ability to: - Launch underwater expendables.	Practical	14/14 SATs
PC 006	PO 006	To assess trainee's ability to: - Operate active SONAR equipment.	Practical	10 hours Operating time, 2 hours of In Contact time
PC 007	PO 007	To assess trainee's ability to: - Perform basic passive acoustic analysis.	Practical	60% on 6 training packs
PC 008	EO 008.02 EO 008.03 EO 008.04 EO 008.05	To assess trainee's ability to: - Operate passive acoustic equipment.	Practical	10 hours Operating time, 2 hours of In Contact time
PC 009	PO 009	To assess trainee's ability to: - Operate SVTT weapon systems.	Practical	43/43 SATs

ANNEX E – TEL DEVELOPMENT OPPORTUNITIES

Title	Type	Description of Requirement	Lesson	LSR #
Perform Noise Rounds	Virtual Training	Develop a 3D model of a ship with noise defects that need to be identified and/or corrected	EO 002.01.02	
CPF Bathy Launch Procedures	Video	Develop a video on correct method to launch a Bathy onboard CPF	EO 00.03.01	
UWT Communications	Emulator	Develop an UWT emulator that simulates communicating underwater	EO 003.03.01	
MK 84 ESUS Training video	Video	Obtain the permission to utilize the US Navy YouTube video on MK 84 ESUS	EO 003.03.01	
Son buoy Launch procedures	Video	Develop a video on correct method to launch a Sonobuoy	EO 004.01.02	
ADC Launch procedures	Video	Develop a video on correct method to launch a ADC	EO 004.01.03	
NAEB Launch procedures	Video	Develop a video on correct method to launch a NAEB	EO 004.01.03	
ESUS Launch procedures	Video	Develop a video on correct method to launch a ESUS	EO 004.01.04	
NOAA Oceanography video	Video	Obtain the permission to utilize the NOAA YouTube video on Oceanography	EO 005.02.01	
USN METOC video	Video	Obtain the permission to utilize the USN METOC YouTube video on Oceanography	EO 005.02.01	
Ocean Floor video	Video	Create a video on Ocean Floor	EO 005.02.02	
Ocean Sediments video	Video	Create a video on Ocean Sediments	EO 005.02.02	
Ocean Pressure “Ice Breaker” video	Video	Obtain the permission to utilize the YouTube video on “Opening a Soda Can on the Ocean Floor”	EO 005.02.03	
Sound Speed video	Video	Obtain the permission to utilize the YouTube video on “Speed of Sound through Solids, Liquids and Gasses”	EO 005.02.05	
How SONAR works? video	Video	Create video on How SONAR Works	EO 005.02.07	
Transmission of Sound & How Sound Waves Travel video	Video	Create video or obtain permission to utilize the YouTube video on “Transmission of Sound & How Sound Waves Travel”	EO 005.02.07	

Wave Fronts & Sound Rays video	Video	Create video or obtain permission to utilize the YouTube video on “Wave Fronts & Sound Rays”	EO 005.02.07	
Decibels Explained video	Video	Create video or obtain permission to utilize the YouTube video on “Decibels Explained”	EO 005.02.07	
Can you Hear Me? video	Video	Create video or obtain permission to utilize the YouTube video on “Can you Hear Me?”	EO 005.02.10	
Ocean Currents video	Video	Create video or obtain permission to utilize the YouTube video on “Ocean Currents”	EO 005.02.12	
HMS Flash up video	Video	Create a HMS Flash up procedures video	EO 006.01.01	
Torpedo Load / Unload Procedures	Virtual Training	Develop a 3D model virtual trainer of a SVTT to allow trainees to practice loading/unloading torpedoes	EO 009.02.01	

ANNEX F – SCHEDULING INFORMATION

Scheduling Annex - RQ-SONAR OP-S3													
		Location	IT Assets					Footprint			Instructor		
		Location	DWAN	CSNI	SSA	Proj	TEL / Critical Resources	Sec Class	WB	Desks	Type	Qual	Sec Cl
TOTAL	**850**	**PO 001 - Comply with Security**											
EO 001.01.01	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 001.01.02	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	MIL	RQ-SONAR OP-MS	Sec Cl
EO 001.01.03	250	Classroom	1	0	0	1	Unclassified Classroom with computer	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 001.01	200	Classroom	1	0	0	1	Unclassified Classroom with computer	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 001	200	Classroom	1	0	0	1	Unclassified Classroom with computer	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
TOTAL	**800**	**PO 002 - Perform Noise Rounds**											
EO 002.01.01	200	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 002.01.02	200	Ship	0	0	0	0	HMC Ship (any class)	UNC	0	0	Mil	RQ-SONAR OP-MS	Sec Cl
EC 002.01	200	Ship	0	0	0	0	HMC Ship (any class)	UNC	0	0	Mil	RQ-SONAR OP-MS	Sec Cl
PC 002	200	Ship	0	0	0	0	HMC Ship (any class)	UNC	0	0	Mil	RQ-SONAR OP-MS	Sec Cl
TOTAL	**1700**	**PO 003 - Operate Ancillary Acoustic Equipment**											
EO 003.01.01	200	Classroom	0	0	13	1	NITES2R Trainer or HMC Ship	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 003.01	150	Classroom	0	0	13	1	NITES2R Trainer or HMC Ship	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 003.02.01	250	Classroom	0	0	13	1	NITES2R Trainer or HMC Ship	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 003.02	50	Classroom	0	0	13	1	NITES2R Trainer or HMC Ship	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 003A	200	Classroom	0	0	13	1	NITES2R Trainer or HMC Ship	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 003.03.01	450	Classroom	0	1	0	1	Classified Classroom / HMC Ship	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 003.03	200	Classroom	0	1	0	1	Classified Classroom / HMC Ship	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 003B	200	Classroom	0	1	0	1	Classified Classroom / HMC Ship	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
TOTAL	**1150**	**PO 004 - Launch Underwater Expendables**											
EO 004.01.01	150	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 004.01.02	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 004.01A	100	Classroom	0	0	0	0	Unclassified Classroom	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 004.01.03	100	Classroom	0	0	0	0	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 004.01B	100	Classroom	0	0	0	0	Unclassified Classroom	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 004.01.04	100	Classroom	0	0	0	0	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 004.01C	100	Classroom	0	0	0	0	Unclassified Classroom	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 004.01.05	50	Classroom	1	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 004.01D	100	Classroom	0	0	0	0	Unclassified Classroom	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 004	250	Classroom	0	0	0	0	Unclassified Classroom	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl

TOTAL	**4300**	**PO 005 - Display Intelligence via Under Water Warfare Brief**											
EO 005.01.01	550	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 005.01	250	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 005.02.01	50	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.02	300	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.03	150	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.04	150	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.05	250	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.06	50	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.07	300	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.08	350	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.09	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.10	200	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.11	50	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.12	350	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.13	300	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.14	250	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EO 005.02.15	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
EC 005.02	300	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
PC 005	250	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-PO2	Sec Cl
TOTAL	**4950**	**PO 006 - Operate Active SONAR Equipment**											
EO 006.01.01	350	Classroom	0	1	0	1	HMC Ship or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.01.02	300	SEACOT	0	0	0	1	HMC Ship or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 006.01A	300	SEACOT	0	0	0	1	HMC Ship or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 006.01B	300	SEACOT	0	0	0	1	HMC Ship or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.02.01	200	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.02.02	100	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.03.01	1100	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 006.03	400	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.04.01	350	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 006.04	300	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 006.05.01	350	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 006.05	300	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 006	600	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl

TOTAL	**10850**	**PO 007 - Perform Basic Passive Acoustic Analysis**											
EO 007.01.01	450	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 007.02.01	1000	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 007.02	50	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 007.03.01	1150	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 007.03	200	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 007.04.01	500	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 007.04	50	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 007.05.01	200	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 007.06.01	6650	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 007.06	300	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 007	300	Classroom	0	0	13	1	ADAC or Classified Classroom	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
TOTAL	**5200**	**PO 008 - Operate Passive Acoustic Equipment**											
EO 008.01.01	400	Trainer	0	0	0	1	HMC Ship or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 008.01	200	Trainer	0	0	0	1	HMC Ship or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 008.02.01	350	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 008.02	400	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 008.03.01	1350	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 008.03	600	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 008.04.01	50	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 008.05.01	650	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 008.05	600	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 008	600	Trainer	0	0	0	1	CANTASS Mission Simulator or SEACOT	SEC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
TOTAL	**2150**	**PO 009 - Operate SVTT Weapons System**											
EO 009.01.01	200	Classroom	13	0	0	1	DWAN with projector & speakers	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 009.02.01	400	HMC Ship or Trainer	0	0	0	0	HMC Ship or Trainer	UNC	1	13	Mil	RQ-WENG-PO2 / RQ-SONAR OP-MS	Sec Cl
EO 009.03.01	200	HMC Ship or Trainer	0	0	0	0	HMC Ship or SEACOT	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 009.03	200	HMC Ship or Trainer	0	0	0	0	HMC Ship or SEACOT	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EO 009.04.01	550	HMC Ship or Trainer	0	0	0	0	HMC Ship or SEACOT	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
EC 009.04	300	HMC Ship or Trainer	0	0	0	0	HMC Ship or SEACOT	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl
PC 009	300	SEACOT	0	0	0	1	HMC Ship or SEACOT	UNC	1	13	Mil	RQ-SONAR OP-MS	Sec Cl

Totals	31950
ATT Periods	639
ATT Days	79.875

ANNEX G – PLAN, TRAIN, DO

PO	EO	Maintain Readiness	Plan	Train	Execute
001	**01**	EO 001.01 Maintain Physical Security			
002	**01**			EO 002.01 Perform Noise Rounds	
003	**01**	EO 003.01 Initialize Ancillary Acoustic Equipment			
	02	EO 003.02 Analyze XBT/XSV Forms			
	03			EO 003.03 Communicate Externally on UWT	
004	**01**	EO 004.01 Underwater Expendables			
005	**01**		EO 005.01 Deliver a Platform Brief		
	02		EO 005.02 Interpret Oceanographic & Environmental Conditions & Data		
06	**01**	EO 006.01 Ensure HMS is Operational			
	02			EO 006.02 Operate HMS	
	03			EO 006.03 HMS Contact Procedures	
	04			EO 006.04 Perform Torpedo Procedures	

	05			EO 006.05 Perform Mine Investigation Procedures	
07	**01**	EO 007.01 Interpret LOFAR Gram Acoustic Characteristics			
	02			EO 007.02 Determine Mathematical Relationships of Signature Sources	
	03				EO 007.03 Perform Passive Aural Listening
	04				EO 007.04 Apply the Five Steps of Analysis
	05				EO 007.05 Localize Passive Acoustic Data Contacts by Doppler
	06				EO 007.06 Provide Generic Classification of Passive Contacts
08	**01**	EO 008.01 Set Up SPS			
	02		EO 008.02 Ensure CANTASS is Operational		
	03			EO 008.03 Set up CANTASS Operating Parameter	
	04			EO 008.04 Maintain Passive Acoustic Equipment Log	
	05				EO 008.05 Tracking Contacts and Amplify Track Information

09	**01**	EO 009.01 Torpedo Safety and Preparedness			
	02			EO 009.02 Assist in Loading/ Unloading Torpedoes	
	03				EO 009.03 Operate TSP
	04				EO 009.04 Perform UWW Weapons Firing

ANNEX H - MAIN REFERENCES

Document Number	Document Title
A-P9-050	Canadian Forces Manual of Individual Training and Education (CFITES)
APP-11	NATO Message Catalogue
APP-7	Allied Joint – Brevity Code Words
ATP-1	Allied Maritime Tactical Instructions and Procedures
ATP-28	Allied Anti-Submarine Warfare Manual
ATP-32	NATO Handbook of Military Oceanographic Info and Services
AXP-1	Submarine and Anti-Submarine Exercise Manual
C-52-476-000/MB-001	Operator's Guide SHINCOM applicable to HALIFAX CLASS
C-59-007-006/MB-001	Sonobuoy Reference Guide
C-59-795-002/MB-001	Sonobuoy Processor Subsytem SPSS 3-1
C-69-703-000/MS-001	Operating and Maintenance Instructions with Parts List Sonar Hull Outfit C-5
C-69-729-000/MS-001	Users Reference Manual for MK 46 Mod 5 (AS)
C-69-746-000/MS-001	Towed Array Group
C-69-748-000/MB-001	Operators Manual, CANTASS Book 1
C-69-748-000/MB-001	Operators Manual, CANTASS Book 2
C-69-750-000/MS-001	Launch Accessories, Torpedo MK 46 Mod 5
C-69-789-000/MB-002	Operator’s Manual MK39 EMATT Advanced Acoustic Threat
C-69-799-000/MB-001	Operating Instructions Detecting-Ranging Set, Sonar, AN/SQS-510 (V) 4
C-69-874-000/MS-001	Torpedo Setting Panel Mk 331 Mod 4
C-69-894-001/MS-001	MK8-F Surface Ship Bathythermograph Data Acquisition System Manual
C69-746-000/MS-001	Towed Array Group SQR-19
CFCD 106	Maritime Tactical Instructions
CFCD 109	Naval Vessel Signature Control Manual
CFCD 113 Vol 1	Maritime Command Jezebel Analysis Manual
CFCD 113 Vol 2	Maritime Command Acoustic Information Guide
DUWCS TUUM-6	Digital Underwater Communications System UWT
MILOC	CAF Manual of Military Oceanography (MILOC) March 2018
NAVORD 2115-1	Control and Accounting Procedures for Classified Standard Documents
NAVORD 3137-0	Acoustic Countermeasure (ACM) – Policy and Guidelines
NAVORD 4003-6	Marine Mammal Mitigation Procedures (MMMP)
NAVORD 4995-0	Safety Firings Orders for Ships
NCPM 1001	Drill Book for Surface Vessel Torpedo Tubes (SVTT)
NCPM 600	UWW Procedural Manual for HALIFAX and IROQUOIS Class
NWP 24-1-1	Principles of LOFAR Gram Analysis
RP 33	Fleet Oceanographic and Acoustic Reference Manual

SEMS	Halifax Class Safety and Environmental Management System (SEMS) Manual
SEMS SOP E05A	Halifax Class Safety and Environmental Management System (SEMS) Manual SOP E05A Mitigation Avoidance Zones
CSUM	CMS 330 Combat System User Manual (CSUM)
	Expendable Bathythermograph / Expendable Sound Velocitmeter (XBT/XSV) Expendable Profiling Systems
	DAEME ASSB Technical Letter 10-70
	DAEME ASSB Technical Letter 10-71
	Intermediate Maintenance and Handling Procedures for Acoustic Device, Countermeasure (ADC) MK2 MOD 3
	Intermediate Maintenance and Handling Procedures for Acoustic NAE Beacon MK 3 MOD 3
Jane’s Online	Jane's Online (https://janes.ihs.com/Janes/search?q=delta+stretch&pg=1)
	Naval Integrated Tactical Environmental System (NITES) II Redesign (N2R), Standard Operating Procedures (SOP), Thursday, 8 March 2018, Version 0.8
NDSOD	National Defense Security Orders and Directives
Ocean Navigator	Ocean Navigator Website (http://navigator.oceansdata.ca)
SSOs	Ships Standing Orders
	US Navy Environmental Guide
	TRINITY METOC Website

ANNEX I - ACRONYMS AND ABBREVIATIONS

Acronym or Abbreviation	Term or Definition
ADAC	Acoustic Data Analysis Centre
ADC	Acoustic Device Countermeasures
ADR	Acoustic Data Recorder
AIT	Alternate Instructional Techniques
ARS	Acoustic Response Sheet
ATD	Actual Training Days
ATT	Actual Training Time
BAAL	Basic Ammunition Allocation List
BAT	Base Administrative Time
BB	Broad Band
CNPTG	Commander Naval Personnel Training Group
CCA	Curriculum Control Authority
CCD	Created Contact Data
CFCD	Canadian Forces Controlled Document
CFITES	Canadian Forces Manual of Individual Training and Education
Ch	Chapter
CMP	Chief of Military Personnel
CMS	CANTASS Mission Simulator
CMS 330	Combat Management System 330
CPA	Closest Point of Approach
CSD	Classified Standard Documents
CSNI	Consolidated Secret Network Infrastructure
CW	Continuous Wave
CZ	Convergence Zone
DCA	Display Control Area
DI	Directivity Index
DL	Distributed Learning
DSCA	Deep Sound Channel Axis
DT	Detection Threshold
DUWCS	Digital Underwater Communications System (UWT)
DWAN	Defence Wide Area Network
EC	Enabling Check
EDI	Explain, Demonstrate, Imitate
EMATT	Expendable Mobile ASW Training Target
EMCOM	Emission Control
EO	Enabling Objective
ESUS	Electronic Signal Underwater Sound
EXTORP	Exercise Torpedo
FM	Frequency Modulated
HALT	High Away Low Towards

HMCS	Her Majesty's Canadian Ship
HMS	Hull Mounted System
HOTTORP	Hydrodynamic Operational Training Torpedo
IAW	In Accordance With
IGOSS	Integrated Global Ocean Services System
JBS	Job-Based Specifications
Jr SONAR Op	Junior SONAR Operator
LEP	List of Effective Pages
LOFAR	Low Frequency Array
MILOC	Military Oceanography
MITE	Military Individual Training and Education
MLP	Master Lesson Plan
MMMP	Marine Mammal Mitigation Procedures
Mod	Module
MS	Master Sailor
NAEBs	Naval Acoustic Electromechanical Beacons
NAVORD	Naval Order
NB	Narrow Band
NCM	Non-Commission Member
NCPM	Naval Combat Procedures Manual
NDSODs	National Defence Security Orders and Directives
NFS	Naval Fleet School
NITES2R	Naval Integrated Tactical Environmental System II Redesign
NL	Noise Level
NPTG	Naval Personnel Training Group
NPTORD	Naval Personnel and Training Order
NQual	National Qualification
NTDC(A)	Naval Training Development Center Atlantic
OFA	Ocean Features Analysis
OJPR	On the Job Performance Record
QAB	Quick Action Button
QSP	Qualification Standard Plan
PAL	Passive Aural Listening
Par	Paragraph
PC	Performance Check
PCl	Performance Checklist
Pg	Page
PLAR	Prior Learning Assessment and Recognition
PO	Performance Objective
PRB	Progress Review Board
PRS	Passive Ranging SONAR
PT	Physical Fitness
PTP	Portable Target Programmer
QSP	Qualification Standard and Plan
RCN	Royal Canadian Navy

RF	Radio Frequency
ROW	Rest Of World
RQ	Rank Qualification
S_1	Spiral 1
S_2	Spiral 2
S_3	Spiral 3
S1	Sailor 1st Class
S2	Sailor 2nd Class
S3	Sailor 3rd Class
SAT	Satisfactory
Sect	Section
SEMS	Safety and Environmental Management System
SL	Source Level
SLD	Sonic Layer Depth
SMHT	System Monitoring Health Table
SONAR	Sound Navigation And Ranging
SOP	Standing Operating Procedure
SPS	Sonobuoy Processing System
SS	Search Summary
SSO	Ship Standing Order
SVP	Sound Velocity Profile
SVTT	Surface Vessel Torpedo Tubes
TA	Training Authority
TAT	Training Administrative Time
TBB	Triple Beam Bearing
TD	Target Detail
TE	Training Establishment
TEL	Technology Enabled Learning
TIOW	Transmission Interval of the Watch
TL	Transmission Loss
TP	Training Plan
TRB	Training Review Board
TS	Target Strength
TSP	Torpedo Setting Panel
TTT	Training Travel Time
TVM	Television Monitor
UWT	Underwater Telephone
UWW	Under Water Warfare
VFK	Variable Function Keys
VOSIW	Velocity Of Sound In Water
WENG	Weapons Engineer
XBT	Expendable Bathythermograph
XSV	Expendable Sound Velocity

ANNEX J - RECOMMENDED SCHEDULE

Week #1

TIME	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Course Introduction & Administration (BAT)**	**EO 001.01.03** –Securing CSD Material (EDI)	**PT**	**EO 002.01.01** – Ship Noise, Quiet State & Organization (Lecture)	**EC 002.01** – Perform Noise Rounds
Period 2 0850-0940	**Course Introduction & Administration (BAT)**	**EO 001.01.03** –Securing CSD Material (EDI)	**PT**	**EO 002.01.01** – Ship Noise, Quiet State & Organization (Lecture)	**EC 002.01** – Perform Noise Rounds
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Course Introduction & Administration (BAT)**	**EO 001.01.03** –Securing CSD Material (EDI)	**EC 001.01** – Maintain Physical Security	**EO 002.01.01** – Ship Noise, Quiet State & Organization (Lecture)	**EC 002.01** – Perform Noise Rounds
Period 4 1040-1130	**Course Introduction & Administration (BAT)**	**EO 001.01.03** –Securing CSD Material (EDI)	**EC 001.01** – Maintain Physical Security	**EO 002.01.01** – Ship Noise, Quiet State & Organization (Lecture)	**EC 002.01** – Perform Noise Rounds
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 001.01.01** –Physical Security Protocols (Lecture)	**EO 001.01.03** –Securing CSD Material (EDI)	**PC 001**	**EO 002.01.02** – Conducting Noise Rounds (Tour / EDI)	**PC 002**
Period 6 1320-1410	**EO 001.01.01** –Physical Security Protocols (Lecture)	**EC 001.01** – Maintain Physical Security	**PC 001**	**EO 002.01.02** – Conducting Noise Rounds (Tour / EDI)	**PC 002**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 001.01.02** –Mustering CSD Material (EDI)	**EC 001.01** – Maintain Physical Security	**PC 001**	**EO 002.01.02** – Conducting Noise Rounds (Tour / EDI)	**PC 002**
Period 8 1510-1600	**EO 001.01.02** –Mustering CSD Material (EDI)	**PT**	**PC 001**	**EO 002.01.02** – Conducting Noise Rounds (Tour / EDI)	**PC 002**

Week #2

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**EO 003.01.01** – Operate Oceanographic Equipment (EDI)	**PT**	**PC 003A**	**PT**	**EC 003.03** – Communicate Externally on UWT
Period 2 0850-0940	**EO 003.01.01** – Operate Oceanographic Equipment (EDI)	**EO 003.02.01** – Draft XBT/XSV/IGOSS Forms (EDI)	**PC 003A**	**PT**	**EC 003.03** – Communicate Externally on UWT
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 003.01.01** – Operate Oceanographic Equipment (EDI)	**EO 003.02.01** – Draft XBT/XSV/IGOSS Forms (EDI)	**PC 003A**	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**EC 003.03** – Communicate Externally on UWT
Period 4 1040-1130	**EO 003.01.01** – Operate Oceanographic Equipment (EDI)	**EO 003.02.01** – Draft XBT/XSV/IGOSS Forms (EDI)	**PC 003A**	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**EC 003.03** – Communicate Externally on UWT
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**BAT**	**EO 003.02.01** – Draft XBT/XSV/IGOSS Forms (EDI)	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**PC 003B**
Period 6 1320-1410	**EC 003.01** – Initialize Ancillary Acoustic Equipment	**EO 003.02.01** – Draft XBT/XSV/IGOSS Forms (EDI)	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**PC 003B**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 003.01** – Initialize Ancillary Acoustic Equipment	**EC 003.02** – Analyze XBT/XSV forms	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**PC 003B**
Period 8 1510-1600	**EC 003.01** – Initialize Ancillary Acoustic Equipment	**TAT**	**EO 003.03.01** – Operate & Communicating UWT (EDI)	**TAT**	**PC 003B**

Week #3

TIME	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**EO 004.01.01** – Safe Storage and Handling of Underwater Expendables (Lecture/EDI)	**PT**	**EC 004.01C** – Prepare NAEB for Launch	**PT**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
Period 2 0850-0940	**EO 004.01.01** – Safe Storage and Handling of Underwater Expendables (Lecture/EDI)	**EO 004.01.03** – Prepare ADC for Launch (EDI)	**EO 004.01.05** – Prepare ESUS for Launch (EDI)	**TAT**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.01.01** – Safe Storage and Handling of Underwater Expendables (Lecture/EDI)	**EO 004.01.03** – Prepare ADC for Launch (EDI)	**EC 004.01D** – Prepare ESUS for Launch	**PC 004**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
Period 4 1040-1130	**TAT**	**EC 004.01B** – Prepare ADC for Launch	**EC 004.01D –** Prepare ESUS for Launch	**EO 005.01.01** –Platform Briefs (EDI/Presentation)	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.01.02** – Prepare Sonobuoy for Launch (EDI)	**EC 004.01B** – Prepare ADC for Launch	**PC 004**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
Period 6 1320-1410	**EO 004.01.02** – Prepare Sonobuoy for Launch (EDI)	**EO 004.01.04** – Prepare NAEB for Launch (EDI)	**PC 004**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)	**EO 005.01.01** –Platform Briefs (EDI/Presentation)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 004.01A** – Prepare Sonobuoys for Launch	**EO 004.01.04** – Prepare NAEB for Launch (EDI)	**PC 004**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)	**EC 005.01** – Deliver Platform Brief
Period 8 1510-1600	**EC 004.01A** – Prepare Sonobuoys for Launch	**EC 004.01C** – Prepare NAEB for Launch	**PC 004**	**EO 005.01.01** –Platform Briefs (EDI/Presentation)	**PT**

Week #4

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**EC 005.01** – Deliver Platform Brief	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.04** – Ocean Thermal Structure (Interactive Lecture)	**PT**	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)
Period 2 0850-0940	**EC 005.01** – Deliver Platform Brief	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.04** – Ocean Thermal Structure (Interactive Lecture)	**TAT**	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 005.01** – Deliver Platform Brief	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.04** – Ocean Thermal Structure (Interactive Lecture)	**EO 005.02.06** – Ocean Bottom & Seawater Characteristics (Confirmation Check)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
Period 4 1040-1130	**EC 005.01** – Deliver Platform Brief	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.05** – Sound Velocity Profile (Interactive Lecture)	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 005.02.01** – Introduction to Military Oceanography (Interactive Lecture)	**EO 005.02.03** – Characteristics of Seawater (Interactive Lecture)	**EO 005.02.05** – Sound Velocity Profile (Interactive Lecture)	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
Period 6 1320-1410	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.03** – Characteristics of Seawater (Interactive Lecture)	**EO 005.02.05** – Sound Velocity Profile (Interactive Lecture)	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 005.02.02** – Ocean Bottom Topography & Sediments (Interactive Lecture)	**EO 005.02.03** – Characteristics of Seawater (Interactive Lecture)	**EO 005.02.05** – Sound Velocity Profile (Interactive Lecture)	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)
Period 8 1510-1600	**PT**	**PT**	**EO 005.02.05** – Sound Velocity Profile (Interactive Lecture)	**EO 005.02.07** – Basic Sound Theory (Interactive Lecture)	**EO 005.02.08** – Sound Propagation (Interactive Lecture)**TAT**

Week #5

TIME	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**EO 005.02.09** – Sound Propagation Loss (Interactive Lecture)	**TAT**	**PT**	**PT**	**EC 005.02** – Interpret Oceanographic & Environmental Conditions
Period 2 0850-0940	**EO 005.02.09** – Sound Propagation Loss (Interactive Lecture)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**BAT**	**EC 005.02** – Interpret Oceanographic & Environmental Conditions
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 005.02.10** – Sources of Ambient Noise (Interactive Lecture)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**EO 005.02.14** – SONAR Equations (Interactive Lecture)	**EC 005.02** – Interpret Oceanographic Conditions
Period 4 1040-1130	**EO 005.02.10** – Sources of Ambient Noise (Interactive Lecture)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**EO 005.02.14** – SONAR Equations (Interactive Lecture)	**EC 005.02** – Interpret Oceanographic Conditions
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 005.02.10** – Sources of Ambient Noise (Interactive Lecture)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**EO 005.02.14** – SONAR Equations (Interactive Lecture)	**EC 005.02** – Interpret Oceanographic Conditions
Period 6 1320-1410	**EO 005.02.10** – Sources of Ambient Noise (Interactive Lecture)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**EO 005.02.14** – SONAR Equations (Interactive Lecture)	**EC 005.02** – Interpret Oceanographic Conditions
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 005.02.11** – Foundations of Acoustics & Sound Propagation (Confirmation Check)	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.13** – Shallow Water Propagation (Interactive Lecture)	**EO 005.02.15** – Range Dependant Oceanographic Features & SONAR Equations (Confirmation Check)	**PC 005**
Period 8 1510-1600	**PT**	**EO 005.02.12** – Fronts & Eddies (Interactive Lecture)	**EO 005.02.14** – SONAR Equations (Interactive Lecture)	**EO 005.02.15** – Range Dependant Oceanographic Features & SONAR Equations (Confirmation Check)	**TAT**

Week #6

TIME	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30
Period 1 0800-0850	**PC 005**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**PT**	**PT**	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
Period 2 0850-0940	**PC 005**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 005**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
Period 4 1040-1130	**PC 005**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**ADAC Introduction & Administration (BAT)**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
Period 6 1320-1410	**ADAC Introduction & Administration (BAT)**	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EC 007.02** – Determine Mathematical Relationships
Period 8 1510-1600	**EO 007.01.01** – Interpret LOFAR Grams (OSPREY Lecture)	**PT**	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**EO 007.02.01** – Engineering Relationships (Lecture/EDI)	**TAT**

Week #7

TIME	DAY 31	DAY 32	DAY 33	DAY 34	DAY 35
Period 1 0800-0850	**EO 007.03.01** – PAL Techniques (EDI)	**PT**	**EO 007.03.01** – PAL Techniques (EDI)	**EC 007.03** – Perform PAL	**EO 007.04.01** – Five Steps to Analysis (Lecture)
Period 2 0850-0940	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EC 007.03** – Perform PAL	**EO 007.04.01** – Five Steps to Analysis (Lecture)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EC 007.03** – Perform PAL	**EO 007.04.01** – Five Steps to Analysis (Lecture)
Period 4 1040-1130	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EC 007.03** – Perform PAL	**EO 007.04.01** – Five Steps to Analysis (Lecture)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.04.01** – Five Steps to Analysis (Lecture)	**EO 007.04.01** – Five Steps to Analysis (Lecture)
Period 6 1320-1410	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.04.01** – Five Steps to Analysis (Lecture)	**EO 007.04.01** – Five Steps to Analysis (Lecture)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.04.01** – Five Steps to Analysis (Lecture)	**EO 007.04.01** – Five Steps to Analysis (Lecture)
Period 8 1510-1600	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**EO 007.03.01** – PAL Techniques (EDI)	**PT**	**PT**

Week #8

TIME	DAY 36	DAY 37	DAY 38	DAY 39	DAY 40
Period 1 0800-0850	**EC 007.04** – Five Steps to Analysis	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 2 0850-0940	**EO 007.05.01** – Passive Acoustic Data by Doppler (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**BAT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 007.05.01** – Passive Acoustic Data by Doppler (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 4 1040-1130	**EO 007.05.01** – Passive Acoustic Data by Doppler (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 007.05.01** – Passive Acoustic Data by Doppler (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 6 1320-1410	**EO 007.06.01** – Mixed Contact Labs (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.06.01** – Mixed Contact Labs (EDI)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 8 1510-1600	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**TAT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**

Week #9

TIME	DAY 41	DAY 42	DAY 43	DAY 44	DAY 45
Period 1 0800-0850	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 2 0850-0940	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 4 1040-1130	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 6 1320-1410	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 8 1510-1600	**PT**	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.03** – PAL Techniques (Interactive Lecture)	**TAT**

Week #10

TIME	DAY 46	DAY 47	DAY 48	DAY 49	DAY 50
Period 1 0800-0850	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 2 0850-0940	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 4 1040-1130	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 6 1320-1410	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)
Period 8 1510-1600	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)

Week #11

TIME	DAY 51	DAY 52	DAY 53	DAY 54	DAY 55
Period 1 0800-0850	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**	**EC 007.06** – Classify Passive Acoustic Contacts
Period 2 0850-0940	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EC 007.06** – Classify Passive Acoustic Contacts
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EC 007.06** – Classify Passive Acoustic Contacts
Period 4 1040-1130	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EC 007.06** – Classify Passive Acoustic Contacts
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**PT**
Period 6 1320-1410	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**TAT**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EC 007.06** – Classify Passive Acoustic Contacts (Debrief)
Period 8 1510-1600	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EO 007.06.01** – Mixed Contact Labs (Practice Labs)	**EC 007.06** – Classify Passive Acoustic Contacts

Week #12

TIME	DAY 56	DAY 57	DAY 58	DAY 59	DAY 60
Period 1 0800-0850	**PC 007**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**PT**	**EC 006.01B** – Perform HMS Closing Up Procedure
Period 2 0850-0940	**PC 007**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EC 006.01B** – Perform HMS Closing Up Procedure
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 007**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EC 006.01B** – Perform HMS Closing Up Procedure
Period 4 1040-1130	**PC 007**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EO 006.02.01** – HMS Logging and MMMP (Interactive Lecture)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**BAT**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EO 006.02.01** – HMS Logging and MMMP (Interactive Lecture)
Period 6 1320-1410	**TAT**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EC 006.01A** – Ensure HMS is Operational	**EC 006.01B** – Perform HMS Closing Up Procedure	**EO 006.02.01** – HMS Logging and MMMP (Interactive Lecture)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 007 (Debrief)**	**EO 006.01.01** – Conduct HMS Flash up (Ship Tour/EDI)	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EC 006.01B** – Perform HMS Closing Up Procedure	**EO 006.02.01** – HMS Logging and MMMP (Interactive Lecture)
Period 8 1510-1600	**PC 007 (Debrief)**	**PT**	**EO 006.01.02** – Conduct HMS Closing Up Procedure (EDI)	**EC 006.01B** – Perform HMS Closing Up Procedure	**PT**

Week #13

TIME	DAY 61	DAY 62	DAY 63	DAY 64	DAY 65

Period 1 0800-0850	**EO 006.02.02** – Operating HMS in Listen Mode (EDI)	**PT**	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**PT**	**EC 006.03** – HMS Contact Procedures
Period 2 0850-0940	**EO 006.02.02** – Operating HMS in Listen Mode (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**BAT**	**EC 006.03** – HMS Contact Procedures
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures
Period 4 1040-1130	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures	**EO 006.04.01** – Torpedo Procedures (EDI)
Period 6 1320-1410	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures	**EO 006.04.01** – Torpedo Procedures (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures	**EO 006.04.01** – Torpedo Procedures (EDI)
Period 8 1510-1600	**TAT**	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EO 006.03.01** – Initial, Lost Contact & Bearing Investigation (EDI)	**EC 006.03** – HMS Contact Procedures	**PT**

Week #14

TIME	DAY 66	DAY 67	DAY 68	DAY 69	DAY 70

Period 1 0800-0850	**EO 006.04.01** – Torpedo Procedures (EDI)	**PT**	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**PT**	**PC 006**
Period 2 0850-0940	**EO 006.04.01** – Torpedo Procedures (EDI)	**EC 006.04** – Perform Torpedo Procedures	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 006.04.01** – Torpedo Procedures (EDI)	**EC 006.04** – Perform Torpedo Procedures	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**
Period 4 1040-1130	**EO 006.04.01** – Torpedo Procedures (EDI)	**EC 006.04** – Perform Torpedo Procedures	**TAT**	**PC 006**	**PC 006**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 006.04** – Perform Torpedo Procedures	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**	**PC 006**
Period 6 1320-1410	**EC 006.04** – Perform Torpedo Procedures	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**	**PC 006**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 006.04** – Perform Torpedo Procedures	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**	**PC 006**
Period 8 1510-1600	**TAT**	**EO 006.05.01** – Mine Investigation Procedures (EDI)	**EC 006.05** – Perform Mine Investigation Procedures	**PC 006**	**PT**

Week #15

TIME	DAY 71	DAY 72	DAY 73	DAY 74	DAY 75
Period 1 0800-0850	**EO 008.01.01** – Set Up SPS (EDI)	**EC 008.01** – Set Up SPS	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**PT**	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
Period 2 0850-0940	**EO 008.01.01** – Set Up SPS (EDI)	**EC 008.01** – Set Up SPS	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**TAT**	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 008.01.01** – Set Up SPS (EDI)	**EC 008.01** – Set Up SPS	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
Period 4 1040-1130	**EO 008.01.01** – Set Up SPS (EDI)	**EC 008.01** – Set Up SPS	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 008.01.01** – Set Up SPS (EDI)	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**EC 008.02** – Ensure CANTASS is Operational	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
Period 6 1320-1410	**EO 008.01.01** – Set Up SPS (EDI)	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**EC 008.02** – Ensure CANTASS is Operational	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 008.01.01** – Set Up SPS (EDI)	**EO 008.02.01** – CANTASS Flash-up/Function Check (EDI)	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)
Period 8 1510-1600	**EO 008.01.01** – Set Up SPS (EDI)	**PT**	**EC 008.02** – Ensure CANTASS is Operational	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**PT**

Week #16

TIME	DAY 76	DAY 77	DAY 78	DAY 79	DAY 80
Period 1 0800-0850	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**PT**	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**PT**	**EC 008.03** – Set Up CANTASS Operating Parameters
Period 2 0850-0940	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**BAT**	**EC 008.03** – Set Up CANTASS Operating Parameters
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EO 008.04.01** – CANTASS Log (EDI)
Period 4 1040-1130	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EC 008.03** – Set Up CANTASS Operating Parameters	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)
Period 6 1320-1410	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EC 008.03** – Set Up CANTASS Operating Parameters	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EC 008.03** – Set Up CANTASS Operating Parameters	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)
Period 8 1510-1600	**TAT**	**TAT EO 008.03.01** – CANTASS Operating Parameters & Search Procedures (EDI)	**EC 008.03** – Set Up CANTASS Operating Parameters	**EC 008.03** – Set Up CANTASS Operating Parameters	**PT**

Week #17

TIME	DAY 81	DAY 82	DAY 83	DAY 84	DAY 85
Period 1 0800-0850	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**PT**	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PT**	**PC 008**
Period 2 0850-0940	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**BAT**	**PC 008**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**PC 008**
Period 4 1040-1130	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**TAT**	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**PC 008**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**PC 008**
Period 6 1320-1410	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**PC 008**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 008.05.01** – Track & Amplify Acoustic Contacts (EDI)	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**EO 009.01.01** – Torpedo Safety, Readiness & Voice Reports (Lecture)
Period 8 1510-1600	**PT**	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**EC 008.05** – Tracking Contacts & Amplifying Track Information	**PC 008**	**EO 009.01.01** – Torpedo Safety, Readiness & Voice Reports (Lecture)

Week #18

TIME	DAY 86	DAY 87	DAY 88	DAY 89	DAY 90
Period 1 0800-0850	**PT**	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EC 009.03** – Operate Torpedo Setting Panel	**PT**	**EO 009.04.01** – UWW Weapons Firing (EDI)
Period 2 0850-0940	**EO 009.01.01** – Torpedo Safety, Readiness & Voice Reports (Lecture)	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EC 009.03** – Operate Torpedo Setting Panel	**TAT**	**EC 009.04** – Perform UWW Weapon Firings
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 009.01.01** – Torpedo Safety, Readiness & Voice Reports (Lecture)	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EC 009.03** – Operate Torpedo Setting Panel	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EC 009.04** – Perform UWW Weapon Firings
Period 4 1040-1130	**TAT**	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EC 009.03** – Operate Torpedo Setting Panel	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EC 009.04** – Perform UWW Weapon Firings
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EO 009.03.01** – Torpedo Settings Panel (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EC 009.04** – Perform UWW Weapon Firings
Period 6 1320-1410	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EO 009.03.01** – Torpedo Settings Panel (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EC 009.04** – Perform UWW Weapon Firings
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EO 009.03.01** – Torpedo Settings Panel (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EC 009.04** – Perform UWW Weapon Firings
Period 8 1510-1600	**EO 009.02.01** – Assist in Loading/Unloading Torpedo (EDI)	**EO 009.03.01** – Torpedo Settings Panel (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**EO 009.04.01** – UWW Weapons Firing (EDI)	**PT**

Week #19

TIME	DAY 86	DAY 87	DAY 88	DAY 89	DAY 90
Period 1 0800-0850	**PC 009**	**BAT (Out Routine)**			
Period 2 0850-0940	**PC 009**	**BAT (Out Routine)**			
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 009**	**Course Graduation (BAT)**			
Period 4 1040-1130	**PC 009**	**Course Graduation (BAT)**			
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 009**	**BAT (Out Routine)**			
Period 6 1320-1410	**PC 009**				
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**BAT (Out Routine)**				
Period 8 1510-1600	**BAT (Out Routine)**				

ANNEX K –SONAR OP RQ S2 OJPR

CERTIFICATE OF QUALIFICATION RQ S2 SONAR Op

Name and Initials	Rank	SN

Date Training Started	Required Date of Completion (12 months from start date)

NOTES

- The OJPR must be completed in accordance with the instructions below.
- Incorrect or incomplete OJPR will be returned for re-submission.

INSTRUCTIONS

1. The current SONAR Operator RQ-SONAR OP-S2 OJPR is found within the RQ-SONAR OP-S3 QSP ANNEX K and is applicable to Surface Fleet SONAR Operators. Unit Training Petty Officers requiring an OJPR copy shall print off one from the QSP Annex or submit a request to their Naval Fleet School (Atlantic / Pacific) or the Naval Training Development Centre (Atlantic).

2. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the trainee and monitored regularly by the supervisor. The trainee's Training PO or Officer is responsible for the member's completion of the OJPR and must maintain Training Tracker fields as outlined in Monitor Mass (MM).

3. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO or Officer will date and sign the applicable item on the master copies of the OJPR sheet and inform the individual of the action taken.

4. Upon posting or transfer, the losing unit will ensure the following:

   a. The trainee's OJPR is reviewed to ensure that all of the objectives which have been successfully completed are credited;
   b. A duplicate copy of the original should be made and entered into the member's PERs file for safe keeping; and
   c. The master copy OJPR is forwarded with the member to the new unit.

5. The gaining unit shall review both the master and duplicate copies of the OJPR package to become familiar with:

a. Percentage of objectives completed;
b. Remarks; and
c. OJT time elapsed.

6. The Training PO will develop an appropriate training plan to meet the Performance Objectives and assist the trainee in attaining ALFU qualification. The Training PO is to regularly review the plan (minimum once every 30 days) to ensure that the trainee is progressing appropriately. The trainee shall be counselled on this review and any training decisions.

PROGESS REVIEW

7. Progress review shall be carried out monthly by the Training Petty Officer and Section Head (NCMs).

8. Unsatisfactory progress is indicated by:

a. The trainee experiencing difficulty successfully completing the objectives; and/or
b. The trainee failing to complete this OJPR package within the allotted 12 month time period.

9. Trainees experiencing difficulty will be counselled on their performance.

10. Should a member require additional time to complete the OJPR, the Training Petty Officer will submit a request for approval on an extension of time to their Chain of Command and appropriate TE to include:

a. Justification for delay in completion;

b. Estimation of time of completion; and

c. Training Plan to meet the new time estimate.

11. Trainees experiencing difficulty will be counselled on their performance.

OJPR Reviewed By:	Comment	Signature	Date

INSTRUCTIONS AND CHECKLISTS

12. The following checklists are to be used to evaluate the trainee's ability to perform the tasks and demonstrate the skills necessary to fulfil the requirements of this OJPR:

a. OJPR PC 002 – Perform Noise Rounds;

b. OJPR PC 003A – Operate Ancillary Acoustic Equipment;

c. OJPR PC 003B – Communicate Externally on Underwater Telephone;

d. OJPR PC 004 – Launch Underwater Expendables (Sonobuoy & ESUS);

e. OJPR PC 006 – Operate Active SONAR Equipment;

f. OJPR PC 007 – Perform Basic Passive Acoustic Analysis;

g. OJPR PC 008 – Operate Passive Acoustic Equipment; and

h. OJPR PC 009 – Operate Surface Vessel Torpedo Tubes (SVTT) Weapon Systems.

13. Within the period of approximately 12 months, the trainee shall be required to perform all of the activities listed on the OJPR PC checklists under the observation, supervision, and evaluation of a supervisor. The OJT supervisor will debrief the trainee following each observed session.

14. The checklists reflect the required performance for the skills/tasks specified in the POs contained in chapter 2. The OJT supervisor shall complete the checklists for all observed activities until the trainee has successfully completed all activities without assistance, once an observed performance reach standard, the supervisor will complete the OJPR record sheet.

15. When for reasons of refits, sailing schedules, etc., it becomes impossible to carry out tasks on the trainee's own ship, he/she should be given every opportunity to carry on with his/her OJPR package on an operational ship or using TE's trainers.

16. A trainee must obtain a pass in all OJT EOs/POs to successfully complete this training package and be granted the Acoustic Sensor Operator (ALFU) qualification.

OJPR COMPLETION INSTRUCTIONS

17. Once the trainee has completed all of the Performance Objectives (POs) within the OJPR, the Training PO will ensure the following personnel sign the trainee's master copy OJPR:

a. UWW Officer / Director;

b. Operations Officer; and

c. Operations Chief / Unit Training Chief.

18. Once all signatures have been received, a duplicate copy of the OJPR will be made. The master copy being entered into the members PERs file and the duplicate copy to the member.

19. The Training PO / Officer or Unit Training Chief will complete the Training Tracker fields in Monitor Mass indicate the date the member completed the OJPR. This will send a message to the appropriate TE (NFS A or P) Scheduling MITE cell who will enter the MITE code 122419 for the completion of the OJPR and the member will be granted the National Qualification code ALFU.

20. The Unit Training Chief will also send an email with the members information (SN, Rank, Name, Trade, OJPR, and completion date) to the following personnel:

a. +NFSA MITE@NPTG NFSA@Halifax NFSAMITE@intern.mil.ca;

b. +ESQNPTG NFSP MITE@NPTG NFSP@Esquimalt

P-ESQ.NPTGNFSPMITE@intern.mil.ca; and

c. DMIL C 2-6-2 (Combat Operator S1 & Below Career Manager).

21. The Training PO / Officer and/or Unit Training Chief should check to ensure the members has received the MITE code in their Monitor Mass/MPRR or Guardian system. If the member has not received the qualification within 6 weeks of completion, contact the Training Establishment MITE cells listed above.

Note: Units at sea may have to wait until return to port for Monitor Mass Training Tracker to update and send the OJPR completion query to the appropriate TE MITE cells.

Performance Check PC 002 Perform Noise Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine.

Arrangements and Procedures

2. PC 002 will be a practical assessment based on content from PO 002.

3. To pass this PC, the trainee must achieve 19/19 SATs.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will be required to have a knowledge of the Noise Monitoring Organization, be aware of publications/logs, perform noise rounds (alongside, daily at sea, assuming a higher state) and correctly log any faults to be corrected. You must pass this PC to pass PO 002.

3. References will be provided.

4. Pass mark is 19/19 SATs IAW PCL 002.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

6. Assessment will occur on Ship or Submarines.

7. Trainees will be required to attend one CRR 12-3 Noise Organization Lecture, be aware of publications/logs, perform noise rounds (alongside, daily at sea, assuming a higher state) and correctly log any faults to be corrected.

PROTECTED B (when completed)

ITEM	PC 002 PERFORM NOISE ROUNDS	SAT	DATE	RANK / NAME (SONAR OP MS & above)	REMARKS
1. Noise Monitoring Organization					
Attend CRR 12-3 Noise Reduction Lecture		☐			
Identifies Department Noise Monitor		☐			
Identifies location(s) of Quiet State signs		☐			
2. Noise Monitoring Logs & Documents					
Knows location of Noise Rounds Log		☐			
Knows location of Noise Defect Log		☐			
Knows location of Ships Artifact Log		☐			
Knows location of Quiet Running Bill		☐			
Fills out Noise Defect form		☐			
3. Conducting Rounds					
Conducts Secure for Sea Rounds (add date)		☐			
Conducts Daily Sea Rounds (add date)		☐			
Conducts Daily Sea Rounds (add date)		☐			
Conducts Daily Sea Rounds (add date)		☐			
Fills out log for Daily Sea Rounds		☐			
Conducts rounds assuming a Higher Quiet state (add date)		☐			
Changes signs for assuming Higher Quiet State		☐			
Fills out log for assuming a Higher Quiet State		☐			
Identifies 3 noise defects during rounds		☐			
Rectifies noise defects during rounds		☐			
Reports rounds completed to DNM		☐			

Score: /19

The trainee must achieve all items on this checklist to pass this PC.

Positive Feedback	Improvement Feedback

PC 002 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 002 with the trainee.		**Trainee** I have been debriefed on the result of PC 002.	
PC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Performance Check – PC 003A Operate Ancillary Acoustic Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine alongside or at sea.

Arrangements and Procedures

2. PC 003 will be a practical assessment based on content from PO 003A.

3. To pass this PC, the trainee must achieve 18/23 SATs.

4. References are allowed.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a practical assessment based on content from PO 003A.

2. During this practical assessment, you will be required to operate ancillary acoustic equipment and correctly draft and upload a Bathy Message. You must pass this PC to receive your signature.

3. References are allowed.

4. No assistance is permitted.

5. Pass mark is 18/23 SATs IAW PCL 003.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. PC Checklist for XBT/XSV launches from Submarine may be amended to reflect their launching SOP's.

PROTECTED B (when completed)

Item	PC 003A ANALYZE XBT/XSV FORMS	SAT	UNSAT	Remarks
1.	**Flash up MK-21 Bathythermograph:**			
a.	Turn on the computer	☐	☐	
b.	Initialize WinMK21 program	☐	☐	
2.	**Set up operating parameters**			
a.	Open WinMK21 Main Window	☐	☐	
b.	Ensure WinMK21 is in Data acquisition mode	☐	☐	
3.	**Initiate a new probe drop**			
a.	Click the New Drop button	☐	☐	
b.	Choose New Drop from the File menu	☐	☐	
4.	**Select the probe type**			
a.	Choose Probe from the Options menu	☐	☐	
b.	Choose Selection	☐	☐	
5.	**Verify the data file name**			
a.	Verify the correct file name for the data and the folder in which the file will be	☐	☐	
6.	**Load/Launch the probe**			
a.	Complete a full Load of probe	☐	☐	
b.	Pulling the release pin out of the canister	☐	☐	
c.	Wait for the probe to complete its descent or Stop data acquisition, by choosing *Terminate* from the *Actions* menu	☐	☐	
7.	**Record XBT/XSV information IAW METOC standards**			
a.	Select correct profile type	☐	☐	
b.	Review data to ensure a successful launch	☐	☐	
8.	**Complete standard Bathymetric forms (C0045)**			
a.	Complete a digital IGOSS form	☐	☐	
b.	Complete a digital VELO form	☐	☐	
9.	**Draft Bathy messages**			
a.	Input data (temperature, sound velocity)	☐	☐	
b.	Input ship call sign	☐	☐	
c.	Submit message for transmission	☐	☐	
10.	**Shut down MK-21 Bathythermograph:**			
a.	Choose Exit from the File menu	☐	☐	
b.	Press Alt+F4	☐	☐	
c.	Click the Windows Close button in the title bar	☐	☐	
d.	Turn off the computer	☐	☐	

Score: /23

The trainee must achieve 18/23 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 003A – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 003A with the trainee.	**Trainee** I have been debriefed on the result of PC 003A.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Performance Check – PC 003B Communicate Externally on Underwater Telephone

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine at sea; and

2. Another unit to communicate with.

Arrangements and Procedures

3. PC 003B will be a practical assessment based on content from SONAR OP RQ-S3 QSP PC 003B.

4. To pass this PC, the trainee must achieve 30/30 SATs.

5. References are allowed.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003B will be a practical assessment based on content from PC 003B.

2. During this practical assessment, you will be required to communicate externally on the Underwater Telephone. You must pass this PC to receive your signature.

3. References are allowed.

4. No assistance is permitted.

5. Pass mark is 30/30 SATs IAW PCL 003B.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. PC Checklist for Submarine UWT communications may be amended to reflect their launching SOP's

PROTECTED B (when completed)

Item	PC 003B COMMUNICATE EXTERNALLY ON UNDERWATER TELEPHONE	SAT	UNSAT	Remarks
1. Conduct Flash-up/Shut-down				
a.	Ensure module is powered and push button "Power on".	☐	☐	
b.	Turn key "ON" in OPS and Bridge.	☐	☐	
c.	Plugged headset and Microphone.	☐	☐	
d.	Ensure Programmable Channel recording is active.	☐	☐	
e.	Selected channel is in passive mode.	☐	☐	
f.	Ensure "Status On" is green.	☐	☐	
2. Conduct messages transmission				
a.	Recording a message than TX to include;			
	• Press RECORD	☐	☐	
	• Push-to-Talk on microphone	☐	☐	
	• Press REPLAY than PLAY	☐	☐	
	• Press FILE than SAVE	☐	☐	
	• Choose DELAY	☐	☐	
	• Push CALL to transmit.	☐	☐	
b.	Choose Pre-set messages to TX to include;			
	• Push MESSAGE mode than OK	☐	☐	
	• Press FILE to choose msg than press OK	☐	☐	
	• Choose DELAY x2	☐	☐	
	• Choose a DELAY value between 30 and 300	☐	☐	
	• Press PLAY to confirm msg	☐	☐	
	• Press OK than CALL to tx	☐	☐	

3. Conduct UWT Communications (with another unit)			
a. Establish Communications (GERTRUDE CHECK)	☐	☐	
b. Utilize 3 letter codes (minimum 3 codes)	☐	☐	
4. Employ Mark-Snap ranging procedures (with another unit if available)			
a. Measure time	☐	☐	
b. Calculate ranges	☐	☐	
c. Calculate range to contact	☐	☐	
5. Conduct Emergency Communication Procedures			
a. Employ Uncle Joe Voice Procedure IAW AXP 1	☐	☐	
b. Employ Hooker Procedure IAW AXP 1	☐	☐	
c. Employ Uniform Procedure IAW AXP 1	☐	☐	
6. Log UWT communications			
a. Enter time of communication	☐	☐	
b. Enter unit(s) conducting communication	☐	☐	
c. Enter communication / response(s)	☐	☐	
7. Shut down UWT			
a. Switch off UWT	☐	☐	

Score: /33

The trainee must achieve all items on this checklist to pass this PC.

Positive Feedback	Improvement Feedback

PC 003B – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 003B with the trainee.		**Trainee** I have been debriefed on the result of PC 003B.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Performance Check – PC 004 Launch Underwater Expendables

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship at sea.

Arrangements and Procedures

2. PC 004 will be a practical assessment based on content from PO 004.

3. To pass this PC, the trainee must achieve 14/14 SATs.

4. Instructors must ensure that the Sonobouy & ESUS are still contained within the packaging to correctly evaluate trainees ability to remove the packaging correctly.

5. References are not allowed.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 will be a practical assessment based on content from PO 004.

2. During this practical assessment, you will be required to launch a Sonobuoy & ESUS. You must pass this EC to receive your signature.

3. References are not allowed.

4. No assistance is permitted.

5. Pass mark is 12/12 SATs IAW PCL 004.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. PC Checklist for Sonobuoy and ESUS launches from Submarine may be amended to reflect their launching SOP's.

Item	PC 004 LAUNCH UNDERWATER EXPENDABLES	SAT	UNSAT	Remarks
LAUNCH SONOBUOY CHECKLIST				
1. Safely Handle Sonobuoy				
(*) Handle and transport Sonobuoy safely.		☐	☐	
2. Prepare Sonobuoy For Launch				
Remove Sonobuoy from over pack		☐	☐	
Remove parachute (if required)		☐	☐	
Input ordered parameters		☐	☐	
Confirm parameter inputs		☐	☐	
(*) Deploy Sonobuoy correctly		☐	☐	
3. Maintain operation logs (B0055)				
Complete Ship Expenditure Log		☐	☐	
LAUNCH ESUS CHECKLIST				
1. Safely handle ESUS				
Handle and transport ESUS safely.		☐	☐	
3. Prepare ESUS for launch				
Select code as directed		☐	☐	
Puncture water seals		☐	☐	
4. Maintain operational logs (B0055)				
Complete Ship Expenditure Log		☐	☐	
5. Verify the shipboard stock against the BAAL entitlement				
Verify the shipboard stock against the BAAL entitlement		☐	☐	

Score: / 12

The trainee must achieve all items on this checklist to pass this PC.

Positive Feedback	Improvement Feedback

PC 004 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 004 with the trainee.	**Trainee** I have been debriefed on the result of PC 004.

EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Performance Check – PO 006 Operate Active SONAR Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine at sea;

2. Embedded Training System (ETS); or

3. SEACOT or Submarine Trainer.

Arrangements and Procedures

4. PC 006 will be a practical assessment based on content from PO 006.

5. To pass this PC, the trainee must achieve 10 accumulative hours of Operating time and 2 accumulative hours of In Contact time IAW PO 006 Active SONAR Time Log.

6. Surface ship Training PO's / Instructor may use RQ S3 PC 006 Checklist as reference for evaluating trainee's ability to operate Ship's Active SONAR Equipment.

7. References are allowed.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 006 will be a practical assessment based on content from PC 006.

2. During this practical assessment, you will be required to set up, operate active SONAR equipment and correctly maintain logs. You must complete the require amount of operating and in contact time and pass this PC to receive your signature.

3. References are allowed.

4. No assistance is permitted.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Nil.

PO 006 - Active SONAR Time Log

_____

Trainee Name and Initials Rank SN

PO 006 Operate Active SONAR Equipment			
OPERATING TIME			
1 Hour	Date:		Remarks:
	Supervisor Initials:		
2 Hour	Date:		Remarks:
	Supervisor Initials:		
3 Hour	Date:		Remarks:
	Supervisor Initials:		
4 Hour	Date:		Remarks:
	Supervisor Initials:		
5 Hour	Date:		Remarks:
	Supervisor Initials:		
6 Hour	Date:		Remarks:
	Supervisor Initials:		
7 Hour	Date:		Remarks:
	Supervisor Initials:		
8 Hour	Date:		Remarks:
	Supervisor Initials:		
9 Hour	Date:		Remarks:
	Supervisor Initials:		
10 Hour	Date:		Remarks:
	Supervisor Initials:		
IN CONTACT TIME			
1 Hour	Date:		Remarks:
	Supervisor Initials:		
2 Hour	Date:		Remarks:
	Supervisor Initials:		

*** The trainee must complete 10 hrs of Operating with a minimum of 2hrs in contact before conducting the Performance Check.**

Performance Check – PO 007 Perform Basic Passive Acoustic Analysis

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Limited Access Area, TRINITY ADAC training packages.

Arrangements and Procedures

2. PC 007 will be a practical assessment based on content from PO 007.

3. To pass this PC, the trainee must achieve 60% on 6 training packs.

4. References are allowed.

5. No assistance is permitted.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 007 will be a practical assessment based on content from PO 007.

2. During this practical assessment, you will be required to perform basic passive acoustic analysis. You must receive the pass mark for 6 training packs to receive your signature for this PC.

3. References are allowed.

4. No assistance is permitted.

5. Pass mark is 60%.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Each training pack will be a mixed lab consisting of 4 Grams, passive aural listening (PAL) and TIES.

2. Logs need to be correctly filled out.

3. Trainee will be required to analyze a Gram IAW the assessment criteria below:

UNCLASSIFIED

Contact Number: _____ **Contact Location:** _____ **Measurement Date / Time:** _____

Sensor: _____ **Sensor Depth (ft):** _____ **SLD (ft):** _____ **AN:** _____

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPPM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry.... ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2%	
tolerance	
Class Indif/D Swath	
1	1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Drive Mode	RR
D/E	

Shaft and Engine Related Sources

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed
Class Indicators		
1	1	1
Modulation/Banding		
1	1	1
Others		
	1	

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1

Transients, Significant Events, Tactical Info

Time	Details

1 pt for each *significant* event listed

Aural Characteristics

Time	Details

1 pt for each significant aural characteristic listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1

1/2 harmonics must be listed

Augmentors 1

for the frequency and calling it an augmentor

Auxiliaries

1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter

Others 1

1pt awarded to other significant sources

Unknowns 0

Classification

4 points total

1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

Performance Check – PO 008 Operate Passive Acoustic Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine at sea;

2. CANTASS Mission Simulator;

3. SEACOT trainer; or

4. Submarine trainer.

Arrangements and Procedures

5. PC 008 will be a practical assessment based on content from PO 008.

6. To pass this PC, the trainee must:

   a. Demonstrate they can set up the Passive Acoustic Equipment; and

   b. Achieve 10 accumulative hours of Operating time and 2 accumulative hours of In Contact time IAW PO 008 Passive SONAR Time Log.

7. Surface ship Training PO's / Instructor may use RQ S3 EC 008.01 and PC 008 Checklists as reference for evaluating trainee's ability to set up and operate Ship's Passive SONAR Equipment.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 008 will be a practical assessment based on content from PO 008.

6. During this practical assessment, you will be required to set up, operate passive SONAR equipment and correctly maintain logs. You must complete the require amount of operating and in contact time and pass this PC to receive your signature.

2. References are allowed.

3. No assistance is permitted.

4. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Total Operating and In Contact time can be accumulated from at sea or trainer time.

PO 008 - Passive SONAR Time Log

_____

Trainee Name and Initials Rank SN

PO 008 Operate Passive SONAR Equipment			
OPERATING TIME			
1 Hour	Date:		Remarks:
	Supervisor Initials:		
2 Hour	Date:		Remarks:
	Supervisor Initials:		
3 Hour	Date:		Remarks:
	Supervisor Initials:		
4 Hour	Date:		Remarks:
	Supervisor Initials:		
5 Hour	Date:		Remarks:
	Supervisor Initials:		
6 Hour	Date:		Remarks:
	Supervisor Initials:		
7 Hour	Date:		Remarks:
	Supervisor Initials:		
8 Hour	Date:		Remarks:
	Supervisor Initials:		
9 Hour	Date:		Remarks:
	Supervisor Initials:		
10 Hour	Date:		Remarks:
	Supervisor Initials:		
IN CONTACT TIME			
1 Hour	Date:		Remarks:
	Supervisor Initials:		
2 Hour	Date:		Remarks:
	Supervisor Initials:		

*** The trainee must complete 10 hrs of Operating with a minimum of 2 hrs in contact before challenging the test.**

Performance Check – PO 009 Operate Surface Vessel Torpedo Tubes (SVTT) Weapon Systems

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Submarine.

Arrangements and Procedures

2. PC 009 will be a practical assessment based on content from PO 009.

3. To pass this PC, the trainee must achieve 43/43 SATs AND all critical items.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 009 will be a practical assessment based on content from PO 009.

2. During this practical assessment, you will be required to operate SVTT weapon systems. You must pass this PC to pass PO 009.

3. References will be provided.

4. Pass mark is 43/43 SATs AND all critical items IAW PCL 009.

5. There is no time limit for this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

7. Assessment will occur as part of the RQ S2 OJPR onboard HMC Ship, with a WEng Tech (SONAR) present.

8. Safety is most important factor, any safety infractions can result in trainee failing to meet Performance Objective.

9. PC Checklist for Weapon launches from Submarine may be amended to reflect their launching SOP's.

Item	PC 009 OPERATE SVTT WEAPON SYSTEMS	SAT	UNSAT	Remarks
CONDUCT TORPEDO LOAD / UNLOAD (under qualified RQ MS WENG supervision)				
1.	Participate in one CRR 11-16 Torpedo Load / Unload training event	☐	☐	
2.	Adhere to Torpedo Loading / Unloading Checklist IAW NCPM 1001	☐	☐	
SYSTEM PREPARATION				
Personnel and Equipment safety maintained throughout				
3.	Possession of TSP keys.	☐	☐	
4.	Ensure power is available.	☐	☐	
5.	Ensure safety equipment is in place and appropriate precautions have been taken.	☐	☐	
6.	Conduct Communication checks	☐	☐	
ASSUMING "STANDING TO STATE" FROM "SECURED" STATE				
7.	Clear away torpedo tubes:			
a.	Check all breech ward access opening covers are secured	☐	☐	
b.	Ensure flex hoses are connected	☐	☐	
c.	Check breech mechanism air pressures are 1600 psi (+ 0 / - 50 psi)	☐	☐	
d.	Check manifold air valve is open (lower barrel) and ship air supply is connected to the tube pneumatic system (manifold gauge indicates between 100 - 150 psi).	☐	☐	
e.	Ensure muzzle door heater switch is turned "ON" when ambient temperature drops below 4.4 degrees Celsius	☐	☐	
f.	Operate manual muzzle door open switches on all tubes. Confirm doors are open and valve operating mechanism move to breech ward end.	☐	☐	
g.	Close muzzle doors —ensure control valve operating mechanism moves to muzzle ward position	☐	☐	
h.	Unlock padlocks on emergency fire levers and leave padlocks in place	☐	☐	
i.	Place tube ready actuator for all tubes to the ON (remote) position	☐	☐	
j.	Ensure AC and DC circuit breakers are ON	☐	☐	
k.	Switch TSP to REMOTE	☐	☐	
l.	Conduct Closing up report to UWWD IAW NCPM 1001	☐	☐	

APPLY ORDERED SETTINGS – LOCAL MODE			
8. Prepare to fire in secondary (select tube and settings as directed) :			
a. Switch TSP to LOCAL when ordered	☐	☐	
b. Report "Standing to in LOCAL"	☐	☐	
c. Selects Tube #	☐	☐	
d. Sets mode (RUNOUT / NO RUNOUT)	☐	☐	
e. Sets Initial Search Depth	☐	☐	
f. Sets Tactical Floor	☐	☐	
g. Sets Gyro Angle	☐	☐	
9. **Trainee will be directed to make tubes "READY":**			
a. Place actuator switch to Open position	☐	☐	
b. Selects STANDBY	☐	☐	
c. Confirms weapon ready to fire from TSP	☐	☐	
10. **Trainee will be directed to "FIRE"**			
a. Selects LAUNCH button	☐	☐	
b. Confirms weapon status on TSP	☐	☐	
11. Conducts Secondary Firing Drill voice reports IAW NCPM 1001	☐	☐	
TORPEDO MISFIRE PROCEDURES "FIRE IN EMERGENCY"			
12. Misfire is reported in secondary mode, directed to "FIRE IN EMERGENCY"			
a. Place actuator switch to Open	☐	☐	
b. Removes Padlock from Emergency Fire Lever	☐	☐	
c. Pulls Firing Lever	☐	☐	
d. Confirms weapon status from tube	☐	☐	
13. Conducts Emergency Firing Drill voice reports IAW NCPM 1001	☐	☐	
SECURE TORPEDO TUBES			
14. When Ordered, make a loaded torpedo tube mechanically safe:			
a. Lock padlocks on the EMERGENCY FIRING LEVERS	☐	☐	
b. Ensure control valve operating lever is in the muzzle ward position and muzzle door indicator light is out	☐	☐	
c. Discharge the breech mechanisms on loaded tubes	☐	☐	
d. Disconnect the flex hose	☐	☐	
e. Switch TSP to LOCAL, and turn the AC Power switch to OFF	☐	☐	
f. Conducts "SECURED" state voice report IAW NCPM 1001	☐	☐	
g. Return keys on completion	☐	☐	
Score: /43			

The trainee must achieve all items on this checklist to pass this PC.			
Positive Feedback		**Improvement Feedback**	
PC 009 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 009 with the trainee.		**Trainee** I have been debriefed on the result of PC 009.	
EC Administrator	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

CERTIFICATION OF COMPLETION

22. This On Job Training Performance Record (OJPR) certifies that the trainee:

SN	Rank	Name	Init	Unit

have successfully completed all of the performances required to receive the qualification ALFU.

On-Job Training Performance Record				
PC	**Title**	**Date**	**Signature**	**Comments**
002	Perform Noise Rounds		**(OJT Supervisor)**	
003A	Operate Ancillary Acoustic Equipment		**(OJT Supervisor)**	
003B	Communicate Externally on Underwater Telephone		**(OJT Supervisor)**	
004	Launch Underwater Expendables (Sonobuoy & ESUS)		**(OJT Supervisor)**	
006	Operate Active SONAR Equipment – Time Log		**(OJT Supervisor)**	
007	Perform Basic Passive Acoustic Analysis		**(OJT Supervisor)**	
008	Operate Passive Acoustic Equipment – Time Log		**(OJT Supervisor)**	
009	Operate Surface Vessel Torpedo Tubes (SVTT) Weapon Systems		**(OJT Supervisor)**	

OJPR Trainee, Name, Rank **Date**

UWWD / UWWO, Name, Rank **Date**

Ops O, Name, Rank	**Date**
Ops Chief, Name, Rank	**Date (Email sent)**
NFS(A) / NFS(P) MITE cell, Name, Rank	**Date**